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LEMKE et al.(10) **Pub. No.: US 2025/0277249 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **ARCHAEAL PYRROLYSYL TRNA
SYNTHETASES FOR ORTHOGONAL USE***C12N 15/113* (2010.01)*C12N 15/85* (2006.01)(71) Applicant: **European Molecular Biology
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The present invention is concerned with efficient orthogonal translation systems for eukaryotic cells. Specifically, the invention relates to eukaryotic cells comprising polynucleotides encoding a prokaryotic tRNA synthetase (pRS) and an RNA molecule comprising a prokaryotic tRNA (ptRNA) and one or more than one ribozyme (termed "ptRNA-ribozyme" herein). It also relates to a method for expressing a polypeptide of interest (POI) comprising one or more non-canonical amino acid (ncAA) residues by such eukaryotic cells, and to kits useful in such method. The invention further relates to polynucleotides encoding the ptRNA-ribozyme.

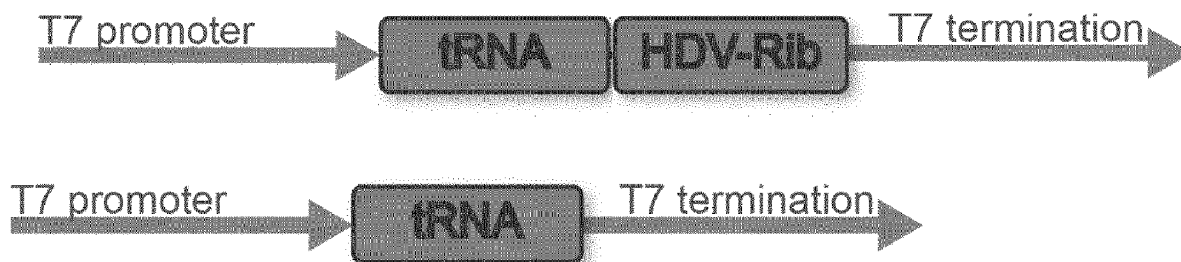
Specification includes a Sequence Listing.

Figure 1

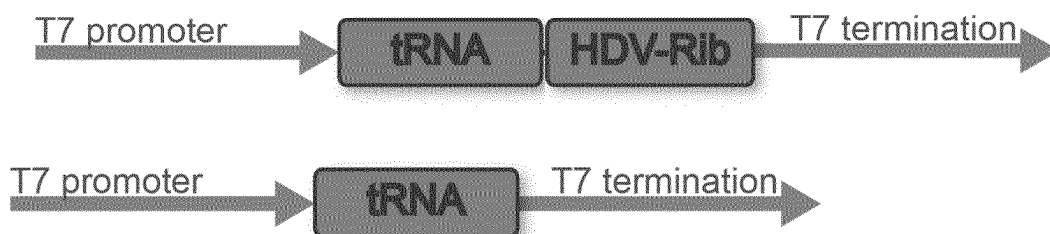


Figure 2

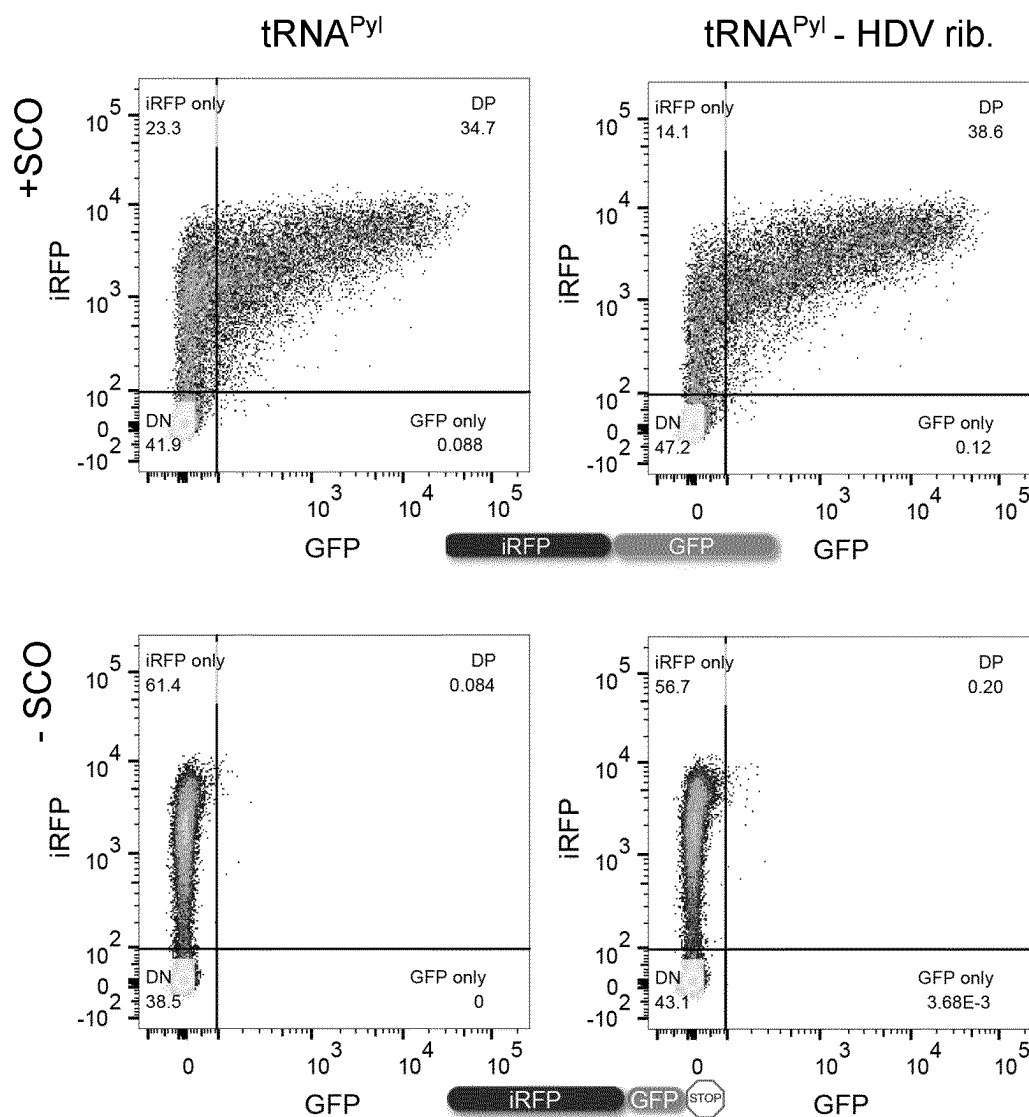


Figure 3

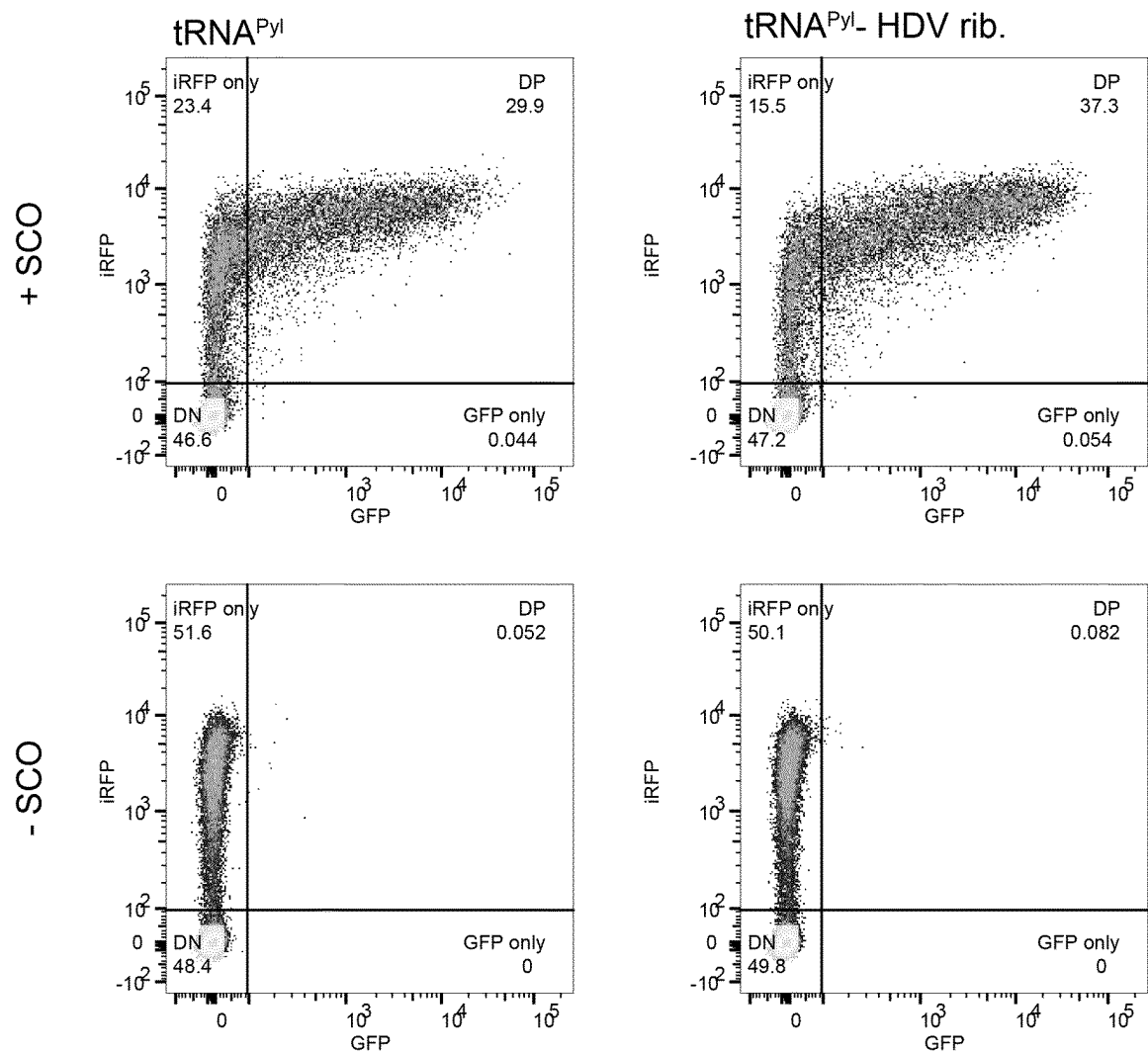
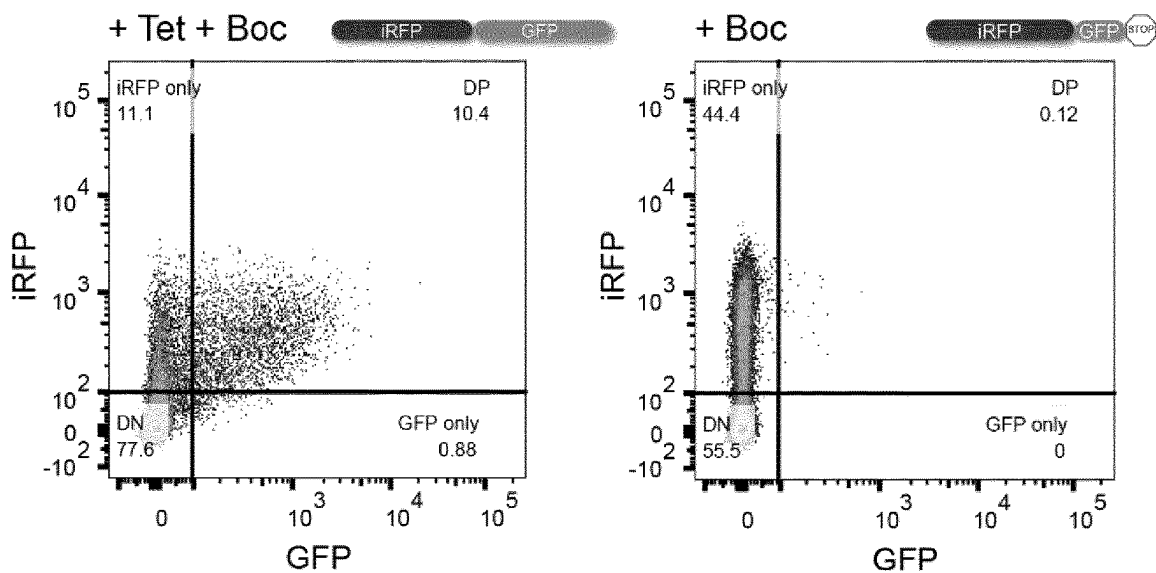


Figure 4

a) T-REx



b) Tet-On

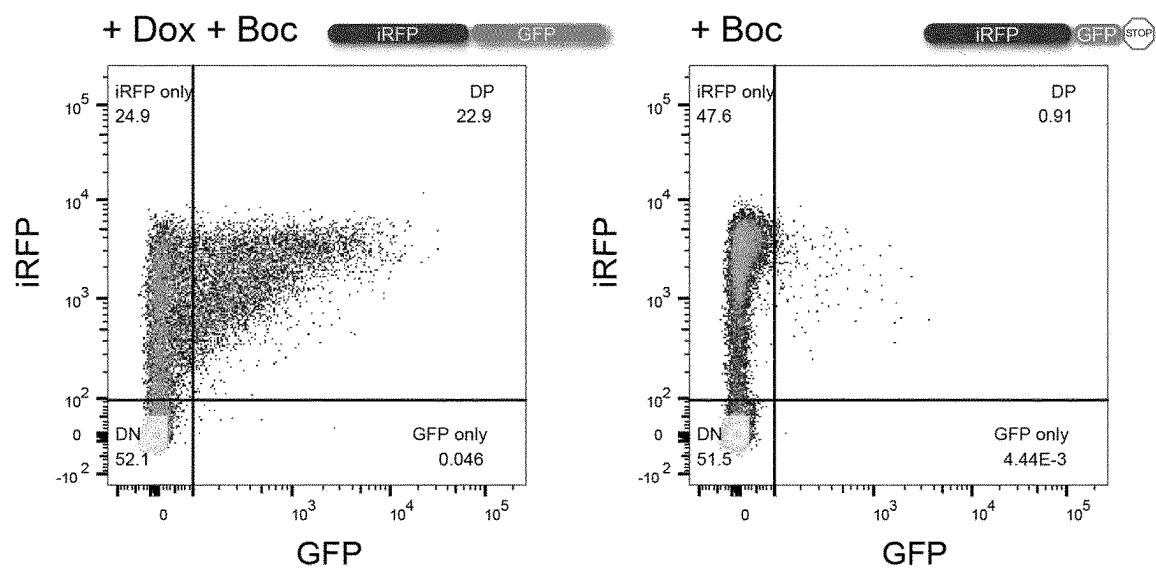
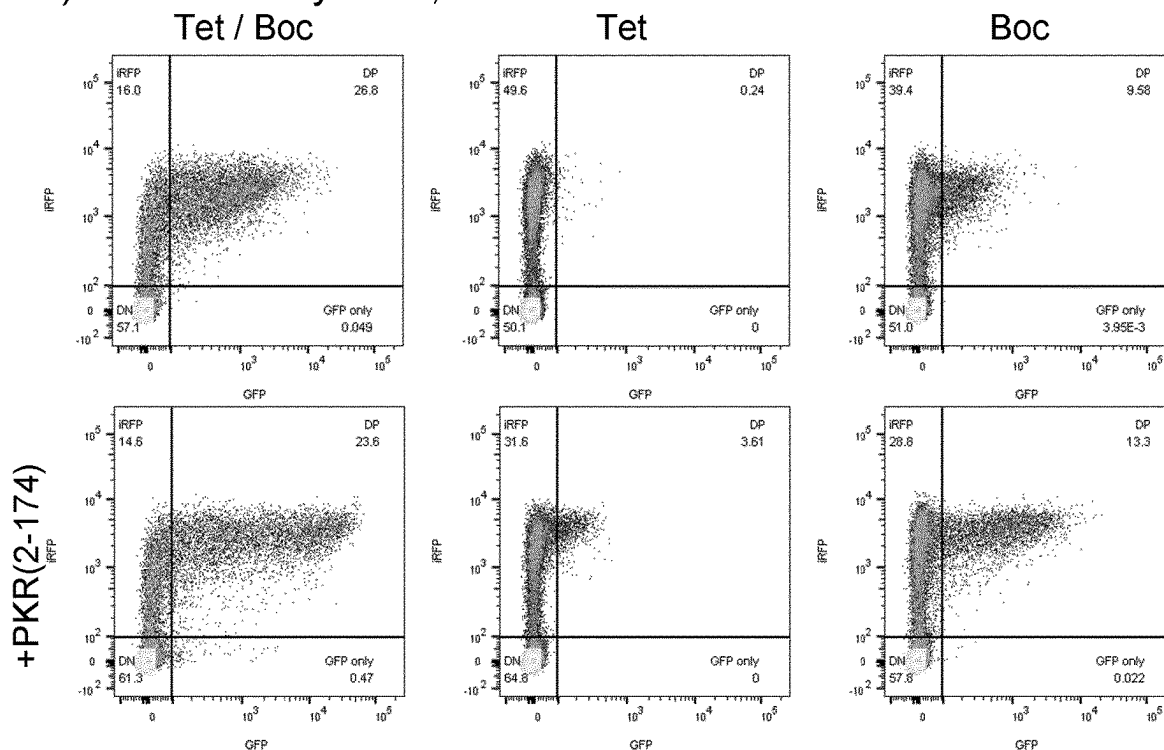


Figure 5

a) CMV-NES-PyIRS^{AF}, 8xtetO-U6-tRNA^{Pyl}



b) CMV-NES-PyIRS^{AF}, H1-tetO-tRNA^{Pyl}

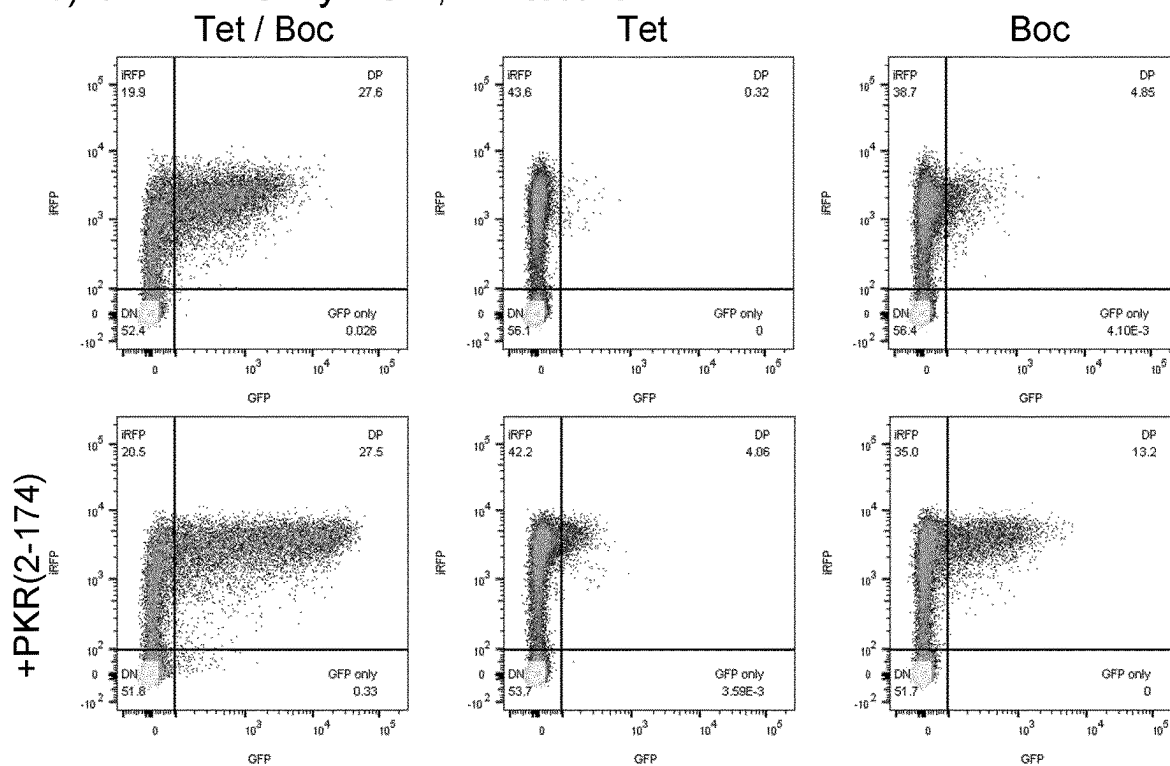


Figure 5 (continued)

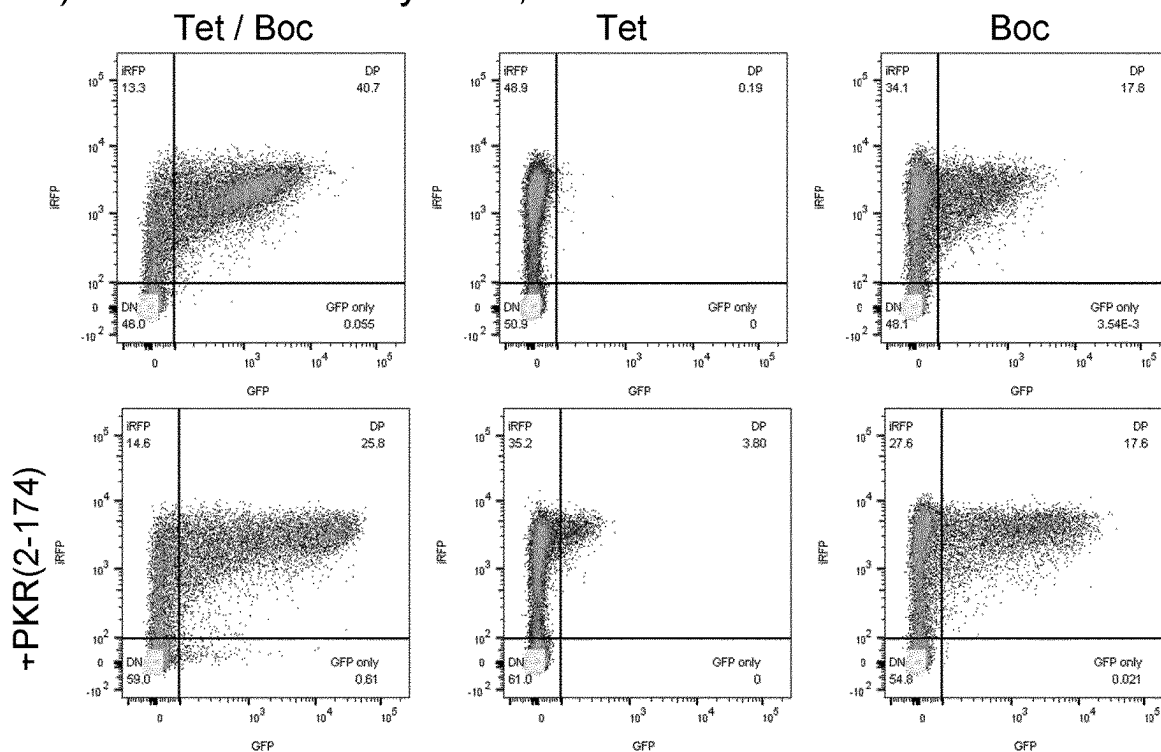
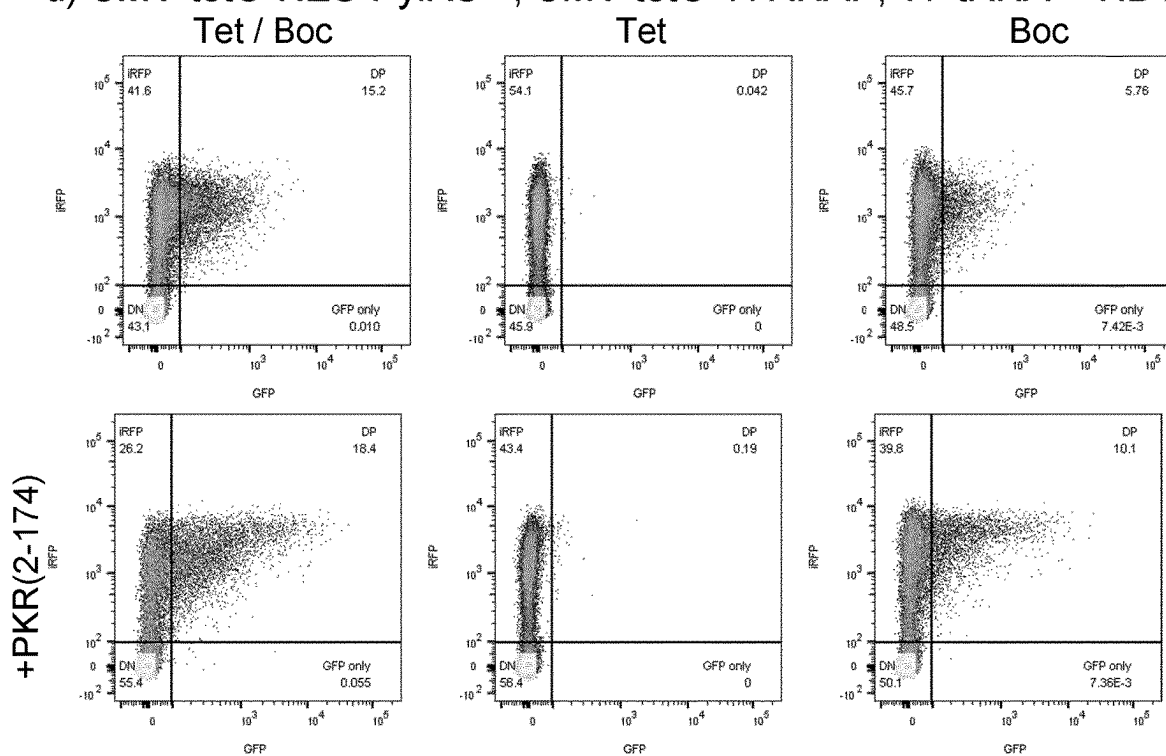
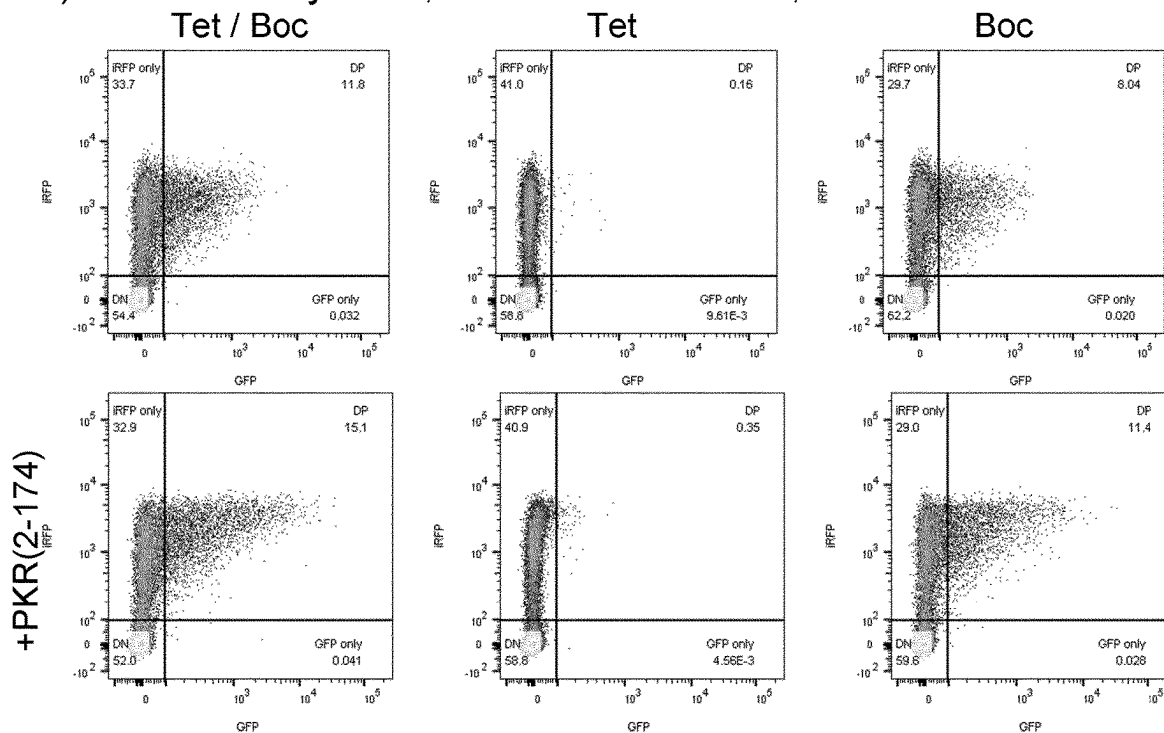
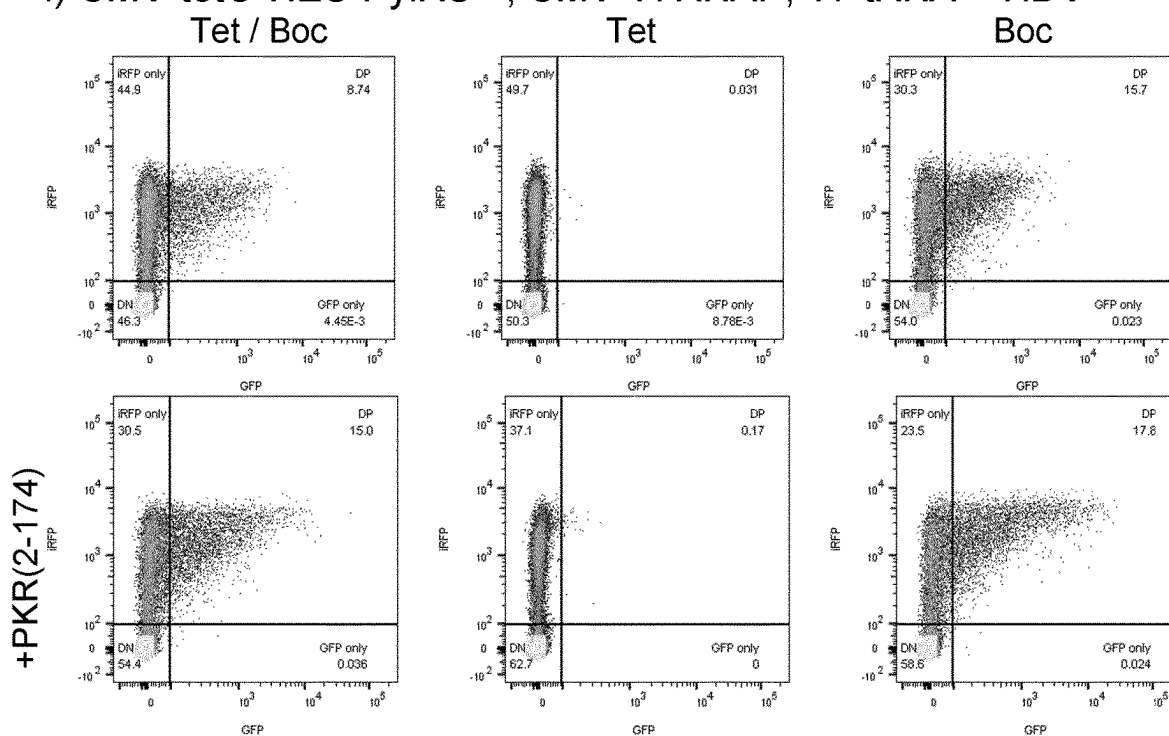
c) CMV-tetO-NES-PyIRS^{AF}, 8xtetO-U6-tetO-tRNA^{Pyl}d) CMV-tetO-NES-PyIRS^{AF}, CMV-tetO-T7RNAP, T7-tRNA^{Pyl}-HDV

Figure 5 (continued)

e) CMV-NES-PylRS^{AF}, CMV-tetO-T7RNAP, T7-tRNA^{Pyl}-HDVf) CMV-tetO-NES-PylRS^{AF}, CMV-T7RNAP, T7-tRNA^{Pyl}-HDV

ARCHAEOAL PYRROLYSYL TRNA SYNTHETASES FOR ORTHOGONAL USE

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application claims the benefit of priority of EP Application No. 20158737.5, filed Feb. 21, 2020.

SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has been submitted in ASCII format via EFS-Web and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Mar. 9, 2023, is named 05710.057US1_ST25 and is 1,23,137 bytes in size.

FIELD OF THE INVENTION

[0003] The present invention is concerned with efficient orthogonal translation systems for eukaryotic cells. Specifically, the invention relates to eukaryotic cells comprising polynucleotides encoding a prokaryotic tRNA synthetase (pRS) and an RNA molecule comprising a prokaryotic tRNA (ptRNA) and one or more than one ribozyme (termed “ptRNA-ribozyme” herein), to a method for expressing a polypeptide of interest (POI) comprising one or more non-canonical amino acid (ncAA) residues by such eukaryotic cells, and to kits useful in such method. The invention further relates to polynucleotides encoding the ptRNA-ribozyme.

BACKGROUND OF THE INVENTION

[0004] Genetic code expansion (GCE) is one of the most powerful tools for protein engineering, including site-specific introduction of post-translational modifications or labeling of proteins with specialized dyes to study protein structure or dynamics. Specifically, GCE allows for the translational modification of polypeptides by direct genetic encoding of ncAA residues, in particular using stop codon suppression, by means of a tRNA/aminoacyl tRNA synthetase (tRNA/RS) pair that is orthogonal to the translation machinery of the host cell. Ideal orthogonal tRNA/RS pairs show no cross-reactivity with the host’s translation machinery, minimizing their impact on the housekeeping translational activity and normal physiology of the cell. Various prokaryotic (bacterial and archaeal) tRNA/RS pairs have been used to genetically encode a variety of ncAA residues in polypeptides including, e.g., engineered *Methanococcus jannaschii* tRNA/tyrosyl-RS, *E. coli* tRNA/leucyl-RS as well as *Methanosarcina mazei* and *M. barkeri* tRNA/pyrrolysyl-RS pairs. See, e.g., Chin et al., J Am Chem Soc 124:9026, 2002; Chin et al., Science 301:964, 2003; Nguyen et al., J Am Chem Soc 131:8720, 2009; Yanagisawa et al., Chem Biol 15:1187, 2008; Liu et al., Annu Rev Biochem 83:379-408, 2010; Lemke, ChemBioChem 15:1691-1694, 2014; Wan et al., Biochim Biophys Acta 1844:1059-170, 2014.

[0005] Over the years, various developments have been made to improve efficiency of GCE. These include, e.g., the use of archaeal RS which lack a nuclear localization signal (NLS) and/or comprise a nuclear export signal (NES) so as to allow for the RS to be localized predominantly in the cytoplasm (rather than in the nucleus) as described in WO 2018/069481. Further approaches for increasing GCE efficiency which have been explored include promoter engi-

neering, better evolution of the orthogonal RS, release-factor engineering and multi-chaining of tRNAs, to name a few (for review see, e.g., Chin et al., Annu Rev Biochem 83:379-408, 2014).

[0006] Nevertheless, there is still a high demand for strategies which improve efficiency of GCE in eukaryotic cells. With more efficient GCE, the cell can express a higher amount of the POI comprising one or more ncAA residues. A higher amount of POI that is available for labeling and imaging purposes is, e.g., advantageous when polypeptides of low abundance are to be labeled. It was therefore an object of the present invention to (further) increase efficiency of GCE.

SUMMARY OF THE INVENTION

[0007] The inventors surprisingly found that expression of the orthogonal tRNA as an RNA molecule that comprises the tRNA together with a ribozyme and the co-expression of a double-stranded RNA-(dsRNA-)binding polypeptide can increase the amount of POI expressed by GCE in eukaryotic cells. Without wishing to be bound by theory it is assumed that the dsRNA-binding polypeptide inhibits the phosphorylation of the alpha subunit of eukaryotic initiation factor 2 α (eIF-2 α) by protein kinase R (PKR) and thus alleviates the impairment of eIF-2 α activity and protein synthesis associated with said PKR activity. Without wishing to be bound by theory, it is further assumed that the ribozyme improves tRNA processing which can be difficult in eukaryotic cells, and thus increases the amount of processed functional orthogonal tRNA that is available for GCE. An ample supply of processed functional orthogonal tRNA is essential for GCE (such as, e.g., amber suppression), in particular considering that orthogonal tRNAs may also interact with non-translational components inside the cells, like tRNA processing machinery and elongation factors.

[0008] Thus, in a first aspect, the present invention relates to a eukaryotic cell comprising:

[0009] (a) a polynucleotide encoding a prokaryotic aminoacyl tRNA synthetase (pRS), and

[0010] (b) a polynucleotide encoding a prokaryotic tRNA (ptRNA) and one or more than one ribozyme.

[0011] In a further aspect, the present invention relates to a eukaryotic cells comprising:

[0012] (a) a polynucleotide encoding a prokaryotic aminoacyl tRNA synthetase (pRS),

[0013] (b) a polynucleotide encoding a prokaryotic tRNA (ptRNA), and

[0014] (c) a polynucleotide encoding a polypeptide capable of binding to dsRNA (dsRNA-binding polypeptide) as described herein.

[0015] In the eukaryotic cells of the present invention, the pRS encoded by the polynucleotide of item (a) is capable of acylating the ptRNA, in particular acylating the ptRNA with non-canonical amino acid (ncAA), and the sequences encoding the ptRNA and the ribozyme(s) comprised by the polynucleotide of item (b) are linked such that transcription produces an RNA molecule comprising both the ptRNA and the ribozyme(s) (ptRNA-ribozyme).

[0016] The dsRNA-binding polypeptide used (such as, e.g., encoded by a polynucleotide used, or expressed) in the context of the present invention, comprises an amino acid sequence having at least 80%, at least 85%, at least 90%, at least 93%, at least 95%, at least 96%, at least 97%, at least

98%, at least 99% or 100% identity to the sequence set forth in any one of SEQ ID NOs:37, and 57-59.

[0017] In a further aspect, the present invention relates to a polynucleotide encoding an RNA molecule comprising a ptRNA and a ribozyme (ptRNA-ribozyme) as described herein.

[0018] In a further aspect, the present invention relates to a polynucleotide, or combination of two or more polynucleotides, comprising a nucleotide sequence encoding an RNA molecule comprising a ptRNA and a ribozyme (ptRNA-ribozyme) as described herein, and:

[0019] (i) a nucleotide sequence encoding a tetO-binding protein as described herein, or

[0020] (ii) a nucleotide sequence encoding a pRS as described herein, or

[0021] (ii) both of (i) and (ii).

[0022] Said two or three nucleotide sequences can be located on the same polypeptide. Or each of said two or three nucleotide sequences can be located on a separate polynucleotide (thus resulting in a combination of two or three polynucleotides, respectively). Or two of said three nucleotide sequences can be located on the same polynucleotide, while the third nucleotide sequence (i.e., the ptRNA-ribozyme-encoding sequence or the pRS-encoding sequence or the tetO-binding protein-encoding sequence) is located on a separate polynucleotide (thus resulting in combination of two polynucleotides).

[0023] In a further aspect, the present invention relates to a method for preparing a polypeptide of interest (POI) having one or more than one non-canonical amino acid (ncAA) residue in its amino acid sequence, wherein the method comprises:

[0024] (a) expressing, in a eukaryotic cell:

[0025] a pRS and

[0026] an RNA molecule comprising a ptRNA and a ribozyme (ptRNA-ribozyme) as described herein;

[0027] (b) expressing the POI in the eukaryotic cell in the presence of one or more than one ncAA or salt thereof corresponding to the one or more than one ncAA residue of the POI; and

[0028] (c) optionally recovering the expressed POI.

[0029] In a further aspect, the present invention relates to a method for preparing a polypeptide of interest (POI) having one or more than one non-canonical amino acid (ncAA) residue in its amino acid sequence, wherein the method comprises:

[0030] (a) expressing, in a eukaryotic cell:

[0031] a pRS,

[0032] a prokaryotic tRNA (ptRNA) and

[0033] a polypeptide capable of binding to dsRNA (dsRNA-binding polypeptide) as described herein;

[0034] and

[0035] (b) expressing the POI in the eukaryotic cell in the presence of one or more than one ncAA or salt thereof corresponding to the one or more than one ncAA residue of the POI; and

[0036] (c) optionally recovering the expressed POI.

[0037] In the methods for preparing a polypeptide of interest (POI) according to the present invention, the pRS is capable of acylating the ptRNA with the ncAA or the salt thereof, and the POI is encoded by a nucleotide sequence that comprises one or more than one selector codon encod-

ing the one or more than one ncAA residue. Said selector codon is the reverse complement of the anticodon of the ptRNA.

[0038] Steps (a) and (b) of said methods can be carried out concomitantly or sequentially. For example, expression step (a) can be carried out first followed by expression step (b). When starting with the expression of pRS and the RNA molecule (ptRNA-ribozyme or ptRNA, respectively) according to step (a), said expression may continue when starting the expression of the POI according to step (b).

[0039] In a further aspect, the present invention relates to a kit for preparing a POI having one or more than one ncAA residue in its amino acid sequence, wherein the kit comprises one or more than one ncAA, or salt thereof, corresponding to the one or more than one ncAA residue of the POI, and:

[0040] (a) a polynucleotide, or combination of two or more polynucleotides, of the invention as described herein, or

[0041] (b) a polynucleotide encoding a dsRNA-binding polypeptide as described herein, or

[0042] (c) a eukaryotic cell of the invention as described herein.

[0043] In GCE, uncontrolled expression of the orthogonal tRNA/RS pair can lead to unwanted side effects. For example, over-excess of unused RS and/or unused tRNA might impair cell viability. The present invention solves this problem by further providing eukaryotic cells, polynucleotides, methods and kits for GCE wherein (i) the transcription of nucleotide sequence(s) encoding the tRNA (ptRNA or ptRNA-ribozyme), the pRS or both is controlled by a tetracycline-responsive promoter element (TRE), or (ii) the transcription of said nucleotide sequence(s) is catalyzed by an RNA polymerase (e.g., a T7 RNA polymerase), wherein expression of said RNA polymerase is controlled by a TRE. Thus, expression of said tRNA and/or pRS becomes inducible by the co-expression of proteins binding to (tetracycline operator (tetO) sequences within) the TRE, such as, e.g., a tetracycline repressor (tetR), a tetracycline-controlled transcriptional activator (tTA) or a reverse tTA (rtTA) and the presence (or absence) of tetracycline or a tetracycline derivative such as, e.g., doxycycline.

BRIEF DESCRIPTION OF THE FIGURES

[0044] FIG. 1 provides a schematic view (5' to 3' direction) of tRNA expression cassettes for T7RNAP-controlled expression of a tRNA with and without a (HDV) ribozyme ("HDV-Rib.").

[0045] FIG. 2 shows the results of a flow cytometry analysis of HEK293T cells carrying expression cassettes for the NLS-iRFP-Flag-GFP^{Y39->TAG}-6His amber suppression reporter, T7RNAP and the amber suppression pair tRNA^{Pyl}/PylRS^{ΔF}, wherein the tRNA^{Pyl} was either expressed together with a HDV ribozyme ("tRNA^{Pyl}-HDV rib.") or without such ribozyme ("tRNA"). The HEK293T cells further carried an expression cassette for HA-tagged human PKR(2-174). Samples include controls where cells were not fed with the ncAA SCO ("SCO", bottom row). The numbers in the quadrants indicate the percentage of cells in the respective population calculated based on the total number of measured cells: cells without reporter gene expression (bottom left, "DN"=double negatives), cells showing only GFP fluorescence (bottom right, "GFP only"), cells expressing only iRFP without amber suppression (top left, "iRFP

only”), cells expressing the full reporter due to amber suppression (top right, “DP”=double positives). See example 1. Experiment with plasmid pcDNA3.1 (containing the CMV promoter).

[0046] FIG. 3 shows the results of a flow cytometry analysis of HEK293T cells carrying expression cassettes for the NLS-iRFP-Flag-GFP^{Y39->TAG}-6His amber suppression reporter, T7RNAP and the amber suppression pair tRNA^{Pyl}/PylRS^{AF}, wherein the tRNA^{Pyl} was either expressed together with a HDV ribozyme (“tRNA^{Pyl}-HDV rib.”) or without such ribozyme (“tRNA^{Pyl}”). The HEK293T cells further carried an expression cassette for HA-tagged human PKR(2-174). Samples include controls where cells were not fed with the ncAA SCO (“-SCO”, bottom row). The numbers in the quadrants indicate the percentage of cells in the respective population calculated based on the total number of measured cells: cells without reporter gene expression (bottom left, “double negatives”), cells showing only GFP fluorescence (bottom right, “GFP only”), cells expressing only iRFP without amber suppression (top left, “iRFP only”), cells expressing the full reporter due to amber suppression (top right, “double positives”). See example 1. Experiment with plasmid pCAGGS (containing CMV enhancer with chicken actin promoter and intron).

[0047] FIG. 4a shows the results of a flow cytometry analysis of HEK Flp-In T-Rex 293 cells carrying expression cassettes for the NLS-iRFP-Flag-GFP^{Y39->TAG}-6His amber suppression reporter, NES-T7RNAP, and the amber suppression pair tRNA^{Pyl}/NES-PylRS^{AF}, wherein the tRNA^{Pyl} was expressed together with a HDV ribozyme (“tRNA^{Pyl}-HDV rib.”) or without such ribozyme (“tRNA^{Pyl}”), and NES-PylRS^{AF} and NES-T7RNAP were inducibly expressed using the T-Rex system (ThermoFisher Scientific). Cells were fed with the ncAA BOC (“+Boc”), and expression of NES-PylRS^{AF} and NES-T7RNAP was either induced with tetracycline (“+Tet”, left chart) or was not induced (control, right chart). The numbers in the quadrants indicate the percentage of cells in the respective population calculated based on the total number of measured cells: cells without reporter gene expression (bottom left, “DN”=double negatives), cells showing only GFP fluorescence (bottom right, “GFP only”), cells expressing only iRFP without amber suppression (top left, “iRFP only”), cells expressing the full reporter due to amber suppression (top right, “DP”=double positives). See example 2.

[0048] FIG. 4b shows the results of a flow cytometry analysis of HEK293T cells carrying expression cassettes for the NLS-iRFP-Flag-GFP^{Y39->TAG}-6His amber suppression reporter, NES-T7RNAP, and the amber suppression pair tRNA^{Pyl}/NES-PylRS^{AF}, wherein the tRNA^{Pyl} was expressed together with a HDV ribozyme (“tRNA^{Pyl}-HDV rib.”) or without such ribozyme (“tRNA^{Pyl}”), and NES-PylRS^{AF} and NES-T7RNAP were inducibly expressed using the Tet-On system. Cells were fed with the ncAA BOC (“+Boc”), and expression of NES-PylRS^{AF} and NES-T7RNAP was either induced with doxycycline (“+Dox”, left chart) or was not induced (control, right chart). The numbers in the quadrants indicate the percentage of cells in the respective population calculated based on the total number of measured cells: cells without reporter gene expression (bottom left, “DN”=double negatives), cells showing only GFP fluorescence (bottom right, “GFP only”), cells expressing only iRFP without amber suppression (top left, “iRFP only”), cells expressing

the full reporter due to amber suppression (top right, “DP”=double positives). See example 2.

[0049] FIG. 5 shows the results of a flow cytometry analysis of HEK293T cells carrying expression cassettes for:

[0050] FIG. 5a: NLS-iRFP-Flag-GFP^{Y39->TAG}-6His amber suppression reporter, tetR, and the amber suppression pair tRNA^{Pyl}/NES-PylRS^{AF} wherein the expression of tRNA^{Pyl} was controlled by a tetracycline-inducible variant of the human U6 promoter comprising eight tetO sequences before the U6 promoter and two tetO sequences after the U6 promoter and expression of NES-PylRS^{AF} was controlled by the (constitutive) CMV promoter;

[0051] FIG. 5b: NLS-iRFP-Flag-GFP^{Y39->TAG}-6His amber suppression reporter, tetR, and the amber suppression pair tRNA^{Pyl}/NES-PylRS^{AF} wherein the expression of tRNA^{Pyl} was controlled by a tetracycline-inducible variant of the human H1 promoter comprising two tetO sequences and expression of NES-PylRS^{AF} was controlled by the (constitutive) CMV promoter;

[0052] FIG. 5c: NLS-iRFP-Flag-GFP^{Y39->TAG}-6His amber suppression reporter, tetR, and the amber suppression pair tRNA^{Pyl}/NES-PylRS^{AF} wherein the expression of tRNA^{Pyl} was controlled by a tetracycline-inducible variant of the human U6 promoter comprising eight tetO sequences before the U6 promoter and two tetO sequences after the U6 promoter and expression of NES-PylRS^{AF} was controlled by a tetracycline-inducible variant of the CMV promoter comprising two tetO sequences;

[0053] FIG. 5d: NLS-iRFP-Flag-GFP^{Y39->TAG}-6His amber suppression reporter, tetR, NES-T7RNAP expressed under the control of a tetracycline-inducible variant of the CMV promoter comprising two tetO sequences, and the amber suppression pair tRNA^{Pyl}/NES-PylRS^{AF} wherein tRNA^{Pyl} was expressed together with a HDV ribozyme under the control of the T7 promoter and expression of NES-PylRS^{AF} was controlled by a tetracycline-inducible variant of the CMV promoter comprising two tetO sequences;

[0054] FIG. 5e: NLS-iRFP-Flag-GFP^{Y39->TAG}-6His amber suppression reporter, tetR, NES-T7RNAP expressed under the control of a tetracycline-inducible variant of the CMV promoter comprising two tetO sequences, and the amber suppression pair tRNA^{Pyl}/NES-PylRS^{AF} wherein tRNA^{Pyl} was expressed together with a HDV ribozyme under the control of the T7 promoter and expression of NES-PylRS^{AF} was controlled by the (constitutive) CMV promoter;

[0055] FIG. 5f: NLS-iRFP-Flag-GFP^{Y39->TAG}-6His amber suppression reporter, tetR, NES-T7RNAP expressed under the control of the (constitutive) CMV promoter, and the amber suppression pair tRNA^{Pyl}O/NES-PylRS^{AF} wherein tRNA^{Pyl} was expressed together with a HDV ribozyme under the control of the T7 promoter and expression of NES-PylRS^{AF} was controlled by a tetracycline-inducible variant of the CMV promoter comprising two tetO sequences.

[0056] The HEK293T cells further carried an expression cassette for HA-tagged human PKR(2-174) (“+PKR(2-174)”, histograms in the bottom row of each of FIGS. 5a-f) or lacked said expression cassette (=control, histograms in the top row of each of FIGS. 5a-f). The cells were fed with the ncAA BOC (“Boc”, histograms in the left and right column of each of FIGS. 5a-f) or not (histograms in the middle column), and expression of the inducible elements (tRNA^{Pyl}, NES-PylRS^{AF} and/or NES-T7RNAP) was induced with tetracycline (“Tet” histograms in the left and middle

column of each of FIGS. 5a-f) or was not induced (histograms in the right column). The numbers in the quadrants indicate the percentage of cells in the respective population calculated based on the total number of measured cells: cells without reporter gene expression (bottom left, “DN”=double negatives), cells showing only GFP fluorescence (bottom right, “GFP only”), cells expressing only iRFP without amber suppression (top left, “iRFP only”), cells expressing the full reporter due to amber suppression (top right, “DP”=double positives). See example 3.

DETAILED DESCRIPTION OF THE INVENTION

[0057] Unless otherwise stated or required by context, singular terms shall include pluralities and plural terms shall include the singular. Thus, for example, eukaryotic cells defined as comprising “a polynucleotide encoding a pRS” include eukaryotic cells comprising one or more than one of such polynucleotide as well as eukaryotic cells wherein said polynucleotide(s) encode(s) one or more than one pRS.

[0058] If not otherwise stated, nucleotide sequences are depicted herein in the 5' to 3' direction. If not otherwise stated, amino acid sequences are depicted herein in the direction from N-terminus to C-terminus.

1. Prokaryotic Aminoacyl tRNA Synthetase/Prokaryotic tRNA (pRS/ptRNA)

[0059] The eukaryotic cells of the invention comprise a polynucleotide encoding a prokaryotic RS/tRNA (pRS/ptRNA) pair, wherein the pRS is capable of acylating the ptRNA. In particular, the pRS is capable of acylating the tRNA with an amino acid or amino acid analog (such as, e.g., compounds which differ from amino acids such that the α -amino group is replaced by a hydroxyl group and/or the carboxylic acid function forms an ester), and preferably with a non-canonical amino acid (ncAA).

[0060] The ptRNA and the pRS used in the context of the present invention are orthogonal to the translation machinery of the eukaryotic (host) cell (recombinantly) expressing said ptRNA and pRS. The term “orthogonal” as used herein indicates that the thus specified molecule (e.g., an orthogonal tRNA (O-tRNA) and/or an orthogonal RS (O-RS)) is used with reduced efficiency by the translation system (e.g., the host cell) of interest.

[0061] The term “translation system” generally refers to a set of components necessary to incorporate a naturally occurring amino acid in a growing polypeptide chain (protein). Components of a translation system include, e.g., ribosomes, tRNAs, aminoacyl tRNA synthetases, mRNA and the like. Translations systems used in the context of the present invention are preferably eukaryotic cells.

[0062] Specifically, “orthogonal” refers to the inability or reduced efficiency (such as, e.g., less than 20% efficient less than 10% efficient, less than 5% efficient, or less than 1% efficient) of an O-tRNA or an O-RS to function with the endogenous RSs or endogenous tRNAs, respectively, of the translation system of interest (wherein said orthogonal RS or orthogonal tRNA is used). For example, an O-tRNA in a translation system of interest is (amino-)acylated by any endogenous RS of the translation system with reduced or even zero efficiency, when compared to (amino-)acylation of an endogenous tRNA by the endogenous RS. For example, an O-RS (amino-)acylates any endogenous tRNA in the

translation system of interest with reduced or even zero efficiency, as compared to (amino-)acylation of the endogenous tRNA by an endogenous RS. An “orthogonal translation system” is used herein to refer to a translation system using an RS/O-tRNA^{ncAA} pair that allows for introducing ncAA residues into a growing polypeptide chain.

[0063] Unless indicated differently, the terms “endogenous tRNA” and “endogenous aminoacyl tRNA synthetase” (“endogenous RS”) used therein refer to a tRNA and an RS, respectively, that was present in the cell ultimately used as translation system prior to introducing the pRS and the ptRNA, respectively, used in the context of the present invention.

[0064] The ptRNAs and pRSs used in the methods and/or fusion proteins of the invention can be naturally occurring (i.e. are ptRNAs and pRSs as found in prokaryotic organisms in nature) or can be derived by mutation of a naturally occurring ptRNA or pRS, respectively. In various embodiments, the ptRNA and pRS are derived from a prokaryote, e.g., from an eubacterium or an archaeobacterium (e.g., *E. coli*; a *Methanococcus* species such as *M. jannaschii*; a *Methanosarcina* species such as *M. mazei*, *M. barkeri*, *M. acetivorans*, *M. thermophila*; a *Methanococcoides* species such as *M. burtonii*; a *Desulfotobacterium* species such as *D. hafniense*). In another embodiment, the ptRNA is a naturally occurring, or a mutant of a naturally occurring, ptRNA from a first prokaryote and the RS is derived from naturally occurring, or a mutant of a naturally occurring, pRS from a second prokaryote.

[0065] Suitable (orthogonal) ptRNA/pRS pairs can be selected from libraries of mutant ptRNAs and pRSs, e.g. based on the results of a library screening. Methods for evolving tRNA/RS pairs are described, e.g., in WO 02/085923 and WO 02/06075.

[0066] Several prokaryotic RSs have been used for genetic code expansion including the *Methanococcus jannaschii* tyrosyl-tRNA synthetase, *E. coli* tyrosyl-tRNA synthetase, *E. coli* leucyl-tRNA synthetase, and pyrrolysyl-tRNA synthetases from certain *Methanosarcina* species (such as *M. mazei*, *M. barkeri*, *M. acetivorans*, *M. thermophila*), *Methanococcoides* species (such as *M. burtonii*), *Desulfotobacterium* species (such as *D. hafniense*) or *Methanomethylophilus* species (such as *M. alvus*). Corresponding orthogonal pRS/ptRNA pairs have been used to genetically encode a variety of functionalities in polypeptides (Chin, Annu Rev Biochem 2014, 83:379-408; Chin et al., J Am Chem Soc 2001, 124:9026; Chin et al., Science 2003, 301:964; Nguyen et al., J Am Chem Soc 2009, 131:8720; Willis and Chin, Nat Chem 2018, 10:831-837; Yanagisawa et al., Chem Biol 2008, 15:1187). Such pRSs and ptRNAs can be used in the present invention.

[0067] Pyrrolysyl tRNA synthetases (PylRSs) which can be used as pRSs in the context of the present invention may be wildtype or genetically engineered PylRSs. Examples for wildtype PylRSs include, but are not limited to. PylRSs from archaeobacteria and eubacteria such as *Methanosarcina* maize, *Methanosarcina barkeri*, *Methanococcoides burtonii*, *Methanosarcina acetivorans*, *Methanosarcina thermophila*, *Methanomethylophilus alvus* and *Desulfotobacterium hafniense*. Genetically engineered PylRSs have been described, for example, by Neumann et al. (Nat Chem Biol 2008, 4:232), by Yanagisawa et al. (Chem Biol 2008, 15:1187), and in EP2192185A1. The efficiency of genetic code expansion using PylRS can be increased by modifying

the amino acid sequence of the PyIRS such that it is not directed to the nucleus. To this end, the nuclear localization signal (NLS) can be removed from the PyIRS or can be overridden by introducing a suitable nuclear export signal (NES). Thus, in particular embodiments of the invention, the pRS is a PyIRS, or a functional fragment or mutant, as described herein that lacks the NLS and/or comprises a NES as described, e.g., in WO 2018/069481. For example, such PyIRS may carry a NES, preferably inserted between (the methionine residue at) amino acid position 1 of the pRS and the amino acid position 2 of the pRS.

[0068] The term “nuclear export signal” (abbreviated as “NES”) refers to an amino acid sequence which can direct a polypeptide containing it (such as a NES-containing pRS of the invention or a NES-T7RNAP as used herein) to be exported from the nucleus of a eukaryotic cell. Said export is believed to be mostly mediated by Crm1 (chromosomal region maintenance 1, also known as karyopherin exportin 1). NESs are known in the art. For example, the database ValidNESs (<http://validness.ym.edu.tw/>) provides sequence information of experimentally validated NES-containing proteins. Further, NES databases like, e.g., NESbase 1.0 (www.cbs.dtu.dk/databases/NESbase-1.0/; see Le Cour et al., Nucl Acids Res 31(1), 2003) as well as tools for NES prediction like NetNES (www.cbs.dtu.dk/services/NetNES/; see La Cour et al., La Cour et al., Protein Eng Des Sel 17(6):527-536, 2004), NESpredictor (NetNES, <http://www.cbs.dtu.dk/>; see Fu et al., Nucl Acids Res 41:D338-D343, 2013; La Cour et al., Protein Eng Des Sel 17(6):527-536, 2004) and NESsential (a web interface combined with ValidNESs) are available to the public. Hydrophobic leucine-rich NESs are most common and represent the best characterized group of NESs to date. A hydrophobic leucine-rich NES is a non-conservative motif having 3 or 4 hydrophobic residues. Many of these NESs comprise the conserved amino acid sequence pattern LxxLxL (SEQ ID NO:28) or LxxxLxL (SEQ ID NO:29), wherein each L is independently selected from leucine, isoleucine, valine, phenylalanine and methionine amino acid residues, and each x is independently selected from any amino acid (see La Cour et al., Protein Eng Des Sel 17(6):527-536, 2004).

[0069] Particular NESs which are suitable as signal peptide for the pRS (in particular the PyIRS) used in the present invention are known in the art (e.g., from NES databases) and described, e.g., in WO 2018/069481. In specific embodiments, the pRS used in the invention comprises a hydrophobic leucine-rich NES, in particular a NES comprising the amino acid sequence LxxLxL (SEQ ID NO:28) or LxxxLxL (SEQ ID NO:29), wherein each L is independently selected from leucine, isoleucine, valine, phenylalanine and methionine, and each x is independently selected from any amino acid; more particularly a NES comprising an amino acid sequence selected from L¹xxL²xxL¹xL³ (SEQ ID NO:30), L¹xxxL²xxL¹xL³ (SEQ ID NO:31), L¹xxL²xxxL¹xL³ (SEQ ID NO:32) and L¹xxxL²xxxL¹xL³ (SEQ ID NO:33), wherein L¹ is leucine, L² is selected from leucine, isoleucine, valine, phenylalanine and methionine, L³ is selected from leucine and isoleucine, and each x is independently selected from any amino acid. Preferably, the NES comprises the amino acid sequence LPPLERLTL (SEQ ID NO:34) which is found in the HIV-1 Rev protein or, more preferably, the amino acid sequence ACPVPLQLPLERLTLTD (SEQ ID NO:35).

[0070] A “nuclear localization signal” (abbreviated as “NLS”, also referred to in the art as “nuclear localization sequence”) is an amino acid sequence that “tags” (i.e. can direct) a polypeptide containing it (e.g., NLS-iRFP-Flag-GFP^{T39→T4G}-6His) to be imported into the nucleus of a eukaryotic cell. Said export is believed to be mediated by binding of the NLS-containing polypeptide to importin (also known as karyopherin) so as to form a complex that moves through a nuclear pore. NLSs are known in the art. A multitude of NLS databases and tools for NLS prediction are available to the public, such as NLSdb (see Nair et al., Nucl Acids Res 31(1), 2003), cNLS Mapper (www.nls-mapper.aib.keio.ac.jp; see Kosugi et al., Proc Natl Acad Sci USA. 106(25):10171-10176, 2009; Kosugi et al., J Biol Chem 284(1):478-485, 2009), SeqNLS (see Lin et al., PLoS One 8(10):e76864, 2013), and NucPred (www.sbc.su.se/~macca-llr/nucpred/; see Branmeier et al., Bioinformatics 23(9): 1159-60, 2007). Classical NLSs can be monopartite or bipartite. Bipartite NLSs consist of two basic amino acid clusters and a spacer of about 10 amino acids linking said clusters. A typical core sequence comprised by monopartite NLSs is KX¹X²X³ (SEQ ID NO:20), wherein X¹ and X² are independently selected from K and R, and X³ is any amino acid. Exemplary NLS include the NLSs comprising the amino acid sequence PKKKRKV (SEQ ID NO:21).

[0071] Examples of specific pRSs which can be used in the context of the present invention include, but are not limited to:

- [0072]** *Methanococcus jannaschii* tyrosyl-tRNA synthetase;
- [0073]** *Escherichia coli* tyrosyl-tRNA synthetase;
- [0074]** *Escherichia coli* leucyl-tRNA synthetase;
- [0075]** *Methanosarcina mazei* pyrrolysyl-tRNA synthetase;
- [0076]** *Methanosarcina barkeri* pyrrolysyl-tRNA synthetase;
- [0077]** *Methanosarcina acetivorans* pyrrolysyl-tRNA synthetase;
- [0078]** *Methanosarcina thermophila* pyrrolysyl-tRNA synthetase;
- [0079]** *Methanococcoides burtonii* pyrrolysyl-tRNA synthetase;
- [0080]** *Desulfotobacterium hafniense* pyrrolysyl-tRNA synthetase;
- [0081]** *Methanomethylophilus alvus* pyrrolysyl-tRNA synthetase;

as well as functional (i.e., enzymatically active) fragments and mutants of these polypeptides. Said functional fragments and mutants may comprise at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98% or at least 99% amino acid sequence identity to the aminoacyl tRNA synthetase they are derived from.

[0082] Particular examples of pRSs useful as in the present invention which are *M. mazei* PyIRSs include, but are not limited to:

- [0083]** PyIRS^{4F} (*Methanosarcina mazei* pyrrolysyl tRNA synthetase double mutant: Y306A, Y384F; UniProt Q8PWY1; SEQ ID NO:22);
- [0084]** PyIRS^{4A} (*Methanosarcina mazei* pyrrolysyl tRNA synthetase double mutant: N346A, C348A; UniProt Q8PWY1; SEQ ID NO:23);

[0085] PylRS^{4AA} (*Methanosarcina mazei* pyrrolysyl tRNA synthetase quadruple mutant: Y306A, N346A, C348A, Y384F; Uniprot Q8PWY1; SEQ ID NO:24);

as well as functional (i.e., enzymatically active) fragments and mutants of these polypeptide segments. Said functional fragments and mutants may comprise at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98% or at least 99% amino acid sequence identity to the PylRSs they are derived from.

[0086] According to particular embodiments, wild-type and mutant *M. mazei* PylRSs as described herein are used for aminoacylation of tRNA with ncAAs as described in WO2012/104422 or WO2015/107064. Exemplary ncAAs for this purpose include, but are not limited to, 2-amino-6-(tert-butoxycarbonylamino)hexanoic acid (BOC), 2-amino-6-(cyclooct-2-yn-1-yloxy carbonylamino)hexanoic acid (SCO), 2-amino-6-(cyclooct-2-yn-1-yloxyethoxycarbonylamino)hexanoic acid, 2-amino-6-[(4E-cyclooct-4-en-1-yl)oxycarbonylamino]hexanoic acid (TCO), 2-amino-6-[(2E-cyclooct-2-en-1-yl)oxycarbonylamino]hexanoic acid (TCO*), 2-amino-6-(prop-2-ynoxycarbonylamino)hexanoic acid (PrK), 2-amino-6-(9-biocylo[6.1.0]non-4-ynylmethoxycarbonylamino)hexanoic acid (BCN) and particularly the 2S-(L-) enantiomer of these amino acids.

[0087] Functional fragments and mutants of pRS described herein are functional in that they have the acylating enzymatic activity of the parent pRS, wherein the degree of said enzymatic activity of the fragment or mutant may be higher or less than, or about the same as, that of the parent pRS. Such fragments and mutants can be characterized by a minimum degree of sequence identity as described herein.

[0088] Amino acid or nucleotide sequence identity as used herein means identity over the entire length of the thus characterized amino acid or nucleotide sequence, respectively. The percentage identity values can be determined as known in the art on the basis of BLAST alignments, blastp algorithms (protein-protein BLAST), or using the Clustal method (Higgins et al., Comput Appl. Biosci. 1989, 5(2): 151-1).

[0089] Functional fragments and mutants of particular pRSs which are useful in the present invention can be obtained, e.g., by conservative amino acid substitution, i.e. the replacement of an amino acid residue with different amino acid residues having similar biochemical properties (e.g. charge, hydrophobicity and size) as known in the art. Typical examples are substitution of Leu by Ile or vice versa, substitution of Asp by Glu or vice versa, substitution of Asn by Gln or vice versa, and others.

[0090] ptRNAs which can be used in the context of the present invention include, but are not limited to, pyrrolysyl tRNA of *M. mazei* and functional mutants thereof. Expediently, the anticodon of the ptRNA is the anticodon to a selector codon such as, e.g., the CUA anticodon to the Amber stop codon TAG, the anticodon UCA to the Opal stop codon TGA, or the anticodon UUA to the Ochre stop codon TAA. Examples for such pyrrolysyl tRNAs include, but are not limited to, those encoded by the nucleotide sequence of SEQ ID NO:25 (tRNA^{Pyl, CUA}), SEQ ID NO:26 (tRNA^{Pyl, UCA}) or SEQ ID NO:27 (tRNA^{Pyl, UUA}).

[0091] A ptRNA used in the context of the present invention can act as a suppressor tRNA which alters the reading of a messenger RNA (mRNA) in a given translation system

(e.g., a eukaryotic cell of the present invention). A suppressor tRNA can read through, e.g., a stop codon, a four base codon, or a rare codon.

[0092] O-RS/O-tRNA (pRS/ptRNA) pairs used in the invention preferably have the following properties: the O-tRNA is preferentially acylated (with an ncAA) by the O-RS. In addition, the orthogonal pair functions in the translation system of interest (e.g. the eukaryotic cell of the invention) such that the O-tRNA (that has been acylated with the ncAA) is used to incorporate said amino acid or ncAA residue into the growing polypeptide chain of a POI. Incorporation occurs in a site specific manner. Specifically, the O-tRNA recognizes (specifically binds via its anticodon to) a selector codon (e.g., an Amber, Ochre or Opal stop codon) in the mRNA coding for the POI.

[0093] The term “preferentially acylates” refers to an efficiency of, e.g., about 50% efficient, about 70% efficient, about 75% efficient, about 85% efficient, about 90% efficient, about 95% efficient, or about 99% or more efficient, at which an O-RS acylates an O-tRNA with a ncAA compared to an endogenous tRNA or amino acid of a translation system of interest. The ncAA residue is then incorporated into a growing polypeptide chain with high fidelity, e.g., at greater than about 75% efficiency for a given selector codon, at greater than about 80% efficiency for a given selector codon, at greater than about 90% efficiency for a given selector codon, at greater than about 95% efficiency for a given selector codon, or at greater than about 99% or more efficiency for a given selector codon.

2. Ribozymes and ptRNA-Ribozyme

[0094] In one aspect of the present invention, the ptRNA is expressed in the form of an RNA molecule comprising the ptRNA sequence and one or more than one ribozyme (ptRNA-ribozyme).

[0095] The term “ribozyme” as used herein refers to an RNA molecule, or a part thereof, that is capable of site-specific self-cleavage. Specifically, a ribozyme is capable of catalyzing the site-specific cleavage of the phosphodiester bond between two ribonucleotides within the ribonucleotide sequence of (same) the ribozyme.

[0096] Ribozymes are well known in the art. Examples of ribozymes which can be used in the ptRNA-ribozyme of the present invention include, but are not limited to, a hammerhead ribozyme (e.g., a hammerhead ribozyme of SEQ ID NO:4), a hairpin ribozyme (e.g., the hairpin ribozyme of SEQ ID NO:5), a Varkud satellite (VS) ribozyme (e.g., the VS ribozyme of SEQ ID NO:6), a glucosamine-6-phosphate synthetase (glmS) riboswitch (e.g., the glmS riboswitch of SEQ ID NO:7) and a hepatitis delta virus (HDV) ribozyme (e.g., a HDV ribozyme of SEQ ID NO:1, 2 or 3). Information on these ribozymes can be found, e.g., in Ferré-D'Amaré et al. (Cold Spring Harb Perspect Biol 2010, 2:a003574), Cochrane et al. (Chem Biol 2007, 14(1):97-105) and Schürer et al. (Nucl Acids Res 2002, 30(12):e56). In preferred embodiments of the invention, the ribozyme is a HDV ribozyme of SEQ ID NO:1.

[0097] The RNA comprising the ptRNA and the ribozyme used in the present invention comprises one or more than one ribozyme which is/are located relative to the ptRNA such that the ribozyme-catalyzed site-specific (self-)cleavage of the RNA releases the 5'-end and/or the 3'-end of the ptRNA. Thus, said ribozyme-catalyzed (self-)cleavage yields an RNA molecule comprising the ptRNA sequence

wherein the 5'-end and/or the 3'-end of the ptRNA sequence coincides with the 5'-end or 3'-end, respectively, of said RNA molecule. In a particular embodiment, ribozyme-catalyzed (self-)cleavage of the RNA comprising the ptRNA and the ribozyme yields an RNA molecule that consists of the ptRNA. In a preferred embodiment, the ribozyme is a HDV ribozyme, in particular a HDV ribozyme of SEQ ID NO:1, that is directly (i.e. without any interjacent ribonucleotides) covalently linked (fused) to the 3'-end of the ptRNA so as to form a continuous ptRNA-ribozyme fusion.

3. Inducible and Constitutive Expression of ptRNA and/or pRS

[0098] As a further aspect, the present invention provides eukaryotic cells, methods for preparing a POI and related subject matter as described herein, wherein the expression of the ptRNA (or the ptRNA-ribozyme) or the pRS or both is inducible (in contrast to constitutive). Specifically, said inducibility is achieved in that transcription of the nucleotide sequences encoding the pRS or the ptRNA/ptRNA-ribozyme or both is controlled (directly) by a tetracycline-responsive promoter element or its transcription is catalyzed by an RNA polymerase, wherein expression of said RNA polymerase is controlled by a tetracycline-responsive promoter element.

[0099] Suitable systems for TRE-regulated transcription are well known in the art and have become standard tools for inducible gene expression. See, e.g., Gossen and Bujard, *Proc Natl Acad Sci USA*. 1992, 89(12):5547-5551; Yao et al., *Hum Gene Ther*. 1998, 9(13):1939-1950; Das et al., *Curr Gene Ther*, 2016, 16(3):156-167.

[0100] The term “tetracycline-responsive promoter element” (abbreviated “tetracycline response element” or “TRE”) refers to a promoter sequence comprising one, or more (such as, e.g., two, three, four, five, six, seven or more, for instance two or seven) tetracycline operator (tetO) sequences. A tetO is a polynucleotide segment having the sequence of SEQ ID NO:8 or a functional variant thereof that is capable of binding to a tetO-binding protein such as, e.g., a tetracycline repressor (tetR), tetracycline-controlled transcriptional activator (tTA) or a reverse tTA (rtTA) protein. A TRE may further comprise at least part of the minimal CMV promoter sequence set forth in SEQ ID NO:9.

[0101] In the context of the present invention, the transcription of a TRE-controlled sequence is in particular the transcriptional synthesis of a herein described RNA molecule (ptRNA or ptRNA-ribozyme), the transcription of the pRS-encoding sequence, or the transcriptional synthesis of the RNA polymerase (e.g., T7 RNA polymerase) catalyzing the transcriptional synthesis of a herein described ptRNA or ptRNA-ribozyme.

[0102] The eukaryotic cells of the invention wherein expression of one or more of the pRS, ptRNA/ptRNA-ribozyme and T7RNAP is controlled by a TRE as described herein expediently further comprise a polynucleotide encoding a tetO-binding protein (tetR, rtTA or tTA) as described herein to allow for inducing the expression said TRE-controlled coding sequences in the presence or absence, respectively, of tetracycline, doxycycline or a functional analogue thereof as described herein.

[0103] Particular examples of TREs which can be used in the context of the present invention include those comprising (or having) the sequence of any one of SEQ ID NOs: 10-12. A TRE comprising (or having) the sequence of SEQ

ID NO:10 or SEQ ID NO:11 is useful in combination with a rtTA, as well as for tTA. A TRE comprising (or having) the sequence of SEQ ID NO:12 is particularly useful in combination with a tetR. The tetO-binding protein and the TRE chosen to control the transcriptional synthesis of a herein described RNA molecule (ptRNA or ptRNA-ribozyme), pRS-mRNA, and/or RNA polymerase mRNA are thus expediently selected to match (bind to) each other so as to allow for inducible transcriptional control.

[0104] rtTAs and tTAs are generally known in the art, e.g., from Das et al. (*Curr Gene Therapy* 2016, 16:156-167) and the references cited therein. Exemplary rtTA proteins may be selected from those listed in FIG. 2B of Das et al. (supra).

[0105] rtTAs are fusion proteins comprising a first polypeptide which binds to a tetO sequence in the presence of tetracycline or a tetracycline analogue (in particular in the presence of doxycycline). Said first polypeptide is (operatively) linked to a second polypeptide which activates transcription in eukaryotic cells. The first polypeptide is preferably a mutant of the 207aa TetR selected from, e.g., the list of rtTAs shown in FIG. 2 of Das et al. (supra). The second polypeptide is preferably the 127aa transcription activation domain of the herpes simplex virus VP16 protein. The first and second polypeptide are preferably covalently linked so as to form a fusion protein, such as a fusion protein comprising the sequence of SEQ ID NO:14 and functional equivalents thereof.

[0106] An rtTA binds to (tetO sequences of) the TRE in the presence of a doxycycline or a functional analogue thereof. Said binding of the rtTA facilitates or enhances the binding of RNA polymerase and thereby the transcription of the TRE-controlled sequence. In such so-called “Tet-On” system, transcription of the TRE-controlled sequence is thus induced in the presence of an rtTA by adding doxycycline, or a functional analogue thereof. Accordingly, methods of the present invention, wherein transcription (and thus expression) of one or more of the pRS, ptRNA/ptRNA-ribozyme and T7RNAP is controlled by such Tet-On system, expediently further comprise the steps of:

[0107] expressing an rtTA in the eukaryotic cell and

[0108] contacting said cell with doxycycline, or a functional analogue thereof (thereby inducing transcription of the TRE-controlled sequence).

[0109] The rtTA is expressed in the cell at least concomitantly with (i.e., prior to and/or concomitantly with) the expression of the TRE-controlled sequence.

[0110] tTAs are fusion proteins comprising a first polypeptide which binds to a tetO sequence in the absence of tetracycline, doxycycline or a functional analogue thereof. Said first polypeptide is (operatively) linked to a second polypeptide which activates transcription in eukaryotic cells. The first polypeptide is preferably the 207aa TetR or a functional derivative thereof. The second polypeptide is preferably the 127aa transcription activation domain of the herpes simplex virus VP16 protein. The first and second polypeptides are preferably covalently linked so as to form a fusion protein.

[0111] A tTA binds to (tetO sequences of) the TRE in the absence (or at reduced concentrations) of a tetracycline, doxycycline or a functional analogue thereof. Said binding of the tTA facilitates or enhances the binding of RNA polymerase and thereby the transcription of the TRE-controlled sequence. In such so-called “Tet-Off” system, transcription of the TRE-controlled sequence is thus induced in

the presence of a tTA by removing (or reducing the concentration of) previously present tetracycline, doxycycline, or a functional analogue thereof. Accordingly, methods of the present invention, wherein transcription (and thus expression) of one or more of the pRS, ptRNA/ptRNA-ribozyme and T7RNAP is controlled by such Tet-Off system, expediently further comprise the steps of:

[0112] expressing an tTA in the eukaryotic cell and

[0113] keeping said cell prior to the expression of the TRE-controlled sequence in contact with tetracycline, doxycycline, or a functional analogue thereof (thereby suppressing the binding of tTA to the TRE and thus the (trans-)activation of the transcriptional synthesis of said TRE-controlled sequence) and then reducing the concentration of, or preferably removing, said tetracycline, doxycycline, or analogue thereof so as to induce transcription of the TRE-controlled sequence.

[0114] The tTA is expressed in the cell at least concomitantly with (i.e., prior to and/or concomitantly with) the expression of the TRE-controlled sequence.

[0115] Tetracycline repressor (tetR) proteins are generally known in the art. See, e.g., Hillen and Berens, *Annu Rev Microbiol* 1994, 48:345-369. Exemplary tetR proteins which can be suitably used in the context of the present invention include polypeptides comprising (or having) the amino acid sequence set forth in SEQ ID NO:13 and functional equivalents thereof.

[0116] A tetR binds to (tetO sequences of) the TRE in the absence (or at reduced concentrations) of a doxycycline or a functional analogue thereof. Said binding of the tetR inhibits or prevents the binding of RNA polymerase and thereby the transcription of the TRE-controlled sequence. In such so-called "T-REx" system, transcription of the TRE-controlled sequence is thus induced in the presence of a tetR by adding tetracycline, doxycycline, or a functional analogue thereof. Accordingly, methods of the present invention, wherein transcription (and thus expression) of one or more of the pRS, ptRNA/ptRNA-ribozyme and T7RNAP is controlled by such T-REx system, expediently further comprise the steps of:

[0117] expressing a tetR in the eukaryotic cell and

[0118] contacting said cell with tetracycline, doxycycline, or a functional analogue thereof (thereby inducing transcription of the TRE-controlled sequence).

[0119] The tetR is expressed in the cell at least prior to (i.e., prior to, or prior to both prior to and concomitantly with) the expression of the TRE-controlled sequence.

[0120] The transcription of a herein described RNA molecule (ptRNA or ptRNA-ribozyme) and/or the transcription of the pRS-encoding sequence can be controlled by a constitutive promoter. Constitutive promoters and the RNA polymerases binding to such promoters are well known in the art and include eukaryotic promoters such as, for example, the CMV promoter (from human cytomegalovirus), SV40 promoter (from simian vacuolating virus 40), U6 promoter (from the human small U6 small nuclear promoter), and H1 promoter (from the human polymerase III RNA promoter), as well as prokaryotic promoters such as, for example, the T7 promoter (from T7 bacteriophage), the Sp6 promoter (from Sp6 bacteriophage).

[0121] Transcription of a herein described RNA molecule (ptRNA or ptRNA-ribozyme) and/or the transcription of the pRS-encoding sequence that is controlled by a constitutive promoter transcription can be constitutive or inducible.

Constitutive transcription can be achieved in particular if the RNA polymerase which catalyzes transcription via said promoter is constitutively expressed in the eukaryotic cell. Alternatively, transcription can be inducible if the RNA polymerase which catalyzes transcription via said promoter is inducibly expressed in the eukaryotic cell, for example under the (direct) control of a TRE. In particular embodiments, wherein expression of the herein described RNA molecule (ptRNA or ptRNA-ribozyme) and/or the transcription of the pRS-encoding sequence is controlled by a constitutive promoter, and is inducible by inducible expression of the RNA polymerase which catalyzes transcription via said promoter, the constitutive promoter is a prokaryotic promoter, in particular a promoter which requires an RNA polymerase that does not naturally occur in eukaryotic cells. Particular examples include the Sp6 promoter and, preferably, the T7 promoter.

[0122] The term "T7 promoter" refers to a region of DNA that initiates transcription of a particular gene by a T7 RNA polymerase (T7RNAP). According to particular embodiments, the T7 promoter used in the context of the present invention comprises (or consists of) a nucleotide sequence having at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% or 100% identity to the sequence set forth in any one of SEQ ID NOs:15-17. In preferred embodiments of invention, the T7 promoter comprises the amino acid sequence of SEQ ID NO:15 or 16.

[0123] In preferred embodiments of the invention, where transcription (and thus expression) of one or both of the pRS and the ptRNA or ptRNA-ribozyme, respectively, is controlled by a constitutive promoter, said constitutive promoter is a T7 promoter and preferably comprises (or consists of) the nucleotide sequence of any one of SEQ ID NOs:15-17. In embodiments where a T7 promoter is used, the eukaryotic cell expediently further expresses a T7RNAP. Said T7RNAP may be constitutively expressed or inducibly expressed, e.g. under the (direct) control of a TRE.

[0124] A "T7 RNA polymerase" (T7RNAP) is a protein that binds to a T7 promoter and catalyzes transcription of sequences downstream of the T7 promoter. The amino acid sequence of the naturally occurring T7RNAP is known in the art, see, e.g., UniProtKB/Swiss-Prot accession no.: P00573.2. According to particular embodiments, the T7RNAP used in the context of the present invention comprises an amino acid sequence having at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% or 100% identity to the sequence set forth in SEQ ID NO:18 or the sequence set forth in SEQ ID NO:19 (includes a NES covering amino acid positions 2-17).

[0125] To direct its localizations within the eukaryotic cell, the T7RNAP (or other RNA polymerase recombinantly expressed in the eukaryotic cell) may carry a signal peptide, in particular at its N-terminus, e.g., inserted between amino acid positions 1 and 2 of the RNA polymerase. In one embodiment, the T7RNAP may carry a nuclear localization signal (NLS) or a nuclear export signal (NES) each as defined herein above (see section 1). The amino acid sequence of such NES-T7RNAP is set forth in SEQ ID NO:19. In another embodiment, the T7RNAP may not carry such nuclear localization or export tag/signal.

4. dsRNA-Binding Polypeptides

[0126] One aspect of the present invention provides eukaryotic cells and methods for preparing a POI as described herein, wherein the eukaryotic cells comprise polynucleotides encoding a ptRNA/pRS pair (wherein the ptRNA can be expressed as ptRNA-ribozyme) and further comprise a polynucleotide encoding a dsRNA-binding polypeptide, and wherein the method for preparing a POI comprises expressing such ptRNA/pRS pair in a eukaryotic cell and expressing a dsRNA-binding polypeptide.

[0127] The dsRNA-binding polypeptide used in the present invention is a polypeptide that is capable of binding to a double stranded RNA molecule (dsRNA) having 30 or more, preferably 80 or more base pairs, and wherein the binding is not dependent on the particular nucleotide sequence of the dsRNA. Such binding is conferred by the dsRNA-binding domain of protein kinase R (PKR) which comprises two dsRNA-binding motifs (each containing an α - β - β - α fold) flanking a linker region. A preferred dsRNA-binding polypeptide used in the present invention is a polypeptide comprising the amino acid sequence of SEQ ID No 36.

[0128] Accordingly, the dsRNA-binding polypeptide used in the present invention comprises, or consists of, an amino acid sequence having at least 80% (such as at least 85%, at least 90%, at least 93%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% or 100%) identity to the sequence set forth in any one of SEQ ID NOs:37, and 57-59. More preferably, the dsRNA-binding polypeptide used in the present invention comprises, or consists of, an amino acid sequence having at least 80% (such as at least 85%, at least 90%, at least 93%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% or 100%) identity to the sequence set forth in any one of SEQ ID NOs:38, and 57-59.

[0129] Expediently, the dsRNA-binding polypeptide lacks a functional PKR kinase domain and preferably is incapable of phosphorylating eukaryotic initiation factor 2 α (eIF-2 α).

[0130] In the methods of the invention for preparing a POI, the dsRNA-binding polypeptide is expressed at least concomitantly with (i.e. concomitantly with or both prior to and concomitantly with) the expression of the ptRNA/pRS pair.

5. Polynucleotides

[0131] The invention also relates to a polynucleotide, or a combination of two or more polynucleotides, comprising:

[0132] a nucleotide sequence that encodes an RNA molecule as described herein,

[0133] wherein said RNA comprises a ptRNA and a ribozyme (ptRNA-ribozyme), and, optionally:

[0134] (i) a nucleotide sequence encoding a tetO-binding protein as described herein, or

[0135] (ii) a nucleotide sequence encoding a pRS as described herein, or

[0136] (ii) both of (i) and (ii);

and/or (a) nucleotide sequence(s) complementary to said encoding nucleotide sequence(s).

[0137] The invention further relates to a polynucleotide, or a combination of two or more polynucleotides, comprising:

[0138] a nucleotide sequence that encodes a dsRNA-binding polypeptide as described herein,

and:

[0139] (i) a nucleotide sequence that encodes a ptRNA or a ptRNA-ribozyme as described herein, or

[0140] (ii) a nucleotide sequence encoding a pRS as described herein, or

[0141] (iii) both of (i) and (ii),

and, optionally:

[0142] a nucleotide sequence encoding a tetO-binding protein as described herein;

and/or (a) nucleotide sequence(s) complementary to said encoding nucleotide sequence(s).

[0143] Unless otherwise stated or required by context, the term “polynucleotide”, “polynucleotide molecule” or “nucleic acid molecule” as used herein refers to polymers of nucleotides as well as ribonucleotides, i.e. refers to both nucleic acids as well as ribonucleic acids such as single-stranded or double-stranded DNA and RNA molecules, including cDNA and mRNA.

[0144] Unless otherwise stated or required by context, the expression “nucleotide sequence encoding [an expression product]” is intended to refer to a sequence directly encoding said expression product and/or to a nucleotide sequence complementary thereto.

[0145] The encoding nucleotide sequences which are specified to be comprised by a polynucleotide, or a combination of two or more polynucleotides, of the present invention, can be located on the same polypeptide or can be distributed over two or more different polynucleotides thus resulting in a combination of said different polypeptides (for example each encoding nucleotide sequence can be located on a different polypeptide of the combination).

[0146] The invention further relates to, in particular recombinant, expression constructs or expression cassettes, containing, under the genetic control of regulatory nucleotide sequences the (specified encoding) nucleotide sequence of the polynucleotide, or combination of polynucleotides, of the invention as described herein.

[0147] In one aspect of the invention, the expression cassette of the invention may comprise a nucleotide sequence encoding a ptRNA-ribozyme as described herein. In said aspect of the invention, said expression cassette may optionally further comprise, or may optionally be combined with one or more further expression cassettes comprising:

[0148] (i) a nucleotide sequence encoding a tetO-binding protein as described herein, or

[0149] (ii) a nucleotide sequence encoding a pRS as described herein, or

[0150] (ii) both of (i) and (ii).

[0151] In a further aspect of the invention, the expression cassette of the invention may comprise a nucleotide sequence encoding a dsRNA-binding polypeptide as described herein. In said aspect of the invention, said expression cassette may further comprise, or may optionally be combined with one or more further expression cassettes comprising:

[0152] (i) a nucleotide sequence that encodes a ptRNA or a ptRNA-ribozyme as described herein, or

[0153] (ii) a nucleotide sequence encoding a pRS as described herein, or

[0154] (iii) both of (i) and (ii).

[0155] The invention also relates to, in particular recombinant, vectors, comprising one or more than one of these expression cassettes or combination of expression cassettes (expression vectors).

[0156] An expression cassette comprises the nucleotide sequence encoding the expression product is functionally linked to a promoter sequence located 5' (upstream) thereof, and typically a terminator sequence (e.g., a T7 terminator sequence) located 3' (downstream) of said encoding sequence, and optionally further regulatory elements. Examples of such further regulatory elements include, but are not limited to, targeting sequences, enhancers, polyadenylation signals, selectable markers, amplification signals, replication origins and the like. Suitable regulatory sequences are described for example in Goeddel, *Gene Expression Technology: Methods in Enzymology* 185, Academic Press, San Diego, CA (1990).

[0157] A "functional" linkage of elements of polynucleotides (e.g., expression cassettes), such as promoter, encoding sequence, terminator, regulators, etc., means that these elements are arranged such that the encoding sequence can be transcribed and the optional regulatory elements can perform their regulation of said transcription. This can be achieved by a direct linkage of the elements in one and the same nucleic acid molecule. However, such direct linkage is not necessarily required. Genetic control sequences, for example enhancer sequences, can even exert their function on the target sequence from more remote positions or even from other DNA molecules. Arrangements are preferred in which the nucleic acid sequence to be transcribed is positioned downstream (i.e. at the 3'-end of) the promoter sequence, so that the two sequences are joined together covalently. The distance between the promoter sequence and the nucleic acid sequence to be expressed can be smaller than 200 base pairs, or smaller than 100 base pairs or smaller than 50 base pairs.

[0158] For expression in a cell, the expression cassette is advantageously inserted into an expression vector. Expression vectors are chosen according to the cell to be used for expression which makes optimal expression of the encoding nucleotide sequences in the cell possible. Vectors are well known by a person skilled in the art and are given for example in "Cloning vectors" (Pouwels P. H. et al., Ed., Elsevier, Amsterdam-New York-Oxford, 1985). Examples of expression vectors include, but are not limited to, plasmids, viral vectors (phages), e.g. SV40, CMV, baculovirus and adenovirus, transposons, IS elements, phasmids, cosmids, and linear or circular DNA. See, e.g., the book "Cloning Vectors" (Eds. Pouwels P. H. et al. Elsevier, Amsterdam-New York-Oxford, 1985, ISBN 0 444 904018). These vectors can be replicated autonomously in the (host) cell or can be replicated chromosomally. Expression vectors comprising one or more than one expression cassette of the present invention represent a further aspect of the invention.

[0159] For the expression of a POI in a eukaryotic cell according to the present invention, it is possible, e.g., to introduce a polynucleotide encoding the POI (e.g., an expression vector) into the cell. Alternatively, an existing gene of the cell can be modified so as to comprise selector codons at those amino acid positions where the POI is intended to carry ncAA residues. Methods for introducing (recombinant) polypeptide-encoding nucleic acid molecules into, or for modifying existing genes of, a cell are known in the art.

[0160] The term "expression" describes, in the context of the invention, the production of polypeptides encoded by the corresponding nucleotide sequence in a eukaryotic cell. The term "expression" is also used for the production of RNA

molecules (ptRNA or ptRNA-ribozyme as described herein) which are encoded by nucleotide sequences in the eukaryotic cell.

[0161] The polynucleotides of the invention, including the expression cassettes and expression vectors of the invention can be prepared using common cloning techniques known in the art. Common recombination and cloning techniques are used, as described for example in T. Maniatis, E. F. Fritsch and J. Sambrook, *Molecular Cloning: A Laboratory Manual*, Cold Spring Harbor Laboratory, Cold Spring Harbor, NY (1989) and in T. J. Silhavy, M. L. Berman and L. W. Enquist, *Experiments with Gene Fusions*, Cold Spring Harbor Laboratory, Cold Spring Harbor, NY (1984) and in Ausubel, F. M. et al., *Current Protocols in Molecular Biology*, Greene Publishing Assoc. and Wiley interscience (1987).

[0162] The polynucleotides, or combinations of polynucleotides, of the invention, including expression cassettes and expression vectors of the invention, can be isolated, for example by methods known in the art.

[0163] An "isolated" polynucleotide or an "isolated" combination of polynucleotides is separated from other polynucleotides that are present in the natural source of the polynucleotide (e.g., the cell), and moreover can be essentially free of other cellular material or culture medium, when it is produced by recombinant techniques, or free of chemical precursors or other chemicals, when it is chemically synthesized.

[0164] Polynucleotides according to the invention can be isolated by standard techniques of molecular biology and the sequence information provided according to the invention. For example, cDNA can be isolated from a suitable cDNA-bank, using one of the concretely disclosed complete sequences or a segment thereof as hybridization probe and standard hybridization techniques (as described for example in Sambrook, J., Fritsch, E. F. and Maniatis, T. *Molecular Cloning: A Laboratory Manual*. 2nd edition, Cold Spring Harbor Laboratory, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY, 1989). Moreover, a polynucleotide, comprising one of the disclosed sequences or a segment thereof, can be isolated by polymerase chain reaction, using the oligonucleotide primers that were constructed on the basis of this sequence. The polynucleotide thus amplified can be cloned into a suitable vector and can be characterized by DNA sequence analysis.

6. POIs Having One or More than One ncAA Residue

[0165] The present invention provides methods for preparing a POI having one or more than one ncAA residue. The ncAA residues of the POI are encoded by selector codons.

[0166] The term "selector codon" as used herein refers to a codon that is recognized (i.e. bound) by the ptRNA in the translation process and is not recognized by endogenous tRNAs of the eukaryotic cell. The term is also used for the corresponding codons in polypeptide-encoding sequences of polynucleotides which are not messenger RNAs (mRNAs), e.g. DNA plasmids. Preferably, the selector codon is a codon of low abundance in naturally occurring eukaryotic cells. Expediently, the anticodon of the ptRNA used in the present invention specifically binds to (hybridizes with) a selector codon within an mRNA and thus incorporates the ncAA residue site-specifically into the growing chain of the polypeptide encoded by said mRNA. The known 64 genetic

(triplet) codons code for the 20 canonical amino acid residues and three stop codons. Because only one stop codon is needed for translational termination, the other two can in principle be used to encode non-proteinogenic amino acids. For example, the amber codon, UAG, has been successfully used as a selector codon in in vitro and in vivo translation systems to direct the incorporation of ncAAs. Selector codons utilized in methods of the present invention expand the genetic codon framework of the protein biosynthetic machinery of the translation system used. Specifically, selector codons include, but are not limited to: nonsense codons, such as stop codons, e.g., amber (UAG), ochre (UAA), and opal (UGA) codons; codons consisting of more than three bases (e.g., four base codons); and codons derived from natural or unnatural base pairs. For a given system, a selector codon can also include one of the natural three base codons (i.e. natural triplets), wherein the endogenous translation system does not (or only scarcely) use said natural triplet, e.g., a system that is lacking a tRNA that recognizes the natural triplet or a system wherein the natural triplet is a rare codon.

[0167] A recombinant tRNA that alters the reading of an mRNA in a given translation system such that it allows for reading through, e.g., a stop codon, a four base codon, or a rare codon (such as a tRNA in a eukaryotic cell of the present invention), is termed suppressor tRNA. The suppression efficiency for a stop codon serving as a selector codon (e.g., the amber codon) depends upon the competition between the (aminoacylated) recombinant tRNA (ptRNA) which acts as suppressor tRNA and the release factor (e.g. RF1) which binds to the stop codon and initiates release of the growing polypeptide chain from the ribosome. Suppression efficiency of such stop codon can therefore be increased using a release factor-(e.g. RF1)-deficient strain.

[0168] A nucleotide sequence encoding a target polypeptide (also referred to herein as polypeptide of interest or POI) can comprise one or more, e.g., two or more, three or more, etc., codons (e.g. selector codons) which are the reverse complement of the anticodon comprised by the ptRNA. Conventional site-directed mutagenesis can be used to introduce said selector codon(s) at the site of interest into a nucleotide sequence so as to generate a POI-encoding nucleotide sequence.

[0169] The abbreviation “ncAA” refers generally to any non-canonical or non-natural amino acid (or amino acid residue) that is not among the 20 naturally occurring “standard” proteinogenic amino acids (R, H, K, D, E, S, T, N, Q, C, G, P, A, V, I, L, M, F, Y, W) and is also not selenocysteine or pyrrolysine. Numerous ncAAs are well known in the art. ncAAs useful in methods and kits of the present invention have been described in the prior art (for review see e.g. Liu et al., *Annu Rev Biochem* 83:379-408, 2010, Lemke, *Chem-BioChem* 15:1691-1694, 2014).

[0170] The term “ncAA” also refers to amino acid analogs, e.g. compounds which differ from amino acids such that the α -amino group is replaced by a hydroxyl group (α -hydroxyl acid) and/or the carboxylic acid function forms an ester. When translationally incorporated into a polypeptide, an α -hydroxyl acid (residue) is coupled through its α -hydroxyl group to carboxylic acid function of the adjacent amino acid (analogue) residue by an ester bond. When ncAAs which are amino acid analogs wherein the carboxylic acid function forms an ester of formula —C(O)—O—R are used for preparing polypeptides in a translation system (such

as a eukaryotic cell), it is believed that R is removed in situ, for example enzymatically, in the translation system prior of being incorporated in the POI. Accordingly, R is expediently chosen so as to be compatible with the translation system's ability to convert the ncAA or salt thereof into a form that is recognized and processed by the PylRS of the invention.

[0171] Particular preferred ncAAs in the context of the present invention are those which can be post-translationally further modified. For example, ncAAs may comprise a group (herein referred to as “labeling group”) that facilitates reaction with a suitable group (herein referred to as “docking group”) of another molecule (herein termed “conjugation partner molecule”) so as to covalently attach the conjugation partner molecule to the ncAA. When an ncAA comprising a labeling group is translationally incorporated into a POI, the labeling group becomes part of the POI. Accordingly, a POI prepared according to the method of the present invention can be reacted with one or more than one conjugation partner molecule such that the conjugation partner molecules bind covalently to the (labeling groups of the) non-canonical amino acid (ncAA) residue(s) of the POI. Such conjugation reactions may be used for in situ coupling of POIs within a cell or tissue expressing the POI, or for site-specific conjugation of isolated or partially isolated POIs.

[0172] Particular useful choices for combinations of labeling groups and docking groups (of conjugation partner molecules) are those which can react by metal-free click reactions. Such click reactions include strain-promoted inverse-electron-demand Diels-Alder cycloadditions (SPIE-DAC; see, e.g., Devaraj et al., *Angew Chem Int Ed Engl* 2009, 48:7013) as well as cycloadditions between strained cycloalkynyl groups, or strained cycloalkynyl analog groups having one or more of the ring atoms not bound by the triple bond substituted by amino groups), with azides, nitrile oxides, nitrones and diazocarbonyl reagents (see, e.g., Sanders et al., *J Am Chem Soc* 2010, 133:949; Agard et al., *J Am Chem Soc* 2004, 126:15046), for example strain promoted alkyne-azide cycloadditions (SPAAC). Such click reactions allow for ultrafast and biorthogonal covalent site-specific coupling of ncAA labeling groups of POIs with suitable groups of coupling partner molecule.

[0173] Pairs of docking and labeling groups which can react via the above-mentioned click reactions are known in the art. Examples of suitable ncAAs comprising docking groups include, but are not limited to, the ncAAs described, e.g., in WO 20121104422 and WO 2015/107064.

[0174] Examples of particular suitable pairs of docking groups (comprised by the conjugation partner molecule) and labeling groups (comprised by the ncAA residue(s) of the POI) include but are not limited to:

[0175] (a) a docking group comprising (or essentially consisting of) a group selected from an azido group, a nitrile oxide functional group (i.e. a radical of formula, a nitron functional group or a diazocarbonyl group, combined with a labeling group comprising (or essentially consisting of) an optionally substituted strained alkynyl group (such groups can react covalently in a copper-free strain promoted alkyne-azide cycloaddition (SPAAC));

[0176] (b) a docking group comprising (or essentially consisting of) an optionally substituted strained alkynyl group, combined with a labeling group comprising (or essentially consisting of) a group selected from an

azido group, a nitrile oxide functional group (i.e. a radical of formula, a nitron functional group or a diazocarbonyl group (such groups can react covalently in a copper-free strain promoted alkyne-azide cycloaddition (SPAAC)));

[0177] (c) a docking group comprising (or essentially consisting of) a group selected from optionally substituted strained alkynyl groups, optionally substituted strained alkenyl groups and norbornenyl groups, combined with a labeling group comprising (or essentially consisting of) an optionally substituted tetrazinyl group (such groups can react covalently in a copper-free strain promoted inverse-electron-demand Diels-Alder cycloaddition (SPIEDAC)).

[0178] (d) a docking group comprising (or essentially consisting of) an optionally substituted tetrazinyl group, combined with a labeling group comprising (or essentially consisting of) a group selected from optionally substituted strained alkynyl groups, optionally substituted strained alkenyl groups and norbornenyl groups (such groups can react covalently in a copper-free strain promoted inverse-electron-demand Diels-Alder cycloaddition (SPIEDAC)).

[0179] Optionally substituted strained alkynyl groups include, but are not limited to, optionally substituted trans-cyclooctenyl groups, such as those described in. Optionally substituted strained alkenyl groups include, but are not limited to, optionally substituted cyclooctenyl groups, such as those described in WO 2012/104422 and WO 2015/107064. Optionally substituted tetrazinyl groups include, but are not limited to, those described in WO 2012/104422 and WO 2015/107064.

[0180] An azido group is a radical of formula $-\text{N}_3$.

[0181] A nitron functional group is a radical of formula $-\text{C}(\text{R}^x)=\text{N}^+(\text{R}^y)-\text{O}^-$, wherein R^x and R^y are organic residues, e.g., residues independently selected from C_1 - C_6 -alkyl as described herein.

[0182] A diazocarbonyl group is a radical of formula $-\text{C}(\text{O})-\text{CH}=\text{N}_2$.

[0183] A nitrile oxide functional group is a radical of formula $-\text{C}=\text{N}^+-\text{O}^-$ or, preferably, of formula $-\text{C}=\text{N}^+(\text{R}^x)-\text{O}^-$, wherein R^x is an organic residue, e.g., a residue selected from C_1 - C_6 -alkyl as described herein.

[0184] "Cyclooctynyl" is an unsaturated cycloaliphatic radical having 8 carbon atoms and one triple bond in the ring structure.

[0185] "Trans-cyclooctenyl" is an unsaturated cycloaliphatic radical having 8 carbon atoms and one double bond that is in trans configuration in the ring structure.

[0186] "Tetrazinyl" is a 6-membered monocyclic aromatic radical having 4 nitrogen ring atoms and 2 carbon ring atoms.

[0187] Unless indicated otherwise, the term "substituted" means that a radical is substituted with 1, 2 or 3, especially 1 or 2, substituent(s). In particular embodiments, these substituents can be selected independently from hydrogen, halogen, C_1 - C_4 -alkyl, $(\text{R}^a\text{O})_2\text{P}(\text{O})\text{O}-\text{C}_1$ - C_4 -alkyl, $(\text{R}^b\text{O})_2\text{P}(\text{O})-\text{C}_1$ - C_4 -alkyl, CF_3 , CN, hydroxyl, C_1 - C_4 -alkoxy, $-\text{O}-\text{CF}_3$, C_2 - C_5 -alkenoxy, C_2 - C_5 -alkanoyloxy, C_1 - C_4 -alkylaminocarbonyloxy or C_1 - C_4 -alkylthio, C_1 - C_4 -alkylamino, di- $(\text{C}_1$ - C_4 -alkyl)amino, C_2 - C_5 -alkenylamino, $\text{N}-\text{C}_2$ - C_5 -alkenyl- $\text{N}-\text{C}_1$ - C_4 -alkyl-amino and di- $(\text{C}_2$ - C_5 -alkenyl)amino, wherein R^a and R^b are independently hydrogen or C_2 - C_5 -alkanoyloxymethyl.

[0188] The term halogen denotes in each case a fluorine, bromine, chlorine or iodine radical, in particular a fluorine radical.

[0189] C_1 - C_4 -Alkyl is a straight-chain or branched alkyl group having from 1 to 4, in particular from 1 to 3 carbon atoms. Examples include methyl and C_2 - C_4 -alkyl such as ethyl, n-propyl, iso-propyl, n-butyl, 2-butyl, iso-butyl and tert-butyl.

[0190] C_2 - C_5 -Alkenyl is a singly unsaturated hydrocarbon radical having 2, 3, 4 or 5 carbon atoms. Examples include vinyl, allyl (2-propen-1-yl), 1-propen-1-yl, 2-propen-2-yl, methallyl (2-methylprop-2-en-1-yl), 1-methylprop-2-en-1-yl, 2-buten-1-yl, 3-buten-1-yl, 2-penten-1-yl, 3-penten-1-yl, 4-penten-1-yl, 1-methylbut-2-en-1-yl and 2-ethylprop-2-en-1-yl.

[0191] C_1 - C_4 -Alkoxy is a radical of formula $\text{R}-\text{O}-$, wherein R is a C_1 - C_4 -alkyl group as defined herein.

[0192] C_2 - C_5 -Alkenoxy is a radical of formula $\text{R}-\text{O}-$, wherein R is C_2 - C_5 -alkenyl as defined herein.

[0193] C_2 - C_5 -Alkanoyloxy is a radical of formula $\text{R}-\text{C}(\text{O})-\text{O}-$, wherein R is C_1 - C_4 -alkyl as defined herein.

[0194] C_1 - C_4 -Alkylaminocarbonyloxy is a radical of formula $\text{R}-\text{NH}-\text{C}(\text{O})-\text{O}-$, wherein R is C_1 - C_4 -alkyl as defined herein.

[0195] C_1 - C_4 -Alkylthio is a radical of formula $\text{R}-\text{S}-$, wherein R is C_1 - C_4 -alkyl as defined herein.

[0196] C_1 - C_4 -Alkylamino is a radical of formula $\text{R}-\text{NH}-$, wherein R is C_1 - C_4 -alkyl as defined herein.

[0197] Di- $(\text{C}_1$ - C_4 -alkyl)amino is a radical of formula $\text{R}^x-\text{N}(\text{R}^y)-$, wherein R^x and R^y are independently C_1 - C_4 -alkyl as defined herein.

[0198] C_2 - C_5 -Alkenylamino is a radical of formula $\text{R}-\text{NH}-$, wherein R is C_2 - C_5 -alkenyl as defined herein.

[0199] $\text{N}-\text{C}_2$ - C_5 -alkenyl- $\text{N}-\text{C}_1$ - C_4 -alkyl-amino is a radical of formula $\text{R}^x-\text{N}(\text{R}^y)-$, wherein R^x is C_2 - C_5 -alkenyl as defined herein and R^y is C_1 - C_4 -alkyl as defined herein.

[0200] Di- $(\text{C}_2$ - C_5 -alkenyl)amino is a radical of formula $\text{R}^x-\text{N}(\text{R}^y)-$, wherein R^x and R^y are independently C_2 - C_5 -alkenyl as defined herein.

[0201] C_2 - C_5 -Alkanoyloxymethyl is a radical of formula $\text{R}^x-\text{C}(\text{O})-\text{O}-\text{CH}_2-$, wherein R^x is C_1 - C_4 -alkyl as defined herein.

[0202] The ncAAs used in the context of the present invention can be used in the form of their salt. Salts of ncAAs as described herein mean acid or base addition salts, especially addition salts with physiologically tolerated acids or bases. Physiologically tolerated acid addition salts can be formed by treatment of the base form of an ncAA with appropriate organic or inorganic acids. ncAAs containing an acidic proton may be converted into their non-toxic metal or amine addition salt forms by treatment with appropriate organic and inorganic bases. The ncAAs and salts thereof described in the context of the present invention also comprise the hydrates and solvent addition forms thereof, e.g. hydrates, alcoholates and the like.

[0203] Physiologically tolerated acids or bases are in particular those which are tolerated by the translation system used for preparation of POI with ncAA residues, e.g. are substantially non-toxic to living eukaryotic cells.

[0204] ncAAs, and salts thereof, useful in the context of the present invention can be prepared by analogy to methods which are well known in the art and are described, e.g., in the various publications cited herein.

[0205] The nature of the coupling partner molecule depends on the intended use. For example, the POI may be coupled to a molecule suitable for imaging methods or may be functionalized by coupling to a bioactive molecule. For instance, in addition to the docking group, a coupling partner molecule may comprise a group that selected from, but are not limited to, dyes (e.g. fluorescent, luminescent, or phosphorescent dyes, such as dansyl, coumarin, fluorescein, acridine, rhodamine, silicon-rhodamine, BODIPY, or cyanine dyes), molecules able to emit fluorescence upon contact with a reagent, chromophores (e.g., phytochrome, phycobilin, bilirubin, etc.), radiolabels (e.g. radioactive forms of hydrogen, fluorine, carbon, phosphorous, sulphur, or iodine, such as tritium, ^{18}F , ^{11}C , ^{14}C , ^{32}P , ^{33}P , ^{33}S , ^{35}S , ^{11}In , ^{125}I , ^{123}I , ^{131}I , ^{212}Bi , ^{90}Y or ^{186}Re), MRI-sensitive spin labels, affinity tags (e.g. biotin, His-tag, Flag-tag, strep-tag, sugars, lipids, sterols, PEG-linkers, benzylguanines, benzylcytosines, or co-factors), polyethylene glycol groups (e.g., a branched PEG, a linear PEG, PEGs of different molecular weights, etc.), photocrosslinkers (such as p-azidoiodoacetanilide), NMR probes. X-ray probes, pH probes, IR probes, resins, solid supports and bioactive compounds (e.g. synthetic drugs). Suitable bioactive compounds include, but are not limited to, cytotoxic compounds (e.g., cancer chemotherapeutic compounds), antiviral compounds, biological response modifiers (e.g., hormones, chemokines, cytokines, interleukins, etc.), microtubule affecting agents, hormone modulators, and steroidal compounds. Specific examples of useful coupling partner molecules include, but are not limited to, a member of a receptor/ligand pair; a member of an antibody/antigen pair, a member of a lectin/carbohydrate pair, a member of an enzyme/substrate pair; biotin/avidin; biotin/streptavidin and digoxin/antidigoxin.

[0206] The ability of certain (labeling groups of) ncAA residues to be coupled covalently in situ (the docking groups of) conjugation partner molecules, in particular by a click reaction as described herein, can be used for detecting a POI having such ncAA residue(s) within a eukaryotic cell or tissue expressing the POI, and for studying the distribution and fate of the POI. Thus, for example, the method of the present invention for preparing a POI by expression in eukaryotic cells can be combined with super-resolution microscopy (SRM) to detect the POI within the cell or a tissue of such cells. Several SRM methods are known in the art and can be adapted so as to utilize click chemistry for detecting a POI expressed by a eukaryotic cell of the present invention. Specific examples of such SRM methods include DNA-PAINT (DNA point accumulation for imaging in nanoscale topography; described, e.g., by Jungmann et al., Nat Methods 11:313-318, 2014). dSTORM (direct stochastic optical reconstruction microscopy) and STED (stimulated emission depletion) microscopy.

7. Eukaryotic Cells of the Invention

[0207] Eukaryotic cells of the invention can be generated by introducing polynucleotides as described herein, in particular polynucleotide comprising nucleotide sequences encoding a pRS and encoding a ptRNA or ptRNA-ribozyme as described herein. And in one aspect, the polynucleotide(s) introduced into the eukaryotic cell further comprise a nucleotide sequence encoding a dsRNA-binding polypeptide. Optionally, the introduced polynucleotide(s) further comprise nucleotide sequences encoding a tetO binding protein and/or a constitutive RNA polymerase (e.g. a

T7RNAP) as described herein. Said nucleotide sequences can be located on the same polynucleotide or distributed over two or more different polynucleotides.

[0208] Eukaryotic cells of the present invention can be selected from, but are not limited to, mammalian cells (e.g., human, mouse, rat, cynomolgus monkey cells), avian cells (e.g., chicken cells), insect cells, yeast cells and plant cells. The eukaryotic cells of the invention may be present as individual cells or may be part of a tissue (e.g. a cell in a (cultured) tissue, organ or entire organism).

[0209] A POI comprising one or more than one non-canonical amino acid (ncAA) residue in its amino acid sequence can be prepared according to the present invention using a eukaryotic cell, in particular a eukaryotic cell of the present invention. The eukaryotic cell expresses a pRS and a ptRNA which for a pair in that the pRS is capable of acylating the ptRNA. In particular embodiments, the ptRNA is expressed in the form of a ptRNA-ribozyme as described herein. The eukaryotic cell further expresses the POI in the presence of one or more than one ncAA or salt thereof corresponding to the one or more than one ncAA residue of the POI. To this end the eukaryotic cell comprises polynucleotides encoding the to be expressed pRS, ptRNA and POI, and further comprises, e.g. is fed with, said ncAA or salt thereof. Where expression of one of said pRS, ptRNA and/or POI is under the control of a TRE, the eukaryotic cell expediently further expresses a tetO-binding protein that is chosen to match the respective TRE used as described herein, and the eukaryotic cell is contacted with, e.g. cultured in a medium containing, tetracycline, doxycycline, or a functional analogue thereof as described herein.

[0210] Recombinant expression in eukaryotic cells, including the induction of TRE-controlled expression, is a standard tool in the art. The skilled person is therefore well able to choose conditions of culturing the eukaryotic cells so as to allow for the (recombinant) expression of expression products of interest (here: at least ptRNA, pRS and POI). Depending on the type of cell, a liquid medium can be used for culturing. Culture can be batchwise, semi-batchwise or continuous. Nutrients can be present at the beginning of the culturing or can be supplied later, semi-continuously or continuously.

[0211] For producing a POI (target polypeptide) according to a method of the present invention, the eukaryotic cell is cultured under suitable conditions, preferably in the presence of (e.g., in a culture medium containing) the ncAA or salt thereof, for a time suitable to allow translation at a ribosome of the cell. Depending on the polynucleotide(s) encoding the POI (and optionally the other elements pRS, ptRNA, etc.), it may be required to induce expression by adding a compound inducing transcription, such as, e.g., arabinose, isopropyl β -D-thiogalactoside (IPTG) or, in case of TRE-controlled expression, tetracycline, doxycycline, or a functional analog thereof. mRNA that encodes the POI (and comprises one or more than codon that is the reverse complement of the anticodon comprised by the ptRNA) is bound by the ribosome. Then, the POI is formed by stepwise attachment of amino acids and ncAAs at positions encoded by codons which are recognized (bound) by respective aminoacyl tRNAs. Thus, the ncAA(s) is/are incorporated in the POI at the position(s) encoded by the (selector) codon(s) that is/are the reverse complement of the anticodon comprised by the ptRNA.

[02112] The eukaryotic cells used for preparing a POI comprising one or more than one non-canonical amino acid (ncAA) residue as described herein can be prepared by introducing polynucleotides encoding the pRS, the ptRNA, the POI and/or further elements, such as a dsRNA-binding polypeptide and/or a tetO-binding polypeptide as described herein, into a eukaryotic (host) cell. Said nucleotide sequences can be located on separate nucleic acid molecules (vectors) or on the same nucleic acid molecule (e.g., vector), in any combination, and can be introduced into the cell in combination or sequentially.

[02113] Preferably common cloning and transfection techniques, known by a person skilled in the art, are used, for example co-precipitation, protoplast fusion, electroporation, virus-mediated gene delivery, lipofection, microinjection or others, for introducing the stated nucleic acid molecules in the respective cell. Suitable techniques are described for example in Current Protocols in Molecular Biology, F. Ausubel et al., Ed., Wiley Interscience, New York 1997, or Sambrook et al. Molecular Cloning: A Laboratory Manual, 2nd edition, Cold Spring Harbor Laboratory, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY, 1989.

[02114] After translation, the POI prepared according to the present invention may optionally be recovered from the eukaryotic cell or, if secreted, from the culture medium, and purified, either partially or substantially to homogeneity, according to procedures generally known in the art. For this purpose, the POI can be recovered and purified, either partially or substantially to homogeneity, according to procedures known to and used by those of skill in the art. Unless the target polypeptide is secreted into the culture medium, recovery usually requires cell disruption. Methods of cell disruption are well known in the art and include physical disruption, e.g., by (ultrasound) sonication, liquid-shear disruption (e.g., via French press), mechanical methods (such as those utilizing blenders or grinders) or freeze-thaw cycling, as well as chemical lysis using agents which disrupt lipid-lipid, protein-protein and/or protein-lipid interactions (such as detergents), and combinations of physical disruption techniques and chemical lysis. Standard procedures for purifying polypeptides from cell lysates or culture media are also well known in the art and include, e.g., ammonium sulfate or ethanol precipitation, acid or base extraction, column chromatography, affinity column chromatography, anion or cation exchange chromatography, phosphocellulose chromatography, hydrophobic interaction chromatography, hydroxylapatite chromatography, lectin chromatography, gel electrophoresis and the like. Protein refolding steps can be used, as desired, in making correctly folded mature proteins. High performance liquid chromatography (HPLC), affinity chromatography or other suitable methods can be employed in final purification steps where high purity is desired. Antibodies made against the polypeptides of the invention can be used as purification reagents, i.e. for affinity-based purification of the polypeptides. A variety of purification/protein folding methods are well known in the art, including, e.g., those set forth in Scopes, Protein Purification, Springer, Berlin (1993); and Deutscher, Methods in Enzymology Vol. 182: Guide to Protein Purification, Academic Press (1990); and the references cited therein.

[02115] As noted, those of skill in the art will recognize that, after synthesis, expression and/or purification, polypeptides can possess a conformation different from the desired conformations of the relevant polypeptides. For

example, overexpressed polypeptides as well as polypeptides produced by prokaryotic systems often are optimized by exposure to chaotropic agents to achieve proper folding. During purification from, e.g., lysates derived from *E. coli*, the expressed polypeptide is optionally denatured and then renatured. This is accomplished, e.g., by solubilizing the proteins in a chaotropic agent such as guanidine HCl. In general, it is occasionally desirable to denature and reduce expressed polypeptides and then to cause the polypeptides to re-fold into the preferred conformation. For example, guanidine, urea, DTT, DTE, and/or a chaperonin can be added to a translation product of interest. Methods of reducing, denaturing and renaturing proteins are well known to those of skill in the art. Polypeptides can be refolded in a redox buffer containing, e.g., oxidized glutathione and L-arginine.

[02116] The POIs expressed by a method of the present invention can be purified by known techniques, such as, e.g., molecular sieve chromatography (gel filtration), such as Q-sepharose chromatography, ion exchange chromatography and hydrophobic chromatography, and other common protein purification techniques such as ultrafiltration, crystallization, salting-out, dialysis and native gel electrophoresis. Suitable methods are described, for example, in Cooper, T. G., Biochemische Arbeitsmethoden [Biochemical processes], Verlag Walter de Gruyter, Berlin, New York or in Scopes, R., Protein Purification, Springer Verlag, New York, Heidelberg, Berlin.

[02117] For isolating a POI, it can be advantageous to link the POI with a tag that can serve for easier purification. This can be achieved by introducing a corresponding tag-encoding sequence into the POI-encoding nucleotide sequence. Suitable tags for protein purification are well known in the art and include, e.g., histidine tags (e.g., His₆ tag) and epitopes that can be recognized as antigens of antibodies (described for example in Harlow, E. and Lane, D., 1988, Antibodies: A Laboratory Manual, Cold Spring Harbor (N.Y.) Press). These tags can serve for attaching the proteins to a solid carrier, for example a polymer matrix, which can for example be used as packing in a chromatography column, or can be used on a microtiter plate or on some other carrier.

[02118] A tag linked to a POI can also serve for detecting the POI. Tags for protein detection are well known in the art and include, e.g., fluorescent dyes, enzyme markers, which form a detectable reaction product after reaction with a substrate, and others.

[02119] Also described are POIs produced by the methods of the invention. Such POIs can be prepared by a method of the invention that makes use of a eukaryotic cell as described herein.

8. Kits

[0220] The present invention also provides kits for preparing a POI having one or more than one non-canonical amino acid (ncAA) residue in its amino acid sequence. The kit of the invention comprises one or more than one ncAA, or salt thereof, corresponding to the one or more than one ncAA residue of the POI. The kit may further comprise a polynucleotide, or combination of polynucleotides, of the present invention, and/or a polynucleotide encoding a dsRNA-binding polypeptide as described herein and/or an eukaryotic cell of the present invention. The kit may additionally comprise one or more reporter constructs encoding an easily detectable (e.g. fluorescent) reporter polypeptide.

The report construct comprises at least one selector codon that is located in the encoding sequence of the report polypeptide such that successful incorporation of an amino acid or ncAA at the position encoded by said selector codon(s) allows for the translational synthesis of the reporter polypeptide.

[0221] The kits of the present invention can be used in methods of the invention for preparing ncAA-residue containing POIs as described herein.

9. Particular Embodiments

[0222] The present invention refers to the following particular embodiments:

[0223] 1. According to a first embodiment the invention relates to an eukaryotic cell comprising:

[0224] (a) a polynucleotide encoding a prokaryotic aminoacyl tRNA synthetase (pRS),

[0225] (b) a polynucleotide encoding a prokaryotic tRNA (ptRNA) and one or more than one ribozyme,

[0226] wherein the pRS is capable of acylating the ptRNA, and

[0227] wherein the sequences encoding the ptRNA and the ribozyme(s) are linked such that transcription produces an RNA molecule comprising both the ptRNA and the ribozyme(s) (ptRNA-ribozyme).

[0228] A second embodiment relates to an eukaryotic cell of embodiment one, wherein the ribozyme is selected from a hammerhead ribozyme, a hairpin ribozyme, a Varkud satellite ribozyme, a glucosamine-6-phosphate synthetase riboswitch and a hepatitis delta virus (HDV) ribozyme.

[0229] A third embodiment relates to an eukaryotic cell of embodiment one or embodiment two, wherein the ribozyme is a HDV ribozyme which, in the ptRNA-ribozyme, is directly linked to the 3' end of the ptRNA.

[0230] A fourth embodiment relates to an eukaryotic cell of any one of embodiments one to three, wherein:

[0231] (i) transcriptional synthesis of the ptRNA-ribozyme is controlled by a tetracycline-responsive promoter element (TRE), or

[0232] (ii) the ptRNA-ribozyme is transcriptionally synthesized by an RNA polymerase, wherein expression of said RNA polymerase is controlled by a TRE.

[0233] A fifth embodiment relates to an eukaryotic cell of any one of embodiments one to four, wherein the transcription of the pRS-encoding sequence is controlled by a tetracycline-responsive promoter element (TRE).

[0234] A sixth embodiment relates to an eukaryotic cell of embodiment four or embodiment five, further comprising a polynucleotide encoding a protein which binds to a tetracycline operator (tetO) sequence (tetO-binding protein), wherein the tetO-binding protein is selected from the group consisting of

[0235] (i) a tetracycline repressor (tetR).

[0236] (ii) a tetracycline-controlled transcriptional activator (tTA), and

[0237] (iii) a reverse tTA (rtTA).

[0238] A seventh embodiment relates to an eukaryotic cell of any one of embodiments one to six, wherein:

[0239] (i) transcriptional synthesis of the ptRNA-ribozyme is controlled by a T7 promoter, and/or

[0240] (ii) transcription of the pRS-encoding sequence is controlled by a T7 promoter,

[0241] and wherein the eukaryotic cell further comprises

[0242] (c) a polynucleotide encoding a T7 RNA polymerase (T7RNAP).

[0243] An eighth embodiment relates to an eukaryotic cell of embodiment seven, wherein expression of the T7RNAP is controlled by a TRE.

[0244] A ninth embodiment relates to an eukaryotic cell of embodiment seven or embodiment eight, wherein the T7RNAP carries

[0245] (i) a nuclear localization signal (NLS), or

[0246] (ii) a nuclear export signal (NES), or

[0247] (iii) none of NES and NLS.

[0248] A tenth embodiment relates to an eukaryotic cell of any one of embodiments one to nine further comprising a polynucleotide encoding a dsRNA-binding polypeptide as defined in any one of following embodiments eleven to thirteen.

[0249] An eleventh embodiment relates to an eukaryotic cell comprising:

[0250] (a) a polynucleotide encoding a prokaryotic aminoacyl tRNA synthetase (pRS),

[0251] (b) a polynucleotide encoding a prokaryotic tRNA (ptRNA), and

[0252] (c) a polynucleotide encoding a polypeptide that is capable of binding to double-stranded RNA (dsRNA-binding polypeptide) and comprises an amino acid sequence having at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identity to the sequence set forth in any one of SEQ ID NOs:37, and 57-59.

[0253] A twelfth embodiment relates to an eukaryotic cell of embodiment eleven, wherein the amino acid sequence comprised by the dsRNA-binding polypeptide has at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identity to the sequence set forth in SEQ ID NO:38.

[0254] A thirteenth embodiment relates to an eukaryotic cell of embodiment eleven or embodiment twelve, wherein the dsRNA-binding polypeptide is incapable of phosphorylating eukaryotic initiation factor 2 α (eIF-2 α).

[0255] A fourteenth embodiment relates to an eukaryotic cell of any one of embodiments eleven to thirteen, wherein:

[0256] (i) transcriptional synthesis of the ptRNA is controlled by a tetracycline-responsive promoter element (TRE), or

[0257] (ii) the ptRNA is transcriptionally synthesized by an RNA polymerase, wherein expression of said RNA polymerase is controlled by a TRE.

[0258] A fifteenth embodiment relates to an eukaryotic cell of any one of embodiments eleven to fourteen, wherein:

[0259] (i) transcription of the pRS-encoding sequence is controlled by a tetracycline-responsive promoter element (TRE), or

[0260] (ii) transcription of the pRS-encoding sequence is catalyzed by an RNA polymerase, wherein the expression of said RNA polymerase is controlled by a TRE.

[0261] A sixteenth embodiment relates to an eukaryotic cell of embodiment fourteen or embodiment fifteen, further comprising a polynucleotide encoding a protein which binds to a tetracycline operator (tetO) sequence (tetO-binding protein), wherein the tetO-binding protein is selected from the group consisting of:

- [0262] (i) a tetracycline repressor (tetR),
- [0263] (ii) a tetracycline-controlled transcriptional activator (tTA), and
- [0264] (iii) a reverse tTA (rtTA).
- [0265] A seventeenth embodiment relates to an eukaryotic cell of any one of embodiments eleven to sixteen, wherein:
- [0266] (i) transcriptional synthesis of the ptRNA is controlled by a T7 promoter, or
- [0267] (ii) transcription of the pRS-encoding sequence is controlled by a T7 promoter, or
- [0268] (iii) both of (i) and (ii);
- [0269] and wherein the eukaryotic cell further comprises
- [0270] (c) a polynucleotide encoding a T7 RNA polymerase (T7RNAP).
- [0271] An eighteenth embodiment relates to an eukaryotic cell of embodiment seventeen, wherein expression of the T7RNAP is controlled by a TRE.
- [0272] A nineteenth embodiment relates to an eukaryotic cell of embodiment seventeen of embodiment eighteen, wherein the T7RNAP carries
- [0273] (i) a nuclear localization signal (NLS), or
- [0274] (ii) a nuclear export signal (NES), or
- [0275] (iii) none of NES and NLS.
- [0276] A twentieth embodiment relates to an eukaryotic cell of any one of embodiments eleven to thirteen, wherein the eukaryotic cell is a cell of any one of embodiments one to nine.
- [0277] A twenty-first embodiment relates to a polynucleotide encoding an RNA molecule comprising a ptRNA and one or more than one ribozyme (ptRNA-ribozyme) as defined in any one of embodiments one to four and seven; wherein, more particularly,
- [0278] a) the ribozyme is a HDV ribozyme which, in the ptRNA-ribozyme, is directly linked to the 3' end of the ptRNA; or
- [0279] b) the ptRNA is derived from a *Methanosarcina* species, in particular a pyrrolysyl tRNA of *M. mazei*; or
- [0280] c) wherein a ptRNA molecule is encoded missing the 3'-terminal aminoacyl acceptor stem motif CCA; or
- [0281] d) the ptRNA is encoded by the following nucleotide sequence (SEQ ID NO:60) ggaacctgatcatgtagatcgaaatggactxxxaatccgttcagccgggttagat-tcccggggttcg wherein xxx encodes an anticodon;
- [0282] A twenty-second embodiment relates to a polynucleotide, or combination of two or more polynucleotides, comprising a nucleotide sequence encoding an RNA molecule comprising a ptRNA and one or more than one ribozyme (ptRNA-ribozyme) as defined in any one of embodiments one to four and seven, and:
- [0283] (i) a nucleotide sequence encoding a tetO-binding protein as defined in embodiment six, or
- [0284] (ii) a nucleotide sequence encoding a pRS as defined in any one of embodiments one, five and seven, or
- [0285] (ii) both of (i) and (ii).
- [0286] A twenty-third embodiment relates to a method for preparing a polypeptide of interest (POI) having one or more than one non-canonical amino acid (ncAA) residue in its amino acid sequence, wherein the method comprises:
- [0287] (a) expressing, in a eukaryotic cell:
- [0288] a prokaryotic aminoacyl tRNA synthetase (pRS) and
- [0289] an RNA molecule comprising a prokaryotic tRNA (ptRNA) and one or more than one ribozyme (ptRNA-ribozyme); and concomitantly or sequentially
- [0290] (b) expressing the POI in the eukaryotic cell in the presence of one or more than one ncAA or salt thereof corresponding to the one or more than one ncAA residue of the POI; and
- [0291] (c) optionally recovering the expressed POI;
- [0292] wherein the pRS is capable of acylating the ptRNA with the ncAA or the salt thereof,
- [0293] wherein the POI is encoded by a nucleotide sequence that comprises one or more than one selector codon encoding the one or more than one ncAA residue, and
- [0294] wherein the selector codon is the reverse complement of the anticodon of the ptRNA.
- [0295] A twenty-fourth embodiment relates to a method of embodiment twenty-three, wherein the ribozyme is selected from a hammerhead ribozyme, a hairpin ribozyme, a Varkud satellite ribozyme, a glucosamine-6-phosphate synthetase riboswitch and a hepatitis delta virus (HDV) ribozyme.
- [0296] A twenty-fifth embodiment relates to a method of embodiment twenty-three or embodiment twenty-four, wherein the ribozyme is a HDV ribozyme which, in the ptRNA-ribozyme, is directly linked to the 3' end of the ptRNA.

T7-Controlled in General

- [0297] A twenty-sixth embodiment relates to a method of any one of embodiments twenty-three to twenty-five, wherein the transcriptional synthesis of the ptRNA-ribozyme is controlled by a T7 promoter, and wherein the method further comprises expressing a T7 RNA polymerase (T7RNAP) in the eukaryotic cell prior to or concomitantly with the thereby controlled expression of the ptRNA-ribozyme (i) or the pRS (ii) or both (iii), respectively.
- T-REX System (tetR, +tet/dox)
- [0298] A twenty-seventh embodiment relates to a method of any one of embodiments twenty-three to twenty-five, wherein:
- [0299] (i) transcriptional synthesis of the ptRNA-ribozyme is controlled by a tetracycline-responsive promoter element (TRE), or
- [0300] (ii) the ptRNA-ribozyme is transcriptionally synthesized by an RNA polymerase, wherein the expression of said RNA polymerase is controlled by a TRE;
- [0301] wherein the method further comprises expressing a tetracycline repressor (tetR) in the eukaryotic cell at least prior to the expression of the ptRNA-ribozyme, and
- [0302] wherein expression of the ptRNA-ribozyme involves contacting the eukaryotic cell with tetracycline, doxycycline, or a functional analogue thereof, thereby inducing transcriptional synthesis of the ptRNA-ribozyme.
- [0303] A twenty-eighth embodiment relates to a method of any one of embodiments twenty-three to twenty-five and twenty-seven, wherein:
- [0304] (i) transcription of the pRS-encoding sequence is controlled by a tetracycline-responsive promoter element (TRE), or

- [0305] (ii) transcription of the pRS-encoding sequence is catalyzed by an RNA polymerase, wherein the expression of said RNA polymerase is controlled by a TRE;
- [0306] wherein the method further comprises expressing a tetracycline repressor (tetR) in the eukaryotic cell at least prior to the expression of the pRS; and
- [0307] wherein expression of the pRS involves contacting the eukaryotic cell with tetracycline, doxycycline, or a functional analogue thereof, thereby inducing transcription of the pRS-encoding sequence.
- [0308] A twenty-ninth embodiment relates to a method of embodiment twenty-seven or embodiment twenty-eight, wherein:
- [0309] (i) transcriptional synthesis of the ptRNA-ribozyme is controlled by a T7 promoter, or
- [0310] (ii) transcription of the pRS-encoding sequence is controlled by a T7 promoter, or
- [0311] (iii) both of (i) and (ii);
- [0312] and wherein the method further comprises:
- [0313] expressing a T7 RNA polymerase (T7RNAP) in the eukaryotic cell prior to or concomitantly with the thereby controlled expression of the ptRNA-ribozyme (i) or the pRS (ii) or both (iii), respectively, wherein expression of the T7RNAP is controlled by a TRE, and
- [0314] expressing a tetR in the eukaryotic cell at least concomitantly with the expression of the T7RNAP;
- [0315] wherein expression of the T7RNAP involves contacting the eukaryotic cell with tetracycline, doxycycline, or a functional analogue thereof, thereby inducing transcription of the T7RNAP-encoding sequence.

Tet-Off System (TA, -tet/dox)

- [0316] A thirtieth embodiment relates to a method of any one of embodiments twenty-three to twenty-five, wherein:
- [0317] (i) transcriptional synthesis of the ptRNA-ribozyme is controlled by a tetracycline-responsive promoter element (TRE), or
- [0318] (ii) the ptRNA-ribozyme is transcriptionally synthesized by an RNA polymerase, wherein expression of said RNA polymerase is controlled by a TRE;
- [0319] wherein the method further comprises:
- [0320] (1) expressing a tetracycline-controlled transcriptional activator (tTA) in the eukaryotic cell prior to and concomitantly with the expression of the ptRNA-ribozyme, and
- [0321] (2) keeping the eukaryotic cell prior to the expression of the ptRNA-ribozyme in contact with tetracycline, doxycycline, or a functional analogue thereof, thereby suppressing the transcriptional synthesis of the ptRNA-ribozyme, and then reducing the concentration of, or preferably removing, said tetracycline, doxycycline, or functional analogue thereof so as to induce transcriptional synthesis of the ptRNA-ribozyme.
- [0322] A thirty-first embodiment relates to a method of any one of embodiments twenty-three to twenty-five and thirty, wherein:
- [0323] (i) transcription of the pRS-encoding sequence is controlled by a TRE, or

- [0324] (ii) transcription of the pRS-encoding sequence is catalyzed by an RNA polymerase, wherein the expression of said RNA polymerase is controlled by a TRE;

[0325] wherein the method further comprises:

- [0326] (1) expressing a tetracycline-controlled transcriptional activator (tTA) in the eukaryotic cell prior to and concomitantly with the expression of the pRS, and
- [0327] (2) keeping the eukaryotic cell prior to the expression of the pRS in contact with tetracycline, doxycycline, or a functional analogue thereof, thereby suppressing transcription of the pRS-encoding sequence, and then reducing the concentration of, or preferably removing, said tetracycline, doxycycline, or functional analogue thereof so as to induce transcription of the pRS-encoding sequence.

[0328] A thirty-second embodiment relates to a method of embodiment thirty or embodiment thirty-one, wherein:

- [0329] (i) transcriptional synthesis of the ptRNA-ribozyme is controlled by a T7 promoter, or
- [0330] (ii) transcription of the pRS-encoding sequence is controlled by a T7 promoter, or
- [0331] (iii) both of (i) and (ii);
- [0332] wherein the method further comprises:
- [0333] (3) expressing a T7 RNA polymerase (T7RNAP) in the eukaryotic cell prior to or concomitantly with the thereby controlled expression of the ptRNA-ribozyme (i) or the pRS (ii) or both (iii), respectively,
- [0334] wherein expression of the T7RNAP is controlled by a TRE,

- [0335] (4) expressing a tTA in the eukaryotic cell prior to and concomitantly with the expression of the T7RNAP, and

- [0336] (5) keeping the eukaryotic cell prior to the expression of the T7RNAP in contact with tetracycline, doxycycline, or a functional analogue thereof, thereby suppressing transcription of the T7RNAP-encoding sequence, and then reducing the concentration of, or preferably removing, said tetracycline, doxycycline, or functional analogue thereof so as to induce transcription of the T7RNAP-encoding sequence.

Tet-On system (rtTA, +dox)

[0337] A thirty-third embodiment relates to a method of any one of embodiments twenty-three to twenty-five, wherein:

- [0338] (i) transcriptional synthesis of the ptRNA-ribozyme is controlled by a tetracycline-responsive promoter element (TRE), or
- [0339] (ii) the ptRNA-ribozyme is transcriptionally synthesized by an RNA polymerase, wherein the expression of said RNA polymerase is controlled by a TRE;

[0340] wherein the method further comprises expressing a reverse tTA (rtTA) in the eukaryotic cell at least concomitantly with the expression of the ptRNA, and

[0341] wherein expression of the ptRNA-ribozyme involves contacting the eukaryotic cell with doxycycline, or a functional analogue thereof, thereby inducing transcriptional synthesis of the ptRNA-ribozyme.

[0342] A thirty-fourth embodiment relates to a method of any one of embodiments twenty-three to twenty-five and thirty-three, wherein:

- [0343] (i) transcription of the pRS-encoding sequence is controlled by a tetracycline-responsive promoter element (TRE), or
- [0344] (ii) transcription of the pRS-encoding sequence is catalyzed by an RNA polymerase, wherein the expression of said RNA polymerase is controlled by a TRE;
- [0345] wherein the method further comprises expressing a reverse tTA (rtTA) in the eukaryotic cell at least concomitantly with the expression of the pRS; and
- [0346] wherein expression of the pRS involves contacting the eukaryotic cell with doxycycline, or a functional analogue thereof, thereby inducing transcription of the pRS-encoding sequence.
- [0347] A thirty-fifth embodiment relates to a method of embodiment thirty-three or embodiment thirty-four, wherein:
- [0348] (i) transcriptional synthesis of the ptRNA-ribozyme is controlled by a T7 promoter, or
- [0349] (ii) transcription of the pRS-encoding sequence is controlled by a T7 promoter, or
- [0350] (iii) both of (i) and (ii);
- [0351] and wherein the method further comprises:
- [0352] expressing a T7 RNA polymerase (T7RNAP) in the eukaryotic cell prior to or concomitantly with the thereby controlled expression of the ptRNA-ribozyme (i) or the pRS (ii) or both (iii), respectively, wherein expression of the T7RNAP is controlled by a TRE, and
- [0353] expressing a rtTA in the eukaryotic cell at least concomitantly with the expression of the T7RNAP;
- [0354] wherein expression of the T7RNAP involves contacting the eukaryotic cell with doxycycline, or a functional analogue thereof, thereby inducing transcription of the T7RNAP-encoding sequence.
- [0355] A thirty-sixth embodiment relates to a method of any one of embodiments twenty-three to twenty-five further comprising:
- [0356] expressing, in the eukaryotic cell, a dsRNA-binding polypeptide as defined in any one of embodiments eleven to thirteen.
- [0357] A thirty-seventh embodiment relates to a method for preparing a polypeptide of interest (POI) having one or more than one non-canonical amino acid (ncAA) residue in its amino acid sequence, wherein the method comprises:
- [0358] (a) expressing, in a eukaryotic cell:
- [0359] a prokaryotic aminoacyl tRNA synthetase (pRS).
- [0360] a prokaryotic tRNA (ptRNA) and
- [0361] a polypeptide capable of binding to double-stranded RNA (dsRNA-binding polypeptide);
- [0362] and concomitantly or sequentially
- [0363] (b) expressing the POI in the eukaryotic cell in the presence of one or more than one ncAA or salt thereof corresponding to the one or more than one ncAA residue of the POI; and
- [0364] (c) optionally recovering the expressed POI;
- [0365] wherein the pRS is capable of acylating the ptRNA with the ncAA or the salt thereof,
- [0366] wherein the dsRNA-binding polypeptide comprises an amino acid sequence having at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%,

98%, or 99% identity to the sequence set forth in any one of SEQ ID NOs:37, and 57-59;

[0367] wherein the POI is encoded by a nucleotide sequence that comprises one or more than one selector codon encoding the one or more than one ncAA residue, and

[0368] wherein the selector codon is the reverse complement of the anticodon of the ptRNA.

[0369] A thirty-eighth embodiment relates to a method of embodiment thirty-seven, wherein the amino acid sequence comprised by the dsRNA-binding polypeptide has at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identity to the sequence set forth in SEQ ID NO:38.

[0370] A thirty-ninth embodiment relates to a method of embodiment thirty-seven or embodiment thirty-eight, wherein the dsRNA-binding polypeptide is incapable of phosphorylating eukaryotic initiation factor 2 α (eIF-2 α).

T7-Controlled in General

[0371] A fortieth embodiment relates to a method of any one of embodiments thirty-seven to thirty-nine, wherein:

[0372] (i) transcriptional synthesis of the ptRNA is controlled by a T7 promoter, or

[0373] (ii) transcription of the pRS-encoding sequence is controlled by a T7 promoter, or

[0374] (iii) both of (i) and (ii);

[0375] wherein the method further comprises expressing a T7 RNA polymerase (T7RNAP) in the eukaryotic cell prior to or concomitantly with the thereby controlled expression of the ptRNA (i) or the pRS (ii) or both (iii), respectively.

T-Rex System (tetR, +tet/dox)

[0376] A forty-first embodiment relates to a method of any one of embodiments thirty-seven to thirty-nine, wherein:

[0377] (i) transcriptional synthesis of the ptRNA is controlled by a tetracycline-responsive promoter element (TRE), or

[0378] (ii) the ptRNA is transcriptionally synthesized by an RNA polymerase, wherein the expression of said RNA polymerase is controlled by a TRE;

[0379] wherein the method further comprises expressing a tetracycline repressor (tetR) in the eukaryotic cell at least prior to the expression of the ptRNA, and

[0380] wherein expression of the ptRNA involves contacting the eukaryotic cell with tetracycline, doxycycline, or a functional analogue thereof, thereby inducing transcriptional synthesis of the ptRNA.

[0381] A forty-second embodiment relates to a method of any one of embodiments thirty-seven to thirty-nine and forty-one, wherein:

[0382] (i) transcription of the pRS-encoding sequence is controlled by a tetracycline-responsive promoter element (TRE), or

[0383] (ii) transcription of the pRS-encoding sequence is catalyzed by an RNA polymerase, wherein the expression of said RNA polymerase is controlled by a TRE;

[0384] wherein the method further comprises expressing a tetracycline repressor (tetR) in the eukaryotic cell at least prior to the expression of the pRS; and

[0385] wherein expression of the pRS involves contacting the eukaryotic cell with tetracycline, doxycycline,

or a functional analogue thereof, thereby inducing transcription of the pRS-encoding sequence.

[0386] A forty-third embodiment relates to a method of embodiment forty-one or embodiment forty-two, wherein:

[0387] (i) transcriptional synthesis of the ptRNA is controlled by a T7 promoter, or

[0388] (ii) transcription of the pRS-encoding sequence is controlled by a T7 promoter, or

[0389] (iii) both of (i) and (ii);

[0390] and wherein the method further comprises:

[0391] expressing a T7 RNA polymerase (T7RNAP) in the eukaryotic cell prior to or concomitantly with the thereby controlled expression of the ptRNA (i) or the pRS (ii) or both (iii), respectively, wherein expression of the T7RNAP is controlled by a TRE, and

[0392] expressing a tetR in the eukaryotic cell at least concomitantly with the expression of the T7RNAP;

[0393] wherein expression of the T7RNAP involves contacting the eukaryotic cell with tetracycline, doxycycline, or a functional analogue thereof, thereby inducing transcription of the T7RNAP-encoding sequence.

Tet-Off System (tTA, -tet/dox)

[0394] A forty-fourth embodiment relates to a method of any one of embodiments thirty-seven to thirty-nine, wherein:

[0395] (i) transcriptional synthesis of the ptRNA is controlled by a tetracycline-responsive promoter element (TRE), or

[0396] (ii) the ptRNA is transcriptionally synthesized by an RNA polymerase, wherein expression of said RNA polymerase is controlled by a TRE;

[0397] wherein the method further comprises:

[0398] (1) expressing a tetracycline-controlled transcriptional activator (tTA) in the eukaryotic cell prior to and concomitantly with the expression of the ptRNA, and

[0399] (2) keeping the eukaryotic cell prior to the expression of the ptRNA in contact with tetracycline, doxycycline, or a functional analogue thereof, thereby suppressing the transcriptional synthesis of the ptRNA, and then reducing the concentration of, or preferably removing, said tetracycline, doxycycline, or functional analogue thereof so as to induce transcriptional synthesis of the ptRNA.

[0400] A forty-fifth embodiment relates to a method of any one of embodiments thirty-seven to thirty-nine and forty-four, wherein:

[0401] (i) transcription of the pRS-encoding sequence is controlled by a TRE, or

[0402] (ii) transcription of the pRS-encoding sequence is catalyzed by an RNA polymerase, wherein the expression of said RNA polymerase is controlled by a TRE;

[0403] wherein the method further comprises:

[0404] (1) expressing a tetracycline-controlled transcriptional activator (tTA) in the eukaryotic cell prior to and concomitantly with the expression of the pRS, and

[0405] (2) keeping the eukaryotic cell prior to the expression of the pRS in contact with tetracycline, doxycycline, or a functional analogue thereof, thereby suppressing transcription of the pRS-encoding sequence, and then reducing the concentration of, or

preferably removing, said tetracycline, doxycycline, or functional analogue thereof so as to induce transcription of the pRS-encoding sequence.

[0406] A forty-sixth embodiment relates to a method of embodiment forty-four or embodiment forty-five, wherein:

[0407] (i) transcriptional synthesis of the ptRNA is controlled by a T7 promoter, or

[0408] (ii) transcription of the pRS-encoding sequence is controlled by a T7 promoter, or

[0409] (iii) both of (i) and (ii);

[0410] wherein the method further comprises:

[0411] (3) expressing a T7 RNA polymerase (T7RNAP) in the eukaryotic cell prior to or concomitantly with the thereby controlled expression of the ptRNA (i) or the pRS (ii) or both (iii), respectively,

[0412] wherein expression of the T7RNAP is controlled by a TRE,

[0413] (4) expressing a tTA in the eukaryotic cell prior to and concomitantly with the expression of the T7RNAP, and

[0414] (5) keeping the eukaryotic cell prior to the expression of the T7RNAP in contact with tetracycline, doxycycline, or a functional analogue thereof, thereby suppressing transcription of the T7RNAP-encoding sequence, and then reducing the concentration of, or preferably removing, said tetracycline, doxycycline, or functional analogue thereof so as to induce transcription of the T7RNAP-encoding sequence.

Tet-On System (rtTA, +dox)

[0415] A forty-seventh embodiment of a method of any one of embodiments thirty-seven to thirty-nine, wherein:

[0416] (i) transcriptional synthesis of the ptRNA is controlled by a tetracycline-responsive promoter element (TRE), or

[0417] (ii) the ptRNA is transcriptionally synthesized by an RNA polymerase, wherein the expression of said RNA polymerase is controlled by a TRE;

[0418] wherein the method further comprises expressing a reverse tTA (rtTA) in the eukaryotic cell at least concomitantly with the expression of the ptRNA, and

[0419] wherein expression of the ptRNA involves contacting the eukaryotic cell with doxycycline, or a functional analogue thereof, thereby inducing transcriptional synthesis of the ptRNA.

[0420] A forty-eighth embodiment of a method of any one of embodiments thirty-seven to thirty-nine and forty-seven, wherein:

[0421] (i) transcription of the pRS-encoding sequence is controlled by a tetracycline-responsive promoter element (TRE), or

[0422] (ii) transcription of the pRS-encoding sequence is catalyzed by an RNA polymerase, wherein the expression of said RNA polymerase is controlled by a TRE;

[0423] wherein the method further comprises expressing a reverse tTA (rtTA) in the eukaryotic cell at least concomitantly with the expression of the pRS; and

[0424] wherein expression of the pRS involves contacting the eukaryotic cell with doxycycline, or a functional analogue thereof, thereby inducing transcription of the pRS-encoding sequence.

[0425] A forty-ninth embodiment relates to a method of embodiment forty-seven or embodiment forty-eight, wherein:

- [0426] (i) transcriptional synthesis of the ptRNA is controlled by a T7 promoter, or
- [0427] (ii) transcription of the pRS-encoding sequence is controlled by a T7 promoter, or
- [0428] (iii) both of (i) and (ii);
- [0429] and wherein the method further comprises:
- [0430] expressing a T7 RNA polymerase (T7RNAP) in the eukaryotic cell prior to or concomitantly with the thereby controlled expression of the ptRNA (i) or the pRS (ii) or both (iii), respectively, wherein expression of the T7RNAP is controlled by a TRE, and
- [0431] expressing a rtTA in the eukaryotic cell at least concomitantly with the expression of the T7RNAP;
- [0432] wherein expression of the T7RNAP involves contacting the eukaryotic cell with doxycycline, or a functional analogue thereof, thereby inducing transcription of the T7RNAP-encoding sequence.
- [0433] A fiftieth embodiment relates to a method of any one of embodiments thirty-seven to forty-nine, wherein the ptRNA is expressed a part of an RNA molecule further comprising one or more than one ribozyme (ptRNA-ribozyme).
- [0434] A fifty-first embodiment relates to a method of embodiment fifty, wherein the method is a method of any one of embodiments twenty-three to thirty-five.
- [0435] A fifty-second embodiment relates to a kit for preparing a polypeptide of interest (POI) having one or more than one non-canonical amino acid (ncAA) residue in its amino acid sequence, wherein the kit comprises one or more than one ncAA, or salt thereof, corresponding to the one or more than one ncAA residue of the POI, and:
- [0436] (a) a polynucleotide of embodiment twenty-one, or
- [0437] (b) a polynucleotide, or combination of two or more polynucleotides, of embodiment twenty-two, or
- [0438] (c) a polynucleotide encoding a dsRNA-binding polypeptide as defined in any one of embodiments eleven to thirteen, or
- [0439] (d) an eukaryotic cell of any one of embodiments one to twenty.
- [0440] The invention is now explained in more detail by making reference to the particular, non-limiting embodiments of the subsequent experiment section.

EXAMPLES

Material and Methods

[0441] Unless stated otherwise, the cloning and expression of recombinant polypeptides was carried out by standard methods, as described for example in Sambrook, J., Fritsch, E. F. and Maniatis, T., *Molecular Cloning: A Laboratory Manual*, 2nd edition, Cold Spring Harbor Laboratory. Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY, 1989.

A) Cloning

[0442] The reporter gene NLS-iRFP-Flag-GFP^{Y39→TAG}-6His (SEQ ID NO:39) was cloned into a pCI plasmid (Promega) as described in Nikic et al. (*Angew Chem Int Ed Engl* 2016, 55:16172-16176). NLS-iRFP-Flag-GFP^{Y39→}

^{→TAG}-6His contains iRFP fused to a GFP wherein position 39 of the GFP is encoded by the amber stop codon TAG (iRFP-GFP^{Y39TAG}).

[0443] The coding sequence of T7 RNA polymerase gene with and without a NES signal (NES-T7RNAP or T7RNAP) was inserted into the pcDNA3.1 plasmid or pCAGGS plasmid. The pcDNA3.1 plasmid and pCAGGS plasmid differ in the promoter the pcDNA3.1 plasmid comprises the CMV promoter; the pCAGGS plasmid comprises a CMV enhancer with a chicken actin promoter and an intron (SEQ ID NO: 56). For constitutive expression the T7RNAP- or NES-T7RNAP-encoding sequence was inserted downstream of the CMV promoter (pcDNA3.1 plasmid) or downstream of the chicken actin promoter and intron (pCAGGS plasmid). For inducible expression via T-REx system, the NES-T7RNAP-encoding sequence was inserted downstream of a “T-REx” promoter comprising two tetO sequences (SEQ ID NO:12, ThermoFisher Scientific). For inducible expression via Tet-On system, the NES-T7RNAP-encoding sequence was inserted downstream of a “Tet-On” promoter comprising 8 tetO sequences (SEQ ID NO:10). The expression cassettes for constitutive, “T-REx” and “Tet-On” expression of NES-T7RNAP are set forth in SEQ ID NOs:40-43, respectively.

[0444] For constitutive expression the coding sequence for PylRS^{AF} (from *Methanosarcina mazei*) with or without a NES signal (NES-PylRS^{AF} or PylRS^{AF}) was inserted into a pcDNA3.1 plasmid (Invitrogen) downstream of the CMV promoter. For inducible expression via T-REx system, the NES-PylRS^{AF}-encoding sequence was inserted into the plasmid pDEST downstream of a “T-REx” promoter comprising two tetO sequences (SEQ ID NO:12, ThermoFisher Scientific). For inducible expression via Tet-On system, the NES-PylRS^{AF} encoding sequence was inserted into the pcDNA3.1 plasmid downstream of a “Tet-On” promoter comprising 8 tetO sequences (SEQ ID NO:10). The expression cassettes for constitutive, “T-REx”- and “Tet-On”-expression of NES-PylRS^{AF} are set forth in SEQ ID NO:44-47, respectively.

[0445] A T7 promoter-controlled expression cassette for tRNA^{Pyl} (SEQ ID NO:48) was generated by cloning a T7 promoter sequence upstream of the tRNA^{Pyl} encoding sequence, and a T7 termination signal downstream of the tRNA^{Pyl} encoding sequence, and insertion of this expression cassette into a p2RZ plasmid (Avis et al., *Methods Mol Biol.* 941:83-98, 2012; (see FIG. 1)). A variant of this expression cassette having a HDV ribozyme (tRNA^{Pyl}-HDV) was generated by a combination of restriction cloning, site-directed mutagenesis and restriction-free cloning so as to insert the HDV ribozyme encoding sequence directly downstream of the tRNA^{Pyl} (SEQ ID NO:49, cf. FIG. 1). For inducible expression via T-REx system, two different expression cassettes under the control of tet-inducible promoters were generated by inserting the tRNA^{Pyl}-encoding sequence into the plasmids pUC57 (in case of the U6 promoter) and pPBEX (in case of the H1 promoter) downstream of a variant of either a human U6 promoter comprising eight tetO sequences before the U6 [promoter and two tetO sequences after or a human H1 promoter comprising two tetO sequences (SEQ ID NOs:50 and 51).

[0446] Expression cassettes for an rtTA or a tetR contained the rtTA-encoding sequence (SEQ ID NO:14) downstream of the CMV promoter and the tetR-encoding sequence (SEQ ID NO:13) downstream of the CMV promoter.

[0447] The N-terminal dsRNA-binding domain (residues 2-174) of human protein kinase R (“PKR(2-174)”) was cloned into a p1.18 plasmid (U.S. Pat. No. 6,187,759 B1, pUC based plasmid containing a hCMV promoter, intron A, a terminator sequence and a multiple cloning site) for constitutive expression together with an N-terminal HA-tag under the control of the CMV promoter fused to intron A (the biggest intron of the transcribed region of the hCMV major immediate-early (IE1) gene, which enhances expression of protein. The cytomegalovirus (hCMV) is also called human beta-herpes virus 5 (HHV-5); cf. Chapman, B. S. et al NAR, 1991, Vol. 19, NO 14, 2979-3986). The amino acid sequence of HA-PKR(2-174) is set forth in SEQ ID NO:36.

B) Cell Culture and Transfection

[0448] HEK293T cells (ATCC CRL-3216) and HEK Flp-In T-REx 293 cells (ThermoFisher Scientific, #R78007) were maintained in Dulbecco’s modified Eagle’s medium (Life Technologies #41965-039) supplemented with 1% penicillin-streptomycin (Sigma P0781, 10,000 U/ml penicillin, 10 mg/ml streptomycin, 0.9% NaCl), 1% L-glutamine (Sigma #G7513), 1% sodium pyruvate (Life Technologies #11360), and 10% Fetal Bovine Serum (FBS, Sigma #F7524). For HEK Flp-In T-REx 293 cells the medium was supplemented with blasticidin (15 µg/ml) accordingly to the manufacturer’s manual. The HEK Flp-In T-REx 293 cell line is derived from HEK293 cells (ATCC CRL-1573) and stably expresses the tet repressor (tetR) gene under the control of the constitutive CMV promoter. See user guide “Growth and Maintenance of the Flp-In™ T-REx™ Cell Line”, catalog no. R780-07, document part no. 25-0369, publication no. MAN0000187, revision 2.0, 14 Sep. 2015 by Gibco/Life Technologies.

[0449] Cells were cultured at 37° C. in a 5% CO₂ atmosphere and passaged every 2-3 days up to 15-20 passages.

[0450] Cells were seeded 15-20 hours prior to transfection (~110,000 cells/well) and grown in 24-well plate cell culture plates at 37° C. in a 5% CO₂ atmosphere to a confluency of 70-80%. For HEK TetOn-3G cells the cell culture plates had previously been coated with polylysine hydrobromide for 4-10 h. For three and four plasmid transfections 1.2 µg total DNA/well was used. For five or six plasmid transfections 1.5 µg total DNA/well was used. Transfections were done with polyethylenimine (PEI) at a DNA to PEI ratio of 1 to 3.

C) Induction of Expression and Feeding with ncAAs

[0451] Four hours after transfection the medium was exchanged for fresh medium. Depending on the experiment and sample, said fresh medium contained 1 µg/ml tetracycline (for induction of expression in T-REx systems, systems with tetR co-expression) or 1 µg/ml doxycycline (for induction of expression in Tet-On systems) and/or 250 µM of the used ncAA (e.g., BOC), or none of these (indicated concentrations=final concentration in the cell culture).

[0452] The cells were then incubated at 37° C. in 5% CO₂ atmosphere for 2 days and afterwards analyzed as described herein.

D) Analysis by Flow Cytometry Spectroscopy (FCS)

[0453] Cells were washed once with 1×PBS and analyzed via flow cytometry spectroscopy. Data acquisition and analysis were performed using a BD LSRFortessa instrument (BD Bioscience) and the FlowJo software (FlowJo).

GFP fluorescence was acquired using the 488 nm laser with detection at 530/30 nm, and iRFP fluorescence was acquired using 670 nm laser with detection at 730/45 nm. The measured events were gated first in the FSC-A×SSC-A scatter plot for live cells, and then in the FSC-A×SSC-W scatter plot for singlets. In the figures, the events of the singlet gate are shown as a plot of GFP signal vs iRFP signal, which is divided into four areas showing four different species: cell with iRFP fluorescence (lacking GFP) in the top left, cells with both iRFP and GFP fluorescence (expressing the full-length protein iRFP-GFP, called “double positive specimen” or “DPs”) in the top right, cells with GFP fluorescence (lacking iRFP) in the lower right, and cells lacking both iRFP and GFP in the lower left.

[0454] The reporter NLS-iRFP-Flag-GFP^{Y39->TAG}-6His is a fusion of iRFP (infrared fluorescent protein) and GFP (green fluorescent protein) wherein amino acid position 39 of the GFP is encoded by the amber (TAG) stop codon. Cells express iRFP (detectable as red fluorescence) if properly transfected and further express GFP (detectable as green fluorescence) only if the amber codon is suppressed to encode the ncAA (here: BOC or SCO). The ratio of green fluorescence (GFP) to red fluorescence (iRFP) in the cells is therefore an indicator of amber suppression efficiency.

Example 1: Amber Suppression Using tRNA^{Pyl} Expressed as RNA-Ribozyme

[0455] Amber suppression was examined using the reporter gene NLS-iRFP-Flag-GFP^{Y39->TAG}-6His, PylRS^{AF}, tRNA^{Pyl} and the ncAA SCO in the presence of PKR(2-174). PKR(2-174) was expressed together with an N-terminal HA-tag (SEQ ID NO:36) as HA-PKR(2-174). tRNA^{Pyl} was expressed together with a HDV ribozyme as a tRNA-ribozyme RNA molecule (tRNA^{Pyl}-HDV). PylRS^{AF} was expressed under the control of a CMV promoter. The cells further expressed T7 RNA polymerase (T7RNAP).

[0456] PylRS^{AF} and T7RNAP were constitutively expressed using HEK293T cells (ATCC CRL-3216).

[0457] HEK293T cells were transfected with vectors carrying expression cassettes for the reporter gene NLS-iRFP-Flag-GFP^{Y39->TAG}-6His, PylRS^{AF} under the control of the CMV promoter, tRNA^{Pyl}-HDV under the control of a T7 promoter and T7RNAP under the control of the CMV promoter using the respective constructs and transfection method described in sections A) and B) above.

[0458] The controls were prepared identically with the exception that cells were not fed with SCO (“-ncAA”) and/or the tRNA^{Pyl} was expressed without the HDV ribozyme.

[0459] The cells were analyzed by FCS as described in section D) above. The results shown in FIG. 2 (using the pcDNA3.1 plasmid) and FIG. 3 (using the pCAGGS plasmid) indicate that expression of tRNA^{Pyl} together with HDV ribozyme resulted in more efficient amber suppression compared to tRNA^{Pyl} expressed without a ribozyme. Without wishing to be bound by theory it is assumed that this effect results from the ribozyme improving tRNA processing.

Example 2: Amber Suppression Using Different Inducible Expression Systems

[0460] Amber suppression was examined using the reporter gene NLS-iRFP-Flag-GFP^{Y39->TAG}-6His, PylRS^{AF}, tRNA^{Pyl} and the ncAA BOC. tRNA^{Pyl} was expressed

together with a HDV ribozyme as a tRNA-ribozyme RNA molecule (tRNA^{Pyl}-HDV) under the control of a T7 promoter. PylRS^{AF} was expressed together with a NES signal (NES-PylRS^{AF}). The cells further expressed T7 RNA polymerase together with a NES signal (NES-T7RNAP).

[0461] NES-PylRS^{AF} and NES-T7RNAP were inducibly expressed using either the T-REx system or the Tet-On system.

[0462] For the T-REx system. HEK Flp-In T-REx 293 cells were transfected with vectors carrying expression cassettes for the reporter gene NLS-iRFP-Flag-GFP^{T39->TAG}-6His, NES-PylRS^{AF} under the control of a “T-REx” promoter, tRNA^{Pyl}-HDV under the control of a T7 promoter and NES-T7RNAP under the control of a “T-REx” promoter using the respective constructs and transfection method described in sections A) and B) above. HEK Flp-In T-REx 293 cells stably express the tet repressor (tetR) gene. Expression of NES-T7RNAP (and thereby tRNA^{Pyl}-HDV) and NES-PylRS^{AF} was induced by adding tetracycline and the cells were fed with the ncAA BOC as described in section C) above.

[0463] For the Tet-On system, HEK293T cells (ATCC CRL-3216) were co-transfected with vectors carrying expression cassettes for the reporter gene NLS-iRFP-Flag-GFP^{T39->TAG}-6His, NES-PylRS^{AF} under the control of a “Tet-On” promoter, tRNA^{Pyl}-HDV under the control of a T7 promoter, NES-T7RNAP under the control of a “Tet-On” promoter and rtTA using the respective constructs and transfection method described in sections A) and B) above in the presence of PKR(2-174). PKR(2-174) was expressed together with an N-terminal HA-tag (SEQ ID NO:36) as HA-PKR(2-174). Expression of NES-T7RNAP (and thereby tRNA^{Pyl}-HDV) and NES-PylRS^{AF} was induced by adding doxycycline and the cells were fed with the ncAA BOC as described in section C) above.

[0464] The controls were prepared identically with the exception that neither tetracycline nor doxycycline was added.

[0465] The cells were analyzed by FCS as described in section D) above. The results shown in FIG. 4 indicate that amber suppression using BOC together with inducibly expressed tRNA^{Pyl}-HDV and NES-PylRS^{AF} worked effectively in both the T-REx system (FIG. 4a) and the Tet-On system (FIG. 4b).

Example 3: Effect of PKR(2-174) on Amber Suppression in Inducible and Non-Inducible Expression Systems

[0466] Amber suppression was examined using the reporter gene NLS-iRFP-Flag-GFP^{T39->TAG}-6His, PylRS^{AF}, tRNA^{Pyl}, tetR and the ncAA BOC in the presence or absence of PKR(2-174).

[0467] PKR(2-174) was expressed together with an N-terminal TA-tag (SEQ ID NO:36) as HA-PKR(2-174).

[0468] One of three different expression cassettes for tRNA^{Pyl} was used: T7-tRNA^{Pyl}-HDV wherein tRNA^{Pyl} was expressed together with a HDV ribozyme (located at the 3'-end of tRNA^{Pyl}) as a tRNA-ribozyme RNA molecule (tRNA^{Pyl}-HDV) under the control of a T7 promoter; 8×tetO-U6-tetO-tRNA^{Pyl} wherein expression of tRNA^{Pyl} is controlled by a tetracycline-inducible variant of the human U6 promoter comprising eight tetO sequences before the U6 promoter and two tetO sequences after the U6 promoter (SEQ ID NO:50); and H1-TetO-tRNA^{Pyl} wherein expression

of tRNA^{Pyl} is controlled by a tetracycline-inducible variant of the human H1 promoter comprising two tetO sequences (SEQ ID NO:51).

[0469] PylRS^{AF} was expressed together with a NES signal (NES-PylRS^{AF}). One of two different expression cassettes for NES-PylRS^{AF} was used: CMV-tetO-NES-PylRS^{AF} wherein expression of NES-PylRS^{AF} is controlled by a tetracycline-inducible variant of the CMV promoter comprising two tetO sequences (SEQ ID NO:46); CMV-NES-PylRS^{AF} wherein expression of NES-PylRS^{AF} is controlled by the (constitutive) CMV promoter (SEQ ID NO:45).

[0470] Cells wherein expression of tRNA^{Pyl} was controlled by a T7 promoter (expression cassette T7-tRNA^{Pyl}-HDV) further contained one of two different expression cassettes for T7 RNA polymerase to be expressed together with a NES signal (NES-T7RNAP); CMV-tetO-T7RNAP wherein NES-T7RNAP expression is under the control of a tetracycline-inducible variant of the CMV promoter comprising two tetO sequences (SEQ ID NO:12); CMV-NES-T7RNAP wherein NES-T7RNAP expression is under the control of the (constitutive) CMV promoter (SEQ ID NO:43).

[0471] HEK293T cells were transfected with vectors carrying the following combinations of expression cassettes (x marks the presence of the respective expression cassette in the combination):

expression cassette	combinations					
	1	2	3	4	5	6
NLS-iRFP-Flag-GFP ^{T39->TAG} -6His	X	X	X	X	X	X
HA-PKR(2-174)	X	X	X	X	X	X
CMV-tetO-NES-PylRS ^{AF}			X	X		X
CMV-NES-PylRS ^{AF}	X	X			X	
T7-tRNA ^{Pyl} -HDV				X	X	X
8×tetO-U6-tetO-tRNA ^{Pyl}	X		X			
H1-tetO-tRNA ^{Pyl}		X				
CMV-tetO-T7RNAP				X	X	
CMV-NES-T7RNAP						X
results shown in FIG.	5a	5b	5c	5d	5e	5f

[0472] Controls were prepared in the same manner but without the expression cassette for HA-PKR(2-174).

[0473] Tetracycline was added for expression of tet-inducible elements and the cells were fed with the ncAA BOC as described in section C) above. Control samples were prepared identically with the exception that the addition of one or both of tetracycline and Boc was omitted.

[0474] The cells were analyzed by FCS as described in section D) above. The results shown in FIG. 5 indicate that co-expression of PKR(2-174) increases the intensity of GFP fluorescence detected in double positive cells, indicating the presence of an increased amount of full-length reporter gene expression product (resulting from successful amber suppression) per cell.

Abbreviations

[0475] BCN=2-amino-6-(9-biocyclo[6.1.0]non-4-ynyl-methoxycarbonylamino)hexanoid acid

[0476] BOC=2-amino-6-(tert-butoxycarbonylamino)hexanoic acid, in the examples “BOO” specifically

- designates (2S)-2-amino-6-(tert-butoxycarbonylamino)hexanoic acid=Boc-L-Lys-OH=N- α -tert-butyloxycarbonyl-L-lysine
- [0477] CMV=cytomegalovirus
- [0478] Crm1=chromosomal region maintenance 1, also known as karyopherin exportin 1
- [0479] dsRNA=double-stranded RNA
- [0480] dSTORM=direct stochastic optical reconstruction microscopy
- [0481] *E. coli* BL21(DE3)AI=*E. coli* strain B F⁻ ompT gal dcm lon hsdS_B(r_B⁻m_B⁻) λ (DE3 [lacI lacUV5-T7p07 ind1 sam7 nin5]) [malB⁺]_{K-12}(λ ^S) araB::T7RNAP-tetA
- [0482] eIF-2 α =alpha subunit of eukaryotic initiation factor 2
- [0483] FBS=fetal bovine serum
- [0484] FISH=fluorescence in situ hybridization
- [0485] GCE=genetic code expansion
- [0486] GFP=green fluorescent protein
- [0487] glmS=glucosamine-6-phosphate synthetase
- [0488] H1 promoter=human RNA polymerase III promoter
- [0489] HDV=hepatitis delta virus
- [0490] IPTG=isopropyl β -D-1-thiogalactopyranoside
- [0491] iRFP=infrared fluorescent protein
- [0492] ncAA=non-canonical amino acid
- [0493] NES=nuclear export signal
- [0494] NLS=nuclear localization signal
- [0495] O-tRNA=orthogonal tRNA
- [0496] O-RS=orthogonal RS
- [0497] PAINT=point accumulation for imaging in nanoscale topography
- [0498] PBS=phosphate buffered saline
- [0499] PKR=protein kinase R; alternative names: dsRNA-dependent protein kinase, interferon-(IFN-) γ -inducible dsRNA-activated serine/threonine-protein kinase, p68 kinase
- [0500] PMSF=phenylmethylsulfonyl fluoride
- [0501] POI=polypeptide of interest, target polypeptide
- [0502] PrK=2-amino-6-(prop-2-ynoxycarbonylamino)hexanoic acid
- [0503] pRS=prokaryotic RS
- [0504] ptRNA=prokaryotic tRNA
- [0505] ptRNA-ribozyme=prokaryotic RNA molecule comprising a ptRNA and at least one ribozyme
- [0506] PylRS=pyrrolysyl tRNA synthetase
- [0507] PylRS^{4F}=mutant *M. mazei* pyrrolysyl tRNA synthetase comprising amino acid substitutions Y306A and Y384F
- [0508] RP-HPLC=reversed phase high-performance liquid chromatography
- [0509] RS=aminoacyl tRNA synthetase
- [0510] RT=room temperature
- [0511] rtTA=reverse tTA
- [0512] SCO=2-amino-6-(cyclooct-2-yn-1-yl)oxycarbonylamino]hexanoic acid
- [0513] SPAAC=(copper-free) strain promoted alkyne-azide cycloaddition
- [0514] SPIEDAC=(copper-free) strain promoted inverse-electron-demand Diels-Alder cycloaddition
- [0515] SRM=super-resolution microscopy
- [0516] SV40=simian vacuolating virus 40
- [0517] T7RNAP=T7 RNA polymerase
- [0518] TCO=2-amino-6[(2E-cyclooct-2-en-1-yl)oxycarbonylamino]hexanoic acid
- [0519] TCO*=2-amino-6[(2E-cyclooct-2-en-1-yl)oxycarbonylamino]hexanoic acid
- [0520] tetR=tetracycline repressor
- [0521] tetO=tetracycline operator
- [0522] tTA=tetracycline-controlled transcriptional activator
- [0523] TRE=tetracycline-responsive promoter element (also termed tetracycline response element)
- [0524] tRNA^{Pyl}=tRNA that can be acylated with pyrrolysin by a wild-type or modified PylRS and has an anticodon that, for site-specific incorporation of the ncAA into a POI, is preferably the reverse complement of a selector codon. (in the tRNA^{Pyl} used in the examples, the anticodon is CUA.)
- [0525] U6 promoter=promoter that normally controls expression of the U6 RNA (a small nuclear RNA) in mammalian cells
- [0526] VS=Varkud satellite

Listing of Sequence

Listing of sequences				
SEQ ID NO	Description	Source	Type	Features
ribozymes				
1	hepatitis delta virus (HDV) ribozyme (Avis)	hepatitis delta virus	hepatitis delta	HOV ribozyme core sequence (7-68) flanking sequence (1-6) (has to be there, but sequence can be different)
			<u>gcuagccauggucuccagccuccucgucggcgcuagugggcacaacugcuucggauggcgaauaggagac</u>	
2	hepatitis delta virus (HDV) ribozyme (Schuerer)	hepatitis delta virus	hepatitis delta	
			gggtcgcatggcatctccacctcctccggtccggtccgacctgggtacttcggtaggctaaggagaag	
3	hepatitis delta virus (HDV) ribozyme (Akoopie)	hepatitis delta virus	hepatitis delta	
			gagggatagtagacagagcctccccgtgggtccctctggataaccaactgatactgtac	
4	hammerhead ribozyme	artificial sequence synthetic		HH ribozyme core sequence (11-45) flanking sequences (1-10, 46-50) (have to be there, but sequence can be different)
			<u>ucacguuuuccugauguccgugaggagacgaaacgguaccgcgguaccguc</u>	
5	hairpin ribozyme	artificial sequence synthetic		[There are other hairpin sequences existing, but this one seems to be the most used one.]
			ggagagagaagucacaccagagaaacacacacaccccauugcacuccggguuggguauauuaccuggu acggggggaacuuugugggcg	
6	Varkud satellite (VS) ribozyme	Neurospora intermedia		
			gggaaaagcuugcgaaaggcgucgucgccccgagcgguaguaagcaggggaacacucccauuucagu acugaaaauugcugagcagcuuagacuacuguauguauguauguaagggcuuaagucgguauuuggcgua agucaguaugcagcagcacagcagcccgcuugcgagaa	
7	glucosamine-6-phosphate synthetase (glms) riboswitch (glms ribozyme)	Bacillus spec.		
			gggcagaaactggcaccuuugcacuccggcgucgagcuagacgagggggguuuuauagcagaauuucggcg augacucccgguuguuacacacacccgcaagcunuuuacuaaauaauaagugacuuaguggacaaa ggugaaaagugauga	
Elements of the TetOn/Off system				
8	tetracycline operator (tetO)	artificial sequence synthetic		
			tcctatcagtgatagaga	

[illegible]

Listing of sequences				
SEQ ID NO	Description	Source	Type	Features
				<u>ttgttttggcaccacaaataacgggactttccaaaatgtcgtaaacactccggcccatcgacgcaaat</u> <u>ggcggttaggogtgtacggtggagggtctatataaagcagagctctctggctaactagagaacccactg</u> <u>cttactgcttatacgaaatataacgactcaactataggagacccaagctggctagcgttttaaactta</u> <u>agcttctgagcaccatgtctagactggacaagagcaaatgataaaccttctcttggaattactcaa</u> <u>tggagtcggtatcgaagcctgacgacaaggaaactgctcaaaagctgggagtggagcagcctaccc</u> <u>tgtatctggcagctgaagaaacaggggccctgctcgatgcctgcgaatcgagatgctggacaggc</u> <u>atcatacccaactctgcccctggaggcgagtcattgcaagactttctgcggaacaacggccaagtca</u> <u>taccgtgtgctctctctcacatcgcgacggggctaaagtgcactcgccaccgcccacaagagaa</u> <u>acagtacgaaacctggaaaatcagctcgcttctctgtgtcagcaaggcttctccctggagacgcac</u> <u>tgtacgctgttcctcgctggccactttacactgggtgcgtattggaggacagagcatcaagt</u> <u>gcaaaagaggaagagagacactaccacgattctatgccccacttctgaaacaagcaattgagct</u> <u>gttcgaccggcaggagccgaaccttcctcttcttggcctggaaactaatcatatgtggctggaga</u> <u>aacagctaaagtgcgaaaagcggcgccgacgcgccttgacgattttgacttagacatgctccca</u> <u>gccgatgccttgacgacttgaccttgatagtcgcgcgtgacgctcttgacgattttgaccttgac</u> <u>atgctccccccgggtaa</u>
52	TetOn3G protein Zhou et al, Nature, 2006	human immunodeficiency virus (HIV) -rtTA variant, optimized for TetOn-system	Protein: MSRLDKSVINSALELLNGVIEGLTTRKLAQKLGVEQPTLWHVKNKRALLDALPIEMLDRHHTH SCPLGESWQDFLRNNAKSYRCALLSHRDGAKVHLGTRPTKQYETLENQLAFLOQQGFSLENA LYALSAVGHFTLGCVLIEQEHQVAKERTPTTDSMPPLLKQAIELFDQGAEPAPLFLGLELIICGL EKQLKCESGGPTDALDDFDLMLPADALDDFDLMLPADALDDFDLDMLPAG*	
15	T7 promoter class III (core)	artificial sequence derived from enterobacteria phage T7 promoter	taatacgcactcactatag	G at position 18 is preferred nucleotide at the +1 position to promote efficient transcription initiation
16	T7 promoter class III	artificial sequence derived from enterobacteria phage T7 promoter	taatacgcactcactatagg	GG at positions 18 and 19 are preferred nucleotides at the +1 and +2 positions to promote efficient transcription initiation
17	T7 promoter class II (core)	artificial sequence derived from enterobacteria phage T7 promoter	taatacgcactcactattaa	A at position 18 is preferred nucleotide at the +1 position of the gene to promote efficient transcription initiation

Listing of sequences

ESQ ID NO	Description	Source	Type	Listing of sequences	Features
18	T7 RNA polymerase	Enterobacteria phage T7		MMTINIAKNRDSI ELAAIPFNTLADHYGERLARQALALEHESYEMGEARPRKMPERQLKAGEVAD NAAKPLITLLPKMIARINDWFEEVKAKRGKRPATQFLQEIKEPAVAYITIKTTLACLTADNTTV QAVASAI GRAI EDEARFGRIRLEAKHFKKNVEQLNKRVGHVYKKAQMVVVEADMLSKGLGGE AWSSWHKEDSIHVGRVCIEMLIESTGMVSLHRQAGVVGQDSEIT ELAPEYAEAIATRAGALAGIS PMFQCPVPPKWTGITGGYTWANGRRPLALVRTHSKALMRYEDVYMPVEVYKAINIAQNTAW KINKKVLAVANVITKKWCHCPVEDI PAI EREELPMKPEDIDMNPALTAWKRAAAVYRKDKARKSR RISLEFMLEGANKFANHKAIMPPYNNMDWRGVYAVSMFNPGQNDMTKGLLTLAKGKPIGKEGY WLKIHGANCAGVDKVPPEPIKFI BENHENIMACAKSPLENTWMAEQDSPCFLAFCEYAGVQH HGLSYNCSLPLAPDGSCSGIQHFSAMLRDVEGVRANLLPSETVQDIYIGIVAKKVNELIQADAING TDNEWTVTIDENTGEISEKVLGTAKALQWLAAYGVTSVTKRSVMTLAYGSKFGRFQQVLED TIQPAIDSGKGLMFTQNOAQGYMAKLIWESVSVTVAAVEAMNWLKSAAKLLAAEVKDKKTGEI LRKCAVHWTPDGFVWQYKPKIQTRLNLMFLQGRLOPTINTNKDSEIDAHKQESGIAFNFV HSQDGSHLRKTVMWAHEKYGIESFALIHDSFGTIPADAANLFAVRETVMVDTYESCDVLADFYDQ FADQLHESQDKMPALPAKGNLNLRLDILLESDFAF	NES
19	NES-T7 RNA polymerase	artificial sequence derived from Enterobacteria phage T7		MCVPPLQPLPRLRLTLDNTINIAKNRDSIELAAIPFNTLADHYGERLARQALALEHESYEMGEARF RKMFERQLKAGEVADNAAAKPLITLLPKMIARINDWFEEVKAKRGKRPATQFLQEIKEPAVAYIT IKTTLACLTADNTTVQAVASAI GRAI EDEARFGRIRLEAKHFKKNVEQLNKRVGHVYKKAQMVQ WVEADMLSKGLGGEAWSSWHKEDSIHVGRVCIEMLIESTGMVSLHRQAGVVGQDSEITELAP EYAEAIATRAGALAGISPMFQCPVPPKWTGITGGYTWANGRRPLALVRTHSKALMRYEDVY MPEVYKAINIAQNTAWKINKVLAVANVITKKWCHCPVEDI PAI EREELPMKPEDIDMNPALTAWK RAAAVYRKDKARKSRISLEFMLEGANKFANHKAIMPPYNNMDWRGVYAVSMFNPGQNDMTK GLLTLAKGKPIGKEGYMLKIHGANCAGVDKVPPEPIKFI BENHENIMACAKSPLENTWMAEQD SPCFLAFCEYAGVQHGLSYNCSLPLAFDGSCSGIQHFSAMLRDVEGVRANLLPSETVQDIY GIVAKKVNELIQADAINGTDNEWTVTIDENTGEISEKVLGTAKALQWLAAYGVTSVTKRSVMTL AYGSKFGRFQQVLEDITQPAIDS GKGMLFTQNOAQGYMAKLIWESVSVTVAAVEAMNWLK AAKLLAAEVKDKKTGEI LRKCAVHWTPDGFVWQYKPKIQTRLNLMFLQGRLOPTINTNKD SEIDAHKQESGIAFNFVHSQDGSHLRKTVMWAHEKYGIESFALIHDSFGTIPADAANLFAVRETVM VDTYESCDVLADFYDQFADQLHESQDKMPALPAKGNLNLRLDILLESDFAF	NES
20	NLS consensus	artificial sequence synthetic		KX ¹ X ² X ³	X ¹ and X ³ are independently selected from K and R, and X ² is any naturally occurring amino acid
21	NLS of SV40 large T antigen	simian vacuolating virus 40 (SV40)		PKKKRKV	
22	Methanococcus mazelii PyIRS Y306A, Y384F mutant (PyIRS ^{Δ4F})	artificial sequence synthetic		DNA: atggcgctgcccggtgccctgcagctgcccgcgctggaaacgcctgacctggatgataaaaaaacgcct gaataacctgatctctgctactggtctgtgtagatgagtcgtaccggaacattcataaatacaaacacc	ATG (M ¹) may be missing

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Listing of sequences				
SEQ ID NO	Description	Source	Type	Features
53	Methanosarcina mazei PyIRS Y306A, Y384F mutant (PyIRS ^{ΔF}) amino acid sequence	artificial sequence synthetic	Protein:	M ¹ may be missing Y306A, Y384F
23	Methanosarcina mazei PyIRS N346A, C348A mutant (PyIRS ^{Δ4})	artificial sequence synthetic	DNA:	ATG (M ¹) may be missing

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Listing of sequences				
SEQ ID NO	Description	Source	Type	Features
54	Methanosarcina mazei PyIRS N346A, C348A mutant (PyIRS ^{44A}) – amino acid sequence	artificial sequence synthetic	Protein: MACPVLQPLPRLTLDDKKPLNTLISATGLWMSTGTIHKIKHHEVSRSKIYIEMACGDHLVNVN NRSRSTAPLEALRHKKYRTCKRCVSDKEDLNKFLTKANEDQTSVKVSVSAPTRTKAMPKSVARAPKPLENTEAAQAPSSKSPALPVSTQESVSPASVSTSISSISGTATASALVKGNTNPIITSM SAPVQASAPALTKSQDRLVLLNPKDEISLNSGKPPRELESELLSRKKDLQOIYAEERENYLKG LEREITRRFVDRGFLEIKSPILIPLEIYERMGIDNDTELSKQIFRVDKNFCLRPLMAPNLNYLRLKQRLP ALDPDKIPEIGCYRKESDGKEHLEEFMTLAFAPQMGSGCTRENLESIITDFLNHLGIDFKIVGDS C M VYGTGLDVMHGDLELSSAVVGPIPLDREWGIDKPIWAGAGFGLERLLKVKHDFKNIKRAARSESY YNGISTNL	M ⁱ may be missing N346A, C348A
24	Methanosarcina mazei PyIRS Y306A, N346A, C348A, Y384Fmutant (PyIRS ^{44A})	artificial sequence synthetic	Protein: gsgtgcgcgggtgcgctgcagctgcgcgcgcgtggaacgcctgacccctggatgataaaaaacgcgtgaa taccctgatctctgtactggtctgtgagtgagctgcgtaccggaaccattcataaaatacaacacacgcg aggttagccgttcgaaatctattgagatggctgtgctgagatcatctggttgtaacaatagccgc tcttctgtacagcagctgcactggtccacaaatctgtaaacctgtgaaacgttgccgtgtgtc cgtaggagatctgacaaatctctgacaaagccaatgagcccaaacgctgaaagctgaaagctg ttagggctctaccgtactaaaaagcaatccgaaatccgttgctgtgcccctaaaccactggaa aaactgaagcagcacagcagccgtctggaagcaaatctctcggccattctctgttttaacca ggagtcggttctgtccagcaagtgtgagccacgcatagcagattagcaccgggtgccaccgcta ggccctgggttaaggcaataccaatccgattacaagcatgtctgcccggttcaagcatcagctcca gcactgacaaaatcccaaacgcatcgtctggaggtctgtgtaacggaagacgaaatcagcctgaa ttccggcaaacccgttctgtgaactggagagcgaactgctgcagctgtaaaaagacctgcaacaa tctatgcgaagaactgagaaatctctgggaactggaaactgaaatcaccctttttcgtggat cgtggctttctggagatcaaatcccgattctgattctctggagatctcgagcgtatgggcacga caatgataccgaactgagcaacaaatctccggtgtggaataaaactctgtctgcacctatgtag caccaaatctggctaaactatctgcgaactggacgtgcctgcctgattcctatacaaatctctgag atcggcccggtgttactgtaagagtcgcagcgttaagaacatctggaggagtttaccatgctggcctt tgcccaaatgggttcagggtgtactgtgagaaactggaaagcatcatcaccgattttctgaaccac tgggcaatgacttcaaatgtcggcgacagctgtatggtgttgccgacacctggatgtagtcac ggcgacctggaaactgctagtcgctgtggggccaatcccgctggatcgtgagtggggtatcgacaa accttgatcggtcggggttttggctggagcgtgtgtggaagtgaagtaaaacacgacttcaagaacatca aacgtgtgcagcttcacagtcctattacaatgggtattcttactaacctgtaa	May be optionally extended by a M ⁱ (before first A-amino acid) Y306A, N346A, C348A, Y384F
55	Methanosarcina mazei PyIRS Y306A, N346A, C348A, Y384Fmutant (PyIRS ^{44A}) amino acid sequence	artificial sequence synthetic	Protein: ACFPVLQPLPRLTLDDKKPLNTLISATGLWMSTGTIHKIKHHEVSRSKIYIEMACGDHLVNVNS RSTRARALRHKKYRTCKRCVSDKEDLNKFLTKANEDQTSVKVSVSAPTRTKAMPKSVARAPKPLENTEAAQAPSSKSPALPVSTQESVSPASVSTSISSISGTATASALVKGNTNPIITSMAP VQASAPALTKSQDRLVLLNPKDEISLNSGKPPRELESELLSRKKDLQOIYAEERENYLKLER ETRRFVDRGFLEIKSPILIPLEIYERMGIDNDTELKQIFRVDKNFCLRPLMAPNLNYLRLKDRALP DPDKIPEIGCYRKESDGKEHLEEFMTLAFAPQMGSGCTRENLESIITDFLNHLGIDFKIVGDS C M VYGTGLDVMHGDLELSSAVVGPIPLDREWGIDKPIWAGAGFGLERLLKVKHDFKNIKRAARSESY YNGISTNL	M ⁱ may be missing N346A, C348A

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Listing of sequences				
SEQ ID NO	Description	Source	Type	Features
			GDTLQVMHGDLELSSAVVGPIPLDREWIDKPWIGAGFGLERLLKVKHDFKNIKRAARSEYNG ISTNL	
25	pyrrolysyl tRNA of <i>Methanosarcina mazel</i> for Amber codon (tRNA ^{Py, CUA})	artificial sequence synthetic	ggaaacctgacatgtagatcgaatggactc_____aaatccggttcagccgggttagattcccggggtttccg	Underlined: anticodon (31-33)
26	pyrrolysyl tRNA of <i>Methanosarcina mazel</i> for Opal codon (tRNA ^{Py, UCA})	artificial sequence synthetic	ggaaacctgacatgtagatcgaatggactc_____aaatccggttcagccgggttagattcccggggtttccg	Underlined: anticodon (31-33)
27	pyrrolysyl tRNA of <i>Methanosarcina mazel</i> for Ochre codon (tRNA ^{Py, UUA})	artificial sequence synthetic	ggaaacctgacatgtagatcgaatggactc_____aaatccggttcagccgggttagattcccggggtttccg	Underlined: anticodon (31-33)
NES				
28	NES motif	artificial sequence synthetic	XXXXXX _ _ _ _ _	Underlined: independently selected from Leu, Ile, Val Phe and Met Bold: independently selected from any canonical amino acid
29	NES motif	artificial sequence synthetic	XXXXXXX _ _ _ _ _	Underlined: independently selected from Leu, Ile, Val Phe and Met Bold: independently selected from any canonical amino acid
30	NES motif	artificial sequence synthetic	LXXXXLXX _ _ _ _ _	Underlined: independently selected from Leu, Ile, Val, Phe and Met Bold: independently selected from any canonical amino acid Italic: independently selected from Leu and Ile

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Listing of sequences				
SEQ ID NO	Description	Source	Type	Features
31	NES motif	artificial sequence synthetic	LXXXXXXLX	Underlined: independently selected from Leu, Ile, Val, Phe and Met Bold: independently selected from any canonical amino acid
32	NES motif	artificial sequence synthetic	LXXXXXXLX	Underlined: independently selected from Leu, Ile, Val, Phe and Met Bold: independently selected from any canonical amino acid
33	NES motif	artificial sequence synthetic	LXXXXXXLX	Underlined: Independently selected from Leu, Ile, Val, Phe and Met Bold: independently selected from any canonical amino acid Italic: independently selected from Leu and Ile
34	NES motif	artificial sequence synthetic	LPPLERLTL	
35	NES motif	artificial sequence synthetic	ACFVPLQLPPLERLTLD	
dsRNA-binding polypeptides				
36	HA-PKR (2-174)	artificial sequence synthetic	MYPYDVPDYAAGDLSAGPFMEELNTYRKQGVVLKYQELPNSGPPHDPDRFTFOVIIDGREPPEG EGRSKKEAKNPAKLAVEILNKEKKA VSP LLTTTNSSEGLSMGNYIGLINRIAQKKRLTVNYEQCA SGVHGPEGFHYKCFMGQKEYSIGTGSTKQEAQLAALAYLQILSEETSVKS	HA (2-10) human PKR (2-174) (11-183)
37	core of dsRNA-binding domain of PKR	artificial sequence synthetic	FMEELNTYRKQGVVLKYQELPNSGPPHDPDRFTFOVIIDGREPPEGEGRSKKEAKNPAKLAVEILNKEKKA VSP LLTTTNSSEGLSMGNYIGLINRIAQKKRLTVNYEQCASGVHGPBGFHYKCKRMGQ KEYSIGTGSTKQEAQLAALAYLQILSE	conserved; R30, F32, G48, S50, K51, K52, K55, F122, K141, K145 restricted: 1) R9, Q10 ≠ hydrophobic (i.e.):

Listing of sequences				
SEQ ID NO	Description	Source	Type	Features
38	(full) dsRNA-binding domain of PKR-M1	artificial sequence synthetic	AGDLSAGPFMEELNTRYQKQGVVLKYQELPNSGPPHRRFTFOVIIDGREFFPEGRSKKEAKN AAAKLAVELINKEKAVSPLLLTTTNSSEGLSMGYIGLINIAQKKRLTVNYEQCASGVHPEGF HYKCKWGQKEYSIGTGSTKQEAQKLAALKLAYILQLSEETS VKS	K, R, H, D, E, S, T, N, Q, C, preferably, R9 = R, S, C, Q, H, (K), Q10 = Q, (N) 2) A59, L66, A149 = A59D, L66A, A149D (i.e.: A59 = A, G, I, L, M, V; L66 = L, I, V, M; A149 = A, G, I, L, M, V)
constructs used in examples				
39	expression cassette for reporter gene NLS-iRFP-Flag-GFP ^{Y39} -TAG ₆ His	artificial sequence synthetic	ggcattgattatgactagttatttaataagtaataatcacgggggttaattagttcattagagccattatg gggttcgcggttacataaacttaactgggttaaatggccgcgtgggtgacccccaacgaccccccccaatt gacgtcaaatatagcgtatgttcccatagtaaacgcgaatagggaactttccattgacgtcaatgggtgg agattattacgggtaaacctggcccaacttggcagtcacatacgaagtgtatataatgcgaagtccgccccattt ggcgcaatgacggtaaatggccgcgtggcattatgccagtcacatgacattacggtggaactttccctac ttggcagtcacatcgtattagtcacgtattaccatgggtgacgtgttttggcagtcacccaatg ggcgtgtagcgggtttgactcacggggatttccaaagtctccacccattgacgtcaatgggagtttg ttttggcaccaaaatacacgggaactttccaaatgcgtataaaccccgcccggttgacgcaaatggg cggtagcggtgacggtggaggtctatatagcagagctcgttttagtgaacgtcagatcacatgaa gotttattgcggtagtattatcacgttaaatgtctaacgcagtcagtgcttctgacacaaacagtcctg aacttaagctgcagaagtggctcgtgagcactgggcaggtaagtcacaaagtttacaagcaggttta aggagaccaatagaaactgggcttgcagacagagaagactcttgcgtttctctgtaggcaacctattg gtcttactgacatccactttgccttctctccacaggtgtccactcccagttcaattacagctcttaa ggctagagtaacttaatacagactcaactataggctagcccaactgggcgcgtctggaaagcaccgccga aaaaaaacgcaaadtggaagatagcgagcgcggaagatccgtccgcaggcagctgaacctcttg acctg ① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨ ⑩ ⑪ ⑫ ⑬ ⑭ ⑮ ⑯ ⑰ ⑱ ⑲ ⑳	Promoter (1-852) NLS (949-984) IRFP (988-1929) 25 Helix (1930-2004) Flag (2011-2034) GFP ^{Y39} CDS (2035-2748) Y39-encoding triplet (2149-1251) 6His (2749-2766)

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Listing of sequences				
SEQ ID NO	Description	Source	Type	Features
41	NES-T7 RNA polymerase expression cassette containing "T-REx" promoter comprising two tetO sequences	artificial sequence synthetic	②	Promoter (1-584)
			②	tetO (586-604;
			②	607-625)
			②	NES (798-845)
			②	CDS (846-3494)
			②	
			②	
			②	
			②	
			②	
			②	
			②	
			②	
			②	
			②	
			②	
			②	
			②	
			②	
			②	
42	NES-T7 RNA polymerase expression cassette containing "Tet-On" promoter	artificial sequence synthetic	②	Promoter (1-458)
			②	tetO (18-36; 54-72;
			②	90-108; 126-144;
			②	162-180; 198-216;

-continued

Listing of sequences				
SEQ ID NO	Description	Source	Type	Features
	comprising 8 teto sequences		①	234-252: 270-288)
			②	NES (473-520)
			③	CDS (521-3169)
			④	
			⑤	
			⑥	
			⑦	
			⑧	
			⑨	
			⑩	
			⑪	
			⑫	
			⑬	
			⑭	
			⑮	
			⑯	
			⑰	
			⑱	
			⑲	
43	NES-17 RNA polymerase expression cassette for constitutive expression via CMV promoter	artificial sequence synthetic	①	CMV promoter (1-588)
			②	NES (693-740)
			③	CDS (741-3389)
			④	
			⑤	
			⑥	
			⑦	
			⑧	
			⑨	
			⑩	
			⑪	
			⑫	

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Listing of sequences

[illegible]

-continued

Listing of sequences

Listing of sequences					
SEQ ID NO	Description	Source	Type		Features
45	NES-PyIRS ^{ΔF} expression cassette for constitutive expression via CMV promoter	artificial sequence synthetic	② ② ② ② ② ② ② ② ② ② ② ② ② ② ② ② ② ②		CMV promoter (1-588) NES (696-743) CDS (744-2105) 306A, 384F
46	NES-PyIRS ^{ΔF} expression cassette containing "T-Rex" promoter comprising two tetO sequences	artificial sequence synthetic	② ② ② ② ② ② ② ② ② ② ② ② ② ② ② ② ② ②		Promoter (1-584) tetO (586-604; 607-625) NES (785-832) CDS (833-2194) 306A, 384F

[illegible]

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Listing of sequences				
SEQ ID NO	Description	Source	Type	Features
Further dsRNA-binding polypeptides				
57	dsRNA-binding polypeptide (https://www.uniprot.org/uniprot/Q03963)	<i>Mus musculus</i>	⑦	MASDTFGYMDKLNKYRQMHGVAITYKELSTSGPPHRRFTFQVLIDEKEFPEAKGRSKQEARNA AAKLAVIDLDNENKVDCHTSASEQGLFVGNVIGLVNSPAQKKLSVNYEQCEPNSELQRFICK CKIGQWYGTSGVTQKEAKQAAEAYOKLIKSPKTAGTSSVVTSFSGFSSSSMTSNGV SQAPGSFSENVFTNGLGENKRGKGVSPDDVORNKYTLDFARNSDPEDIIEIGLGGFGV KAKHRIDGREYALKRKYNTKAEHEVQALAEINHVNIQYHSCWEGVDYDPEHSMDSYRYKT RCLFIQMEFCDKGTLEQWMRNRNQSKVDKALILDYEQIVTGVYIHSKGLIHRDLKPGNI FLVDE RHIKIGDFGLATALENDGKSRTRTGTLOMSP EOILFKHYGKVDIFALGLILAELLHTCFTESEKI KPFESLRKGD FNDIFDNKEKSLLKLLSEKPKDRPETS EILKTLAEWENISEKKKRNTC
58	dsRNA-binding polypeptide (https://www.uniprot.org/uniprot/89UKH7)	<i>Pan troglodytes</i>	⑦	MAGDLSAGFFMEELNTYRQOGVVLKYQELPNSGPPHRRFTFRVIIDGREPPEGEGRSKKEAK NAAKLAVEILNKEKAVSPLVLTNTNSSEGLSMGNYIGLINRIAQKKELTVNYRQCASGVHPEG FHYCKWGOKEYSIGTSYTKQAKQLAAKLAYLQILSETSVKSYLSSSGSFATTCDSQSNLVT STLASESSSEGDF SADTSINSDSLNSSLLMGLRNORAKRSLAPRFDLPDMKETKYTVD KRGMDPKIELIGSGGFGQVFKAKHRIDGKTYVIKRVKYNNEKAREVKALAKLDHVNIVHYNGC WDGFDYDPTSDDDLESDDYPENSNSRSKTKCLFIQMEFCDKGTLEQWIEDRRGKKLDKVL ALELFEQITKGVDYIHSKKLIHRDLKPSNIFLVDTKOVKIGDFGLVTSLNKNDGKTRSGKTLRYMSP EQISSODYGKVDLYALGLILAELLHVCDTAFETSKFFTDLRDGIISDILFDKKEKTLQKLLSKKPE DRPTSEILRTLTVMWKKSPKNERHTC

-continued

Listing of sequences				
SEQ ID NO	Description	Source	Type	Features
59	dsRNA-binding polypeptide (https://www.uniprot.org/uniprot/G3RTN4)	Gorilla gorilla	MAGDLSAGFFMEELNTYRQKGVLKYQELPNSGPPHRRFTFRVIIDREPEPEGEGRSKKEAK NAAKLAVEILNKEKKAVSPLLLTTTNSSEGLSMGNYIGLINRIAQKKRLTVNYEQCASGVHGPEG FHYKCKMGQKEYSIGTGSTKQEAQLAAKLAYLIQIVSEETSVKSDYLSGSGFATTCDSQSNLSVT STLASESSSEGDFSDATSEINSENSDSLSNLLMGLRNQKAKRSLAPRFDLPDMKETKYTVD KRFQMDFKIEILIGSGGFGQVFKAHRIDGKTYVIKRVKYNNEKAEREVKALAKLDHVNIVHYNGC WDGVDYDPETSDDYLESSDYDPENSKNSRSKTKCLFIQMEFCDKGTLEQWIENRRGEKLDKVL ALELFEQITKGVDYTHSKKLIHRDLKPSNIFLVDTKQVKIGDFGLVTSLKNDGKRTSRSGTLYRMSP EQISSODYGKEVDLYALGLILAEILLHVCDTAFETSKFTDLRDGIISDIFDKKEKTLLOKLLSKKPED RPTSEILRLTLVWKKSPENRHTC	
60	pyrrolysyl tRNA of Methanosarcina mazei for unidentified codon (tRNA ^{Pyl,rem})	Artificial sequence Synthetic cf.: SEQ ID Nos: 25, 26 and 27	ggaaacctgacatgtagatcgaaatggactnnnaatccggtcagccgggttagattcccggggtttccg	Underlined: unidentified anticodon (31-33)

② Indicates text missing or illegible when filed

 SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 60

<210> SEQ ID NO 1
 <211> LENGTH: 68
 <212> TYPE: DNA
 <213> ORGANISM: hepatitis delta virus
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <223> OTHER INFORMATION: hepatitis delta virus (HDV) ribozyme (Avis)
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (1)..(6)
 <223> OTHER INFORMATION: flanking sequence
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (7)..(68)
 <223> OTHER INFORMATION: HDV ribozyme core sequence

<400> SEQUENCE: 1

```
gcuaGCCaUG guccCagccU ccucgcUGgc gGcuagUGgg caacauGcuu cgGcaUGgCG      60
aaUGggGac                                                                    68
```

<210> SEQ ID NO 2
 <211> LENGTH: 67
 <212> TYPE: DNA
 <213> ORGANISM: hepatitis delta virus
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <223> OTHER INFORMATION: hepatitis delta virus (HDV) ribozyme (Schuerer)

<400> SEQUENCE: 2

```
ggGtcGgcat gGcatctcca cctcctcgCG gtccGacctg gGctacttcg gtagGctaag      60
ggagaag                                                                    67
```

<210> SEQ ID NO 3
 <211> LENGTH: 56
 <212> TYPE: DNA
 <213> ORGANISM: hepatitis delta virus
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <223> OTHER INFORMATION: hepatitis delta virus (HDV) ribozyme (Akoopie)

<400> SEQUENCE: 3

```
gaggGatagt acagagcctc cccgtGgctc ccttGgataa ccaactGata ctgtac          56
```

<210> SEQ ID NO 4
 <211> LENGTH: 50
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: hammerhead ribozyme
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (1)..(10)
 <223> OTHER INFORMATION: flanking sequence
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (11)..(45)
 <223> OTHER INFORMATION: ribozyme core sequence
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (46)..(50)
 <223> OTHER INFORMATION: flanking sequence

<400> SEQUENCE: 4

```
ucagguuuCC cUGaUGaguc cGUGaggacG aaacGguacc cGguaccGuc                50
```

-continued

<210> SEQ ID NO 5
<211> LENGTH: 92
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: hairpin ribozyme

<400> SEQUENCE: 5

ggagagagaa gucaaccaga gaaacacacc aaccgauugc acuccggguu ggugguauau 60
uaccugguac gggggaaacu ucguggggc cg 92

<210> SEQ ID NO 6
<211> LENGTH: 176
<212> TYPE: DNA
<213> ORGANISM: Neurospora intermedia
<220> FEATURE:
<221> NAME/KEY: misc_feature
<223> OTHER INFORMATION: Varkud satellite (VS) ribozyme

<400> SEQUENCE: 6

gggaaagcuu gcgaaggcg ucgucgcccc gagcgguagu aagcaggga cucaccucca 60
auuucaguac ugaauuguc guagcaguug acuacugua ugugauuggu agaggcuaag 120
ugacgguaau ggcguaaguc aguauugcag cacagcaca gcccgcuugc gagaau 176

<210> SEQ ID NO 7
<211> LENGTH: 152
<212> TYPE: DNA
<213> ORGANISM: Bacillus sp.
<220> FEATURE:
<221> NAME/KEY: misc_feature
<223> OTHER INFORMATION: glucosamine-6-phosphate synthetase (glmS)
riboswitch (glmS ribozyme)

<400> SEQUENCE: 7

gcgccagaa tggcaccauu gcacuccggu gccaguugac gagguggggu uauccgagau 60
uucggcggau gacucccggu uguucaucac aaccgcaagc uuuuacuaa aucauuagg 120
ugacuuagug gacaaaggug aaagugugau ga 152

<210> SEQ ID NO 8
<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: tetracycline operator (tetO)

<400> SEQUENCE: 8

tccctatcag tgatagaga 19

<210> SEQ ID NO 9
<211> LENGTH: 59
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: minimal cytomegalovirus (CMV) promoter

<400> SEQUENCE: 9

gtaggcgtgt acgggtgggag gtctatataa gcagagctcg tttagtgaac cgtcagatc 59

<210> SEQ ID NO 10

-continued

```

<211> LENGTH: 467
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: "Tet-On" promoter comprising 8 tetO sequences
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (18)..(36)
<223> OTHER INFORMATION: tetO sequence
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (54)..(72)
<223> OTHER INFORMATION: tetO sequence
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (90)..(208)
<223> OTHER INFORMATION: tetO sequence
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (126)..(144)
<223> OTHER INFORMATION: tetO sequence
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (162)..(180)
<223> OTHER INFORMATION: tetO sequence
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (198)..(216)
<223> OTHER INFORMATION: tetO sequence
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (234)..(252)
<223> OTHER INFORMATION: tetO sequence
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (270)..(288)
<223> OTHER INFORMATION: tetO sequence

<400> SEQUENCE: 10

caattgctcg agtttactcc ctatcagtga tagagaacgt atgtcgactt tactccctat      60
cagtgataga gaacgtatga agagtttact cctatcagt gatagagaac gtatgcagac      120
tttactccct atcagtgata gagaacgtat aaggagttta ctccctatca gtgatagaga      180
acgtatgacc agtttactcc ctatcagtga tagagaacgt atctacagtt tactccctat      240
cagtgataga gaacgtatat ccagtttact cctatcagt gatagagaac gtatacgctt      300
taggcgtgta cgggtggcgc ctataaaagc agagctcggt tagtgaaccg tcagatcgcc      360
tggagcaatt ccacaacact ttgtgtctat accaactttc cgtaccactt cctaccctcg      420
aaagtcataa tggctagcct gcagggatcc agatatcaag cttcacc                      467

<210> SEQ ID NO 11
<211> LENGTH: 425
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: "Tet-On" promoter comprising 7 tetO sequences
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (12)..(30)
<223> OTHER INFORMATION: tetO sequences
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (48)..(66)
<223> OTHER INFORMATION: tetO sequences
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (84)..(102)
<223> OTHER INFORMATION: tetO sequences
<220> FEATURE:

```

-continued

```

<221> NAME/KEY: misc_feature
<222> LOCATION: (120)..(138)
<223> OTHER INFORMATION: tetO sequences
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (156)..(174)
<223> OTHER INFORMATION: tetO sequences
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (192)..(210)
<223> OTHER INFORMATION: tetO sequences
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (228)..(246)
<223> OTHER INFORMATION: tetO sequences

<400> SEQUENCE: 11

ctcgacttta ctccctatca gtgatataga acgtatgaag agtttactcc ctatcagtga      60
tagagaacgt atgcagactt tactccctat cagtgataga gaacgtataa ggagtttact      120
ccctatcagt gatagagaac gtagtaccag ttctactccct atcagtgata gagaacgtat      180
ctacagttta ctccctatca gtgatataga acgtatatcc agtttactcc ctatcagtga      240
tagagaacgt atacgcttta ggcgtgtacg gtgggcgccct ataaaagcag agctcgttta      300
gtgaaccgtc agatcgctg gagcaattcc acaacacttt tgtcttatac caactttccg      360
taccacttcc taccctcgaa agtccatag gctagcctgc agggatccag atatcaagct      420
tcacc                                              425

<210> SEQ ID NO 12
<211> LENGTH: 794
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: "T-REx" promoter comprising two tetO sequences
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1)..(380)
<223> OTHER INFORMATION: CMV enhancer
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (381)..(584)
<223> OTHER INFORMATION: CMV promoter
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (586)..(604)
<223> OTHER INFORMATION: tetO sequence
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (607)..(625)
<223> OTHER INFORMATION: tetO sequence

<400> SEQUENCE: 12

gacattgatt attgactagt tattaatagt aatcaattac ggggtcatta gttcatagcc      60
catatatgga gttccgcgtt acataactta cggtaaatgg cccgcctggc tgaccgccca      120
acgacccccg ccattgacg tcaataatga cgtatgttcc catagtaacg ccaataggga      180
ctttccattg acgtcaatgg gtggagtatt tacggtaaac tgcccacttg gcagtacatc      240
aagtgtatca tatgccaagt acgcccccta ttgacgtcaa tgacggtaaa tggcccgcct      300
ggcattatgc ccagtacatg accttatggg actttcctac ttggcagtac atctacgtat      360
tagtcatcgc tattaccatg gtgatgcggt ttgggcagta catcaatggg cgtggatagc      420
ggtttgactc acggggattt ccaagtctcc accccattga cgtcaatggg agtttgtttt      480

```

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```

ggcaccacaaa tcaacgggac ttccaaaat gtcgtaacaa ctccgcccc ttgacgcaaa 540
tgggcggtag gcgtgtacgg tgggaggtct atataagcag agctctccct atcagtgata 600
gagatctccc tatcagtgat agagatcgtc gacgagctcg tttagtgaac cgtcagatcg 660
cctggagacg ccatccacgc tgttttgacc tccatagaag acaccgggac cgatccagcc 720
tccggactct agcgtttaaa catcacaagt ttgtacaaaa aagcaggcgc tagcgtttaa 780
acttaagctt cacc 794

```

```

<210> SEQ ID NO 13
<211> LENGTH: 207
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: tetracycline repressor (tetR)
UniProtKB/Swiss-Prot accession no.: P04483.1

```

```

<400> SEQUENCE: 13

```

```

Met Ser Arg Leu Asp Lys Ser Lys Val Ile Asn Ser Ala Leu Glu Leu
1      5      10      15
Leu Asn Glu Val Gly Ile Glu Gly Leu Thr Thr Arg Lys Leu Ala Gln
20     25     30
Lys Leu Gly Val Glu Gln Pro Thr Leu Tyr Trp His Val Lys Asn Lys
35     40     45
Arg Ala Leu Leu Asp Ala Leu Ala Ile Glu Met Leu Asp Arg His His
50     55     60
Thr His Phe Cys Pro Leu Glu Gly Glu Ser Trp Gln Asp Phe Leu Arg
65     70     75     80
Asn Asn Ala Lys Ser Phe Arg Cys Ala Leu Leu Ser His Arg Asp Gly
85     90     95
Ala Lys Val His Leu Gly Thr Arg Pro Thr Glu Lys Gln Tyr Glu Thr
100    105    110
Leu Glu Asn Gln Leu Ala Phe Leu Cys Gln Gln Gly Phe Ser Leu Glu
115    120    125
Asn Ala Leu Tyr Ala Leu Ser Ala Val Gly His Phe Thr Leu Gly Cys
130    135    140
Val Leu Glu Asp Gln Glu His Gln Val Ala Lys Glu Glu Arg Glu Thr
145    150    155    160
Pro Thr Thr Asp Ser Met Pro Pro Leu Leu Arg Gln Ala Ile Glu Leu
165    170    175
Phe Asp His Gln Gly Ala Glu Pro Ala Phe Leu Phe Gly Leu Glu Leu
180    185    190
Ile Ile Cys Gly Leu Glu Lys Gln Leu Lys Cys Glu Ser Gly Ser
195    200    205

```

```

<210> SEQ ID NO 14
<211> LENGTH: 1442
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: rtTA expression cassette
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1)..(588)
<223> OTHER INFORMATION: CMV
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (696)..(1442)

```

-continued

<223> OTHER INFORMATION: TetOn3G

<400> SEQUENCE: 14

```

gttgacattg attattgact agttattaat agtaatcaat tacgggggtca ttagttcata    60
gcccatatat ggagttccgc gttacataac ttacggtaaa tggcccgccct ggctgaccgc    120
ccaacgaccc cgcgccattg acgtcaataa tgacgtatgt toccatagta acgccaatag    180
ggactttcca ttgacgtcaa tgggtgggact atttacggta aactgcccac ttggcagtac    240
atcaagtgta tcatatgccca agtacgcccc ctattgacgt caatgacggg aaatggcccc    300
cctggcatta tgcccagtac atgaccttat gggactttcc tacttggcag tacatctacg    360
tattagtcat cgctattacc atgggtgatgc ggttttggca gtacatcaat gggcgtaggat    420
agcggtttga ctacggggga ttccaagtc tccaccccat tgacgtcaat gggagtttgt    480
tttggcacca aaatcaacgg gactttccaa aatgtcgtaa caactccgcc ccattgacgc    540
aaatggggcg taggcgtgta cgggtgggag tctatataag cagagctctc tggctaacta    600
gagaaccac tgcttactgg cttatcgaaa ttaatacgac tcatatagg gagaccaag    660
ctggctagcg tttaaactta agcttctcga gcaccatgtc tagactggac aagagcaaag    720
tcataaaact tgctctggaa ttactcaatg gagtcgggat cgaaggcctg acgacaagga    780
aactcgctca aaagctggga gttgagcagc ctaccctgta ctggcactg aagaacaagc    840
gggcctgct cgatgcctcg ccaatcgaga tgctggacag gcatcatacc cactcctgcc    900
ccctggaagg cgagtcattg caagactttc tgcggaacaa cgccaagtca taccgctgtg    960
ctctcctctc acatcgcgac ggggctaaag tgcctctcgg caccgcccc acagagaaac   1020
agtacgaaac cctggaaaat cagctcgctg tcctgtgtca gcaaggcttc tccttgagga   1080
acgcactgta cgctctgtcc gccgtgggcc actttactac gggctgcgta ttggagggaac   1140
aggagcatca agtagcaaaa gaggaagag agacacctac caccgattct atgccccac    1200
ttctgaaaca agcaattgag ctgttcgacc ggcagggagc cgaacctgcc tcctttttcg   1260
gcctggaact aatcatatgt ggcttgagga aacagctaaa gtgcgaaaag ggcggggcga    1320
ccgacgcctt tgacgatttt gacttagaca tgctcccagc cgatgccctt gacgactttg   1380
accttgatat gctgcctgct gacgctcttg acgattttga ccttgacatg ctccccgggt   1440
aa

```

<210> SEQ ID NO 15

<211> LENGTH: 18

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: T7 promoter class III (core)

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (18)..(18)

<223> OTHER INFORMATION: G at position 18 is preferred nucleotide at the +1 position to promote efficient transcription initiation

<400> SEQUENCE: 15

```

taatacgact cactatag

```

<210> SEQ ID NO 16

<211> LENGTH: 19

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

-continued

```

<220> FEATURE:
<223> OTHER INFORMATION: T7 promoter class III
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (18)..(19)
<223> OTHER INFORMATION: GG at positions 18 and 19 are preferred
nucleotides at the +1 and +2 positions to promote efficient
transcription initiation

```

```

<400> SEQUENCE: 16

```

```

taatacgact cactatagg                                     19

```

```

<210> SEQ ID NO 17
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: T7 promoter class II (core)
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (18)..(18)
<223> OTHER INFORMATION: A at position 18 is preferred nucleotide at
the +1 position of the gene to promote efficient transcription
initiation

```

```

<400> SEQUENCE: 17

```

```

taatacgact cactatta                                     18

```

```

<210> SEQ ID NO 18
<211> LENGTH: 883
<212> TYPE: PRT
<213> ORGANISM: Enterobacteria phage T7

```

```

<400> SEQUENCE: 18

```

```

Met Asn Thr Ile Asn Ile Ala Lys Asn Asp Phe Ser Asp Ile Glu Leu
1           5           10          15
Ala Ala Ile Pro Phe Asn Thr Leu Ala Asp His Tyr Gly Glu Arg Leu
20          25          30
Ala Arg Glu Gln Leu Ala Leu Glu His Glu Ser Tyr Glu Met Gly Glu
35          40          45
Ala Arg Phe Arg Lys Met Phe Glu Arg Gln Leu Lys Ala Gly Glu Val
50          55          60
Ala Asp Asn Ala Ala Ala Lys Pro Leu Ile Thr Thr Leu Leu Pro Lys
65          70          75          80
Met Ile Ala Arg Ile Asn Asp Trp Phe Glu Glu Val Lys Ala Lys Arg
85          90          95
Gly Lys Arg Pro Thr Ala Phe Gln Phe Leu Gln Glu Ile Lys Pro Glu
100         105         110
Ala Val Ala Tyr Ile Thr Ile Lys Thr Thr Leu Ala Cys Leu Thr Ser
115         120         125
Ala Asp Asn Thr Thr Val Gln Ala Val Ala Ser Ala Ile Gly Arg Ala
130         135         140
Ile Glu Asp Glu Ala Arg Phe Gly Arg Ile Arg Asp Leu Glu Ala Lys
145         150         155         160
His Phe Lys Lys Asn Val Glu Glu Gln Leu Asn Lys Arg Val Gly His
165         170         175
Val Tyr Lys Lys Ala Phe Met Gln Val Val Glu Ala Asp Met Leu Ser
180         185         190
Lys Gly Leu Leu Gly Gly Glu Ala Trp Ser Ser Trp His Lys Glu Asp

```

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195	200	205
Ser Ile His Val Gly Val Arg Cys Ile Glu Met Leu Ile Glu Ser Thr		
210	215	220
Gly Met Val Ser Leu His Arg Gln Asn Ala Gly Val Val Gly Gln Asp		
225	230	235 240
Ser Glu Thr Ile Glu Leu Ala Pro Glu Tyr Ala Glu Ala Ile Ala Thr		
	245	250 255
Arg Ala Gly Ala Leu Ala Gly Ile Ser Pro Met Phe Gln Pro Cys Val		
	260	265 270
Val Pro Pro Lys Pro Trp Thr Gly Ile Thr Gly Gly Gly Tyr Trp Ala		
	275	280 285
Asn Gly Arg Arg Pro Leu Ala Leu Val Arg Thr His Ser Lys Lys Ala		
	290	295 300
Leu Met Arg Tyr Glu Asp Val Tyr Met Pro Glu Val Tyr Lys Ala Ile		
305	310	315 320
Asn Ile Ala Gln Asn Thr Ala Trp Lys Ile Asn Lys Lys Val Leu Ala		
	325	330 335
Val Ala Asn Val Ile Thr Lys Trp Lys His Cys Pro Val Glu Asp Ile		
	340	345 350
Pro Ala Ile Glu Arg Glu Glu Leu Pro Met Lys Pro Glu Asp Ile Asp		
	355	360 365
Met Asn Pro Glu Ala Leu Thr Ala Trp Lys Arg Ala Ala Ala Val		
370	375	380
Tyr Arg Lys Asp Lys Ala Arg Lys Ser Arg Arg Ile Ser Leu Glu Phe		
385	390	395 400
Met Leu Glu Gln Ala Asn Lys Phe Ala Asn His Lys Ala Ile Trp Phe		
	405	410 415
Pro Tyr Asn Met Asp Trp Arg Gly Arg Val Tyr Ala Val Ser Met Phe		
	420	425 430
Asn Pro Gln Gly Asn Asp Met Thr Lys Gly Leu Leu Thr Leu Ala Lys		
	435	440 445
Gly Lys Pro Ile Gly Lys Glu Gly Tyr Tyr Trp Leu Lys Ile His Gly		
450	455	460
Ala Asn Cys Ala Gly Val Asp Lys Val Pro Phe Pro Glu Arg Ile Lys		
465	470	475 480
Phe Ile Glu Glu Asn His Glu Asn Ile Met Ala Cys Ala Lys Ser Pro		
	485	490 495
Leu Glu Asn Thr Trp Trp Ala Glu Gln Asp Ser Pro Phe Cys Phe Leu		
	500	505 510
Ala Phe Cys Phe Glu Tyr Ala Gly Val Gln His His Gly Leu Ser Tyr		
515	520	525
Asn Cys Ser Leu Pro Leu Ala Phe Asp Gly Ser Cys Ser Gly Ile Gln		
530	535	540
His Phe Ser Ala Met Leu Arg Asp Glu Val Gly Gly Arg Ala Val Asn		
545	550	555 560
Leu Leu Pro Ser Glu Thr Val Gln Asp Ile Tyr Gly Ile Val Ala Lys		
	565	570 575
Lys Val Asn Glu Ile Leu Gln Ala Asp Ala Ile Asn Gly Thr Asp Asn		
	580	585 590
Glu Val Val Thr Val Thr Asp Glu Asn Thr Gly Glu Ile Ser Glu Lys		
595	600	605

-continued

Val Lys Leu Gly Thr Lys Ala Leu Ala Gly Gln Trp Leu Ala Tyr Gly
 610 615 620
 Val Thr Arg Ser Val Thr Lys Arg Ser Val Met Thr Leu Ala Tyr Gly
 625 630 635 640
 Ser Lys Glu Phe Gly Phe Arg Gln Gln Val Leu Glu Asp Thr Ile Gln
 645 650 655
 Pro Ala Ile Asp Ser Gly Lys Gly Leu Met Phe Thr Gln Pro Asn Gln
 660 665 670
 Ala Ala Gly Tyr Met Ala Lys Leu Ile Trp Glu Ser Val Ser Val Thr
 675 680 685
 Val Val Ala Ala Val Glu Ala Met Asn Trp Leu Lys Ser Ala Ala Lys
 690 695 700
 Leu Leu Ala Ala Glu Val Lys Asp Lys Lys Thr Gly Glu Ile Leu Arg
 705 710 715 720
 Lys Arg Cys Ala Val His Trp Val Thr Pro Asp Gly Phe Pro Val Trp
 725 730 735
 Gln Glu Tyr Lys Lys Pro Ile Gln Thr Arg Leu Asn Leu Met Phe Leu
 740 745 750
 Gly Gln Phe Arg Leu Gln Pro Thr Ile Asn Thr Asn Lys Asp Ser Glu
 755 760 765
 Ile Asp Ala His Lys Gln Glu Ser Gly Ile Ala Pro Asn Phe Val His
 770 775 780
 Ser Gln Asp Gly Ser His Leu Arg Lys Thr Val Val Trp Ala His Glu
 785 790 795 800
 Lys Tyr Gly Ile Glu Ser Phe Ala Leu Ile His Asp Ser Phe Gly Thr
 805 810 815
 Ile Pro Ala Asp Ala Ala Asn Leu Phe Lys Ala Val Arg Glu Thr Met
 820 825 830
 Val Asp Thr Tyr Glu Ser Cys Asp Val Leu Ala Asp Phe Tyr Asp Gln
 835 840 845
 Phe Ala Asp Gln Leu His Glu Ser Gln Leu Asp Lys Met Pro Ala Leu
 850 855 860
 Pro Ala Lys Gly Asn Leu Asn Leu Arg Asp Ile Leu Glu Ser Asp Phe
 865 870 875 880
 Ala Phe Ala

<210> SEQ ID NO 19
 <211> LENGTH: 899
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: NES-T7 RNA polymerase
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (2)..(17)
 <223> OTHER INFORMATION: NES

<400> SEQUENCE: 19

Met Cys Pro Val Pro Leu Gln Leu Pro Pro Leu Glu Arg Leu Thr Leu
 1 5 10 15
 Asp Asn Thr Ile Asn Ile Ala Lys Asn Asp Phe Ser Asp Ile Glu Leu
 20 25 30
 Ala Ala Ile Pro Phe Asn Thr Leu Ala Asp His Tyr Gly Glu Arg Leu
 35 40 45

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Ala	Arg	Glu	Gln	Leu	Ala	Leu	Glu	His	Glu	Ser	Tyr	Glu	Met	Gly	Glu	50	55	60
Ala	Arg	Phe	Arg	Lys	Met	Phe	Glu	Arg	Gln	Leu	Lys	Ala	Gly	Glu	Val	65	70	75
Ala	Asp	Asn	Ala	Ala	Ala	Lys	Pro	Leu	Ile	Thr	Thr	Leu	Leu	Pro	Lys	85	90	95
Met	Ile	Ala	Arg	Ile	Asn	Asp	Trp	Phe	Glu	Glu	Val	Lys	Ala	Lys	Arg	100	105	110
Gly	Lys	Arg	Pro	Thr	Ala	Phe	Gln	Phe	Leu	Gln	Glu	Ile	Lys	Pro	Glu	115	120	125
Ala	Val	Ala	Tyr	Ile	Thr	Ile	Lys	Thr	Thr	Leu	Ala	Cys	Leu	Thr	Ser	130	135	140
Ala	Asp	Asn	Thr	Thr	Val	Gln	Ala	Val	Ala	Ser	Ala	Ile	Gly	Arg	Ala	145	150	155
Ile	Glu	Asp	Glu	Ala	Arg	Phe	Gly	Arg	Ile	Arg	Asp	Leu	Glu	Ala	Lys	165	170	175
His	Phe	Lys	Lys	Asn	Val	Glu	Glu	Gln	Leu	Asn	Lys	Arg	Val	Gly	His	180	185	190
Val	Tyr	Lys	Lys	Ala	Phe	Met	Gln	Val	Val	Glu	Ala	Asp	Met	Leu	Ser	195	200	205
Lys	Gly	Leu	Leu	Gly	Gly	Glu	Ala	Trp	Ser	Ser	Trp	His	Lys	Glu	Asp	210	215	220
Ser	Ile	His	Val	Gly	Val	Arg	Cys	Ile	Glu	Met	Leu	Ile	Glu	Ser	Thr	225	230	235
Gly	Met	Val	Ser	Leu	His	Arg	Gln	Asn	Ala	Gly	Val	Val	Gly	Gln	Asp	245	250	255
Ser	Glu	Thr	Ile	Glu	Leu	Ala	Pro	Glu	Tyr	Ala	Glu	Ala	Ile	Ala	Thr	260	265	270
Arg	Ala	Gly	Ala	Leu	Ala	Gly	Ile	Ser	Pro	Met	Phe	Gln	Pro	Cys	Val	275	280	285
Val	Pro	Pro	Lys	Pro	Trp	Thr	Gly	Ile	Thr	Gly	Gly	Gly	Tyr	Trp	Ala	290	295	300
Asn	Gly	Arg	Arg	Pro	Leu	Ala	Leu	Val	Arg	Thr	His	Ser	Lys	Lys	Ala	305	310	315
Leu	Met	Arg	Tyr	Glu	Asp	Val	Tyr	Met	Pro	Glu	Val	Tyr	Lys	Ala	Ile	325	330	335
Asn	Ile	Ala	Gln	Asn	Thr	Ala	Trp	Lys	Ile	Asn	Lys	Lys	Val	Leu	Ala	340	345	350
Val	Ala	Asn	Val	Ile	Thr	Lys	Trp	Lys	His	Cys	Pro	Val	Glu	Asp	Ile	355	360	365
Pro	Ala	Ile	Glu	Arg	Glu	Glu	Leu	Pro	Met	Lys	Pro	Glu	Asp	Ile	Asp	370	375	380
Met	Asn	Pro	Glu	Ala	Leu	Thr	Ala	Trp	Lys	Arg	Ala	Ala	Ala	Ala	Val	385	390	395
Tyr	Arg	Lys	Asp	Lys	Ala	Arg	Lys	Ser	Arg	Arg	Ile	Ser	Leu	Glu	Phe	405	410	415
Met	Leu	Glu	Gln	Ala	Asn	Lys	Phe	Ala	Asn	His	Lys	Ala	Ile	Trp	Phe	420	425	430
Pro	Tyr	Asn	Met	Asp	Trp	Arg	Gly	Arg	Val	Tyr	Ala	Val	Ser	Met	Phe	435	440	445

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Asn	Pro	Gln	Gly	Asn	Asp	Met	Thr	Lys	Gly	Leu	Leu	Thr	Leu	Ala	Lys
450						455					460				
Gly	Lys	Pro	Ile	Gly	Lys	Glu	Gly	Tyr	Tyr	Trp	Leu	Lys	Ile	His	Gly
465					470					475					480
Ala	Asn	Cys	Ala	Gly	Val	Asp	Lys	Val	Pro	Phe	Pro	Glu	Arg	Ile	Lys
				485					490					495	
Phe	Ile	Glu	Glu	Asn	His	Glu	Asn	Ile	Met	Ala	Cys	Ala	Lys	Ser	Pro
		500						505					510		
Leu	Glu	Asn	Thr	Trp	Trp	Ala	Glu	Gln	Asp	Ser	Pro	Phe	Cys	Phe	Leu
		515					520					525			
Ala	Phe	Cys	Phe	Glu	Tyr	Ala	Gly	Val	Gln	His	His	Gly	Leu	Ser	Tyr
	530					535					540				
Asn	Cys	Ser	Leu	Pro	Leu	Ala	Phe	Asp	Gly	Ser	Cys	Ser	Gly	Ile	Gln
545					550					555					560
His	Phe	Ser	Ala	Met	Leu	Arg	Asp	Glu	Val	Gly	Gly	Arg	Ala	Val	Asn
			565						570					575	
Leu	Leu	Pro	Ser	Glu	Thr	Val	Gln	Asp	Ile	Tyr	Gly	Ile	Val	Ala	Lys
			580					585					590		
Lys	Val	Asn	Glu	Ile	Leu	Gln	Ala	Asp	Ala	Ile	Asn	Gly	Thr	Asp	Asn
		595					600					605			
Glu	Val	Val	Thr	Val	Thr	Asp	Glu	Asn	Thr	Gly	Glu	Ile	Ser	Glu	Lys
	610					615					620				
Val	Lys	Leu	Gly	Thr	Lys	Ala	Leu	Ala	Gly	Gln	Trp	Leu	Ala	Tyr	Gly
625					630					635					640
Val	Thr	Arg	Ser	Val	Thr	Lys	Arg	Ser	Val	Met	Thr	Leu	Ala	Tyr	Gly
			645						650					655	
Ser	Lys	Glu	Phe	Gly	Phe	Arg	Gln	Gln	Val	Leu	Glu	Asp	Thr	Ile	Gln
			660					665					670		
Pro	Ala	Ile	Asp	Ser	Gly	Lys	Gly	Leu	Met	Phe	Thr	Gln	Pro	Asn	Gln
		675					680					685			
Ala	Ala	Gly	Tyr	Met	Ala	Lys	Leu	Ile	Trp	Glu	Ser	Val	Ser	Val	Thr
		690				695					700				
Val	Val	Ala	Ala	Val	Glu	Ala	Met	Asn	Trp	Leu	Lys	Ser	Ala	Ala	Lys
705					710					715					720
Leu	Leu	Ala	Ala	Glu	Val	Lys	Asp	Lys	Lys	Thr	Gly	Glu	Ile	Leu	Arg
				725					730					735	
Lys	Arg	Cys	Ala	Val	His	Trp	Val	Thr	Pro	Asp	Gly	Phe	Pro	Val	Trp
			740					745					750		
Gln	Glu	Tyr	Lys	Lys	Pro	Ile	Gln	Thr	Arg	Leu	Asn	Leu	Met	Phe	Leu
		755					760					765			
Gly	Gln	Phe	Arg	Leu	Gln	Pro	Thr	Ile	Asn	Thr	Asn	Lys	Asp	Ser	Glu
	770					775					780				
Ile	Asp	Ala	His	Lys	Gln	Glu	Ser	Gly	Ile	Ala	Pro	Asn	Phe	Val	His
785					790					795					800
Ser	Gln	Asp	Gly	Ser	His	Leu	Arg	Lys	Thr	Val	Val	Trp	Ala	His	Glu
			805						810					815	
Lys	Tyr	Gly	Ile	Glu	Ser	Phe	Ala	Leu	Ile	His	Asp	Ser	Phe	Gly	Thr
			820					825					830		
Ile	Pro	Ala	Asp	Ala	Ala	Asn	Leu	Phe	Lys	Ala	Val	Arg	Glu	Thr	Met
		835					840					845			
Val	Asp	Thr	Tyr	Glu	Ser	Cys	Asp	Val	Leu	Ala	Asp	Phe	Tyr	Asp	Gln

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850	855	860	
Phe Ala Asp Gln Leu His Glu Ser Gln Leu Asp Lys Met Pro Ala Leu			
865	870	875	880
Pro Ala Lys Gly Asn Leu Asn Leu Arg Asp Ile Leu Glu Ser Asp Phe			
	885	890	895

Ala Phe Ala

<210> SEQ ID NO 20

<400> SEQUENCE: 20

000

<210> SEQ ID NO 21

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Simian vacuolating virus 40 (SV40)

<400> SEQUENCE: 21

Pro Lys Lys Lys Arg Lys Val

1 5

<210> SEQ ID NO 22

<211> LENGTH: 1416

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Methanosarcina mazei PylRS Y306A, Y384F mutant (PylRSF)

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (1)..(3)

<223> OTHER INFORMATION: ATG coding for M may be missing

<400> SEQUENCE: 22

atggcgtgcc	cggtgccgct	gcagctgccg	ccgctggaac	gcctgacct	ggatgataaa	60
aaaccgctga	ataccctgat	ctctgtact	ggtctgtgga	tgagtcgtac	cggaaccatt	120
cataaaatca	aacaccacga	ggtagccgt	tcgaaaatct	atattgagat	ggcgtgtggc	180
gatcatctgg	ttgtgaacaa	tagccgctct	tctcgtacag	cacgtgcact	gcgtcaccac	240
aaatatcgta	aaacctgtaa	acgttgccgt	gtgtccgatg	aggatctgaa	caaattcctg	300
acaaaagcca	atgaggacca	aacaagcgtg	aaagtgaag	tcgttagcgc	tcttaccctg	360
actaaaaaag	caatgccgaa	atccgttgct	cgtgcccta	aaccactgga	aaacactgaa	420
gcagcacagg	cacagccgtc	tggaagcaaa	ttctctccgg	ccattcctgt	ttctaccag	480
gagtcggttt	ctgttccagc	aagtgtgagc	accagcatta	gcagtattag	caccggtgcc	540
accgctagcg	ccctggttaa	aggcaatacc	aatccgatta	caagcatgtc	tgccccggtt	600
caagcatcag	ctccagcact	gacaaaatcc	caaaccgata	gtctggaggt	tctgctgaat	660
ccgaaagacg	aatcagcct	gaattccggc	aaaccgtttc	gtgaactgga	gagcgaaactg	720
ctgtcacgtc	gtaaaaaaga	cctgcaacaa	atctatgccg	aagaacgtga	gaactatctg	780
gggaaactgg	aacgtgaat	caccgccttt	tctgtggatc	gtggctttct	ggagatcaaa	840
tccccgattc	tgattcctct	ggagtatatc	gagcgtatgg	gcacgcacaa	tgataccgaa	900
ctgagcaaac	aaattttccg	tgtggataaa	aacttctgtc	tgccgcctat	gctagcacca	960
aatctggcta	actatctgcg	caaaactggac	cgtgccctgc	ctgatcctat	caaaatcttc	1020

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gagatcggcc cgtgttatcg taaagagtcc gacggtaaag aacatctgga ggagtttacc 1080
atgctgaact tttgccaaat gggttcaggt tgtactcgtg agaacctgga aagcatcatc 1140
accgatatttc tgaaccacct gggcattgac ttcaaaattg tgggcgacag ctgtatgggtg 1200
tttggcgaca ccctggatgt catgcacggc gacctggaac tgtctagtgc cgttggtggc 1260
ccaatcccg cggatcgtga gtgggggtatc gacaaacctt ggatcgggtgc gggtttttgt 1320
ctggagcgctc tgctgaaagt aaaacacgac ttcaagaaca tcaaacgtgc tgcacgttcc 1380
gagtcctatt acaatgggat ttctactaac ctgtaa 1416

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<210> SEQ ID NO 23

<211> LENGTH: 1416

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Methanosarcina mazei PylRS N346A, C348A mutant (PylRSAA)

<400> SEQUENCE: 23

```

atggcgtgcc cggtgccgct gcagctgccg ccgctggaac gcctgacctt ggatgacaaa 60
aaaccgctga atacctgat ctctgtactt ggtctgtgga tgagtcgtac cggaaccatt 120
cataaaatca aacaccacga ggttagccgt tcgaaaatct atattgagat ggcgtgtggc 180
gatcatctgg ttgtgaacaa tagccgctct tctcgtacag cacgtgcact gcgtcaccac 240
aaatatcgta aaacctgtaa acgttgccgt gtgtccgatg aggatctgaa caaattcctg 300
acaaaagcca atgaggacca aacaagcgtg aaagtgaaag tcgttagcgc tcctaccgt 360
actaaaaaag caatgccgaa atccgttgct cgtgccccta aaccactgga aaacactgaa 420
gcagcacagg cacagccgtc tggagcaaaa ttctctccgg ccattcctgt ttctaccag 480
gagtcggttt ctgttccagc aagtgtgagc accagcatta gcagtattag caccggtgcc 540
accgctagcg ccctgggtta aggcaatacc aatccgatta caagcatgct tgcgccggtt 600
caagcatcag ctccagcact gacaaaatcc caaacggatc gtctggaggt tctgtgtaat 660
ccgaaagacg aaatcagcct gaattccggc aaaccgtttc gtgaactgga gagcgaactg 720
ctgtcacgctc gtaaaaaaga cctgcaacaa atctatgccg aagaacgtga gaactatctg 780
gggaaactgg aacgtgaaat caccgcgttt ttcgtggatc gtggccttct ggagatcaaa 840
tccccgattc tgattcctct ggagtatatc gagcgtatgg gcacgcacaa tgataccgaa 900
ctgagcaaac aaattttccg tgtggataaa aacttctgtc tgcgccctat gctggcacca 960
aatctgtata actatctgcg caaactggac cgtgccctgc ctgacccat caaaatcttc 1020
gagatcggcc cgtgttatcg taaagagtcc gacggtaaag aacatctgga ggagtttacc 1080
atgctggcct ttgccaaat gggttcaggt tgtactcgtg agaacctgga aagcatcatc 1140
accgatatttc tgaaccacct gggcattgac ttcaaaattg tgggcgacag ctgtatgggtg 1200
tatggcgaca ccctggatgt catgcacggc gacctggaac tgtctagtgc cgttggttga 1260
ccaattccgc tggaccgtga gtgggggtatc gacaaacctt ggatcggagc aggatcgggt 1320
ctggaacgcc tgctgaaagt gaaacacgac ttcaaaaaca tcaaacgtgc cgcgcgttct 1380
gaatcgtatt ataacgggat ctctacgaac ctgtaa 1416

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<210> SEQ ID NO 24

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<211> LENGTH: 1413
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Methanosarcina mazei PylRS Y306A, N346A, C348A, Y384Fmutant (PylRSAAAF)

<400> SEQUENCE: 24

```

gcgtgccccg tgccgctgca gctgccgccg ctggaacgcc tgaccctgga tgataaaaaa    60
ccgctgaata cccgatctc tgctactggt ctgtggatga gtcgtaccgg aaccattcat    120
aaaatcaaac accacgaggt tagccgttcg aaaatctata ttgagatggc gtgtggcgat    180
catctggttg tgaacaatag ccgctcttct cgtacagcac gtgcactgcg tcaccacaaa    240
tatcgtaaaa cctgtaaacg ttgccgtgtg tccgatgagg atctgaacaa attcctgaca    300
aaagccaatg aggacaaaac aagcgtgaaa gtgaaagtcg ttagcgctcc taccctgact    360
aaaaaagcaa tgccgaaatc cgttgctcgt gccctaaac cactggaaaa cactgaagca    420
gcacaggcac agccgtctgg aagcaaattc tctccggcca ttctgtttc taccaggag    480
tccgtttctg ttccagcaag tgtgagcacc agcattagca gtattagcac cggtgccacc    540
gctagcgccc tggttaaagg caataccaat ccgattacaa gcatgtctgc cccggttcaa    600
gcatcagctc cagcactgac aaaatcccaa accgatcgct tggaggttct gctgaatccg    660
aaagacgaaa tcagcctgaa ttccggcaaa ccgtttcgtg aactggagag cgaactgctg    720
tcacgtcgta aaaaagacct gcaacaaatc tatgccgaag aacgtgagaa ctatctgggg    780
aaactggaac gtgaaatcac ccgctttttc gtggatcgct gctttctgga gatcaaatcc    840
ccgattctga ttctctgga gtatatcgag cgtatgggca tcgacaatga taccgaactg    900
agcaaaaaaa ttttccgtgt ggataaaaac ttctgtctgc gccctatgct agcaccaaat    960
ctggctaact atctgcgcaa actggaccgt gccctgcctg atcctatcaa aatcttcgag   1020
atcggcccggt gttatcgtaa agagtccgac ggtaaagaac atctggagga gtttaccatg   1080
ctggcctttg cccaaatggg ttcaggttgt actcgtgaga acctggaaaag catcaccacc   1140
gattttctga accacctggg cattgacttc aaaattgtgg gcgacagctg tatgggtgtt   1200
ggcgacaccc tggatgtcat gcacggcgac ctggaactgt ctagtgccgt tgtgggcca   1260
atcccgctgg atcgtgagtg gggtatcgac aaaccttgga tcggtgcggg ttttgtctg   1320
gagcgtctgc tgaagtaaaa acacgacttc aagaacatca aacgtgctgc acgttccgag   1380
tcctattaca atggtatttc tactaacctg taa                                1413

```

<210> SEQ ID NO 25
 <211> LENGTH: 69
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: pyrrolsyl tRNA of Methanosarcina mazei for
 Amber codon (tRNA^{Pyl}, CUA)
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (31)..(33)
 <223> OTHER INFORMATION: anticodon

<400> SEQUENCE: 25

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ggaaacctga tcatttagat cgaatggact ctaaaccgt tcagccgggt tagattcccg    60
gggtttccg                                69

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<210> SEQ ID NO 26
<211> LENGTH: 69
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: pyrrolysyl tRNA of Methanosarcina mazei for
Opal codon (trNAPyl,UCA)
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (31)..(33)
<223> OTHER INFORMATION: anticodon

<400> SEQUENCE: 26

ggaaacctga tcatttagat cgaatggact tcaaatccgt tcagccgggt tagattccg 60
gggtttccg 69

<210> SEQ ID NO 27
<211> LENGTH: 69
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: pyrrolysyl tRNA of Methanosarcina mazei for
Ochre codon (trNAPyl,UUA)
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (31)..(33)
<223> OTHER INFORMATION: anticodon

<400> SEQUENCE: 27

ggaaacctga tcatttagat cgaatggact ttaaatccgt tcagccgggt tagattccg 60
gggtttccg 69

<210> SEQ ID NO 28
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: NES motif
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(1)
<223> OTHER INFORMATION: Xaa is selected from Leu, Ile, Val, Phe and Met
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (2)..(3)
<223> OTHER INFORMATION: Xaa is selected from any canonical amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (4)..(4)
<223> OTHER INFORMATION: Xaa is selected from Leu, Ile, Val, Phe and Met
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (5)..(5)
<223> OTHER INFORMATION: Xaa is selected from any canonical amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (6)..(6)
<223> OTHER INFORMATION: Xaa is selected from Leu, Ile, Val, Phe and Met

<400> SEQUENCE: 28

Xaa Xaa Xaa Xaa Xaa Xaa
1 5

<210> SEQ ID NO 29
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
<223> OTHER INFORMATION: NES motif
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(1)
<223> OTHER INFORMATION: Xaa is selected from Leu, Ile, Val, Phe and Met
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (2)..(4)
<223> OTHER INFORMATION: Xaa is selected from any canonical amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (5)..(5)
<223> OTHER INFORMATION: Xaa is selected from Leu, Ile, Val, Phe and Met
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (6)..(6)
<223> OTHER INFORMATION: Xaa is selected from any canonical amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (7)..(7)
<223> OTHER INFORMATION: Xaa is selected from Leu, Ile, Val, Phe and Met

<400> SEQUENCE: 29

Xaa Xaa Xaa Xaa Xaa Xaa Xaa
1 5

<210> SEQ ID NO 30
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: NES motif
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (2)..(3)
<223> OTHER INFORMATION: Xaa is selected from any canonical amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (4)..(4)
<223> OTHER INFORMATION: Xaa is selected from Leu, Ile, Val, Phe and Met
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (5)..(6)
<223> OTHER INFORMATION: Xaa is selected from any canonical amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (8)..(8)
<223> OTHER INFORMATION: Xaa is selected from any canonical amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (9)..(9)
<223> OTHER INFORMATION: Xaa is selected from Leu and Ile

<400> SEQUENCE: 30

Leu Xaa Xaa Xaa Xaa Xaa Leu Xaa Xaa
1 5

<210> SEQ ID NO 31
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: NES motif
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (2)..(3)
<223> OTHER INFORMATION: Xaa is selected from any canonical amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (4)..(4)
<223> OTHER INFORMATION: Xaa is selected from Leu, Ile, Val, Phe and Met

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<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (5)..(6)
<223> OTHER INFORMATION: Xaa is selected from any canonical amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (8)..(8)
<223> OTHER INFORMATION: Xaa is selected from any canonical amino acid

<400> SEQUENCE: 31

Leu Xaa Xaa Xaa Xaa Xaa Leu Xaa
1 5

<210> SEQ ID NO 32
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: NES motif
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (2)..(3)
<223> OTHER INFORMATION: Xaa is selected from any canonical amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (4)..(4)
<223> OTHER INFORMATION: Xaa is selected from Leu, Ile, Val, Phe and Met
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (5)..(7)
<223> OTHER INFORMATION: Xaa is selected from any canonical amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (9)..(9)
<223> OTHER INFORMATION: Xaa is selected from any canonical amino acid

<400> SEQUENCE: 32

Leu Xaa Xaa Xaa Xaa Xaa Xaa Leu Xaa
1 5

<210> SEQ ID NO 33
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: NES motif
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (2)..(4)
<223> OTHER INFORMATION: Xaa is selected from any canonical amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (5)..(5)
<223> OTHER INFORMATION: Xaa is selected from Leu, Ile, Val, Phe and Met
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (6)..(8)
<223> OTHER INFORMATION: Xaa is selected from any canonical amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (10)..(10)
<223> OTHER INFORMATION: Xaa is selected from any canonical amino acid
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (11)..(11)
<223> OTHER INFORMATION: Xaa is selected from Leu and Ile

<400> SEQUENCE: 33

Leu Xaa Xaa Xaa Xaa Xaa Xaa Xaa Leu Xaa Xaa
1 5 10

-continued

<210> SEQ ID NO 34
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: NES motif

<400> SEQUENCE: 34

Leu Pro Pro Leu Glu Arg Leu Thr Leu
1 5

<210> SEQ ID NO 35
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: NES motif

<400> SEQUENCE: 35

Ala Cys Pro Val Pro Leu Gln Leu Pro Pro Leu Glu Arg Leu Thr Leu
1 5 10 15

Asp

<210> SEQ ID NO 36
<211> LENGTH: 183
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: HA-PKR(2-174)
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (2)..(10)
<223> OTHER INFORMATION: HA
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (11)..(183)
<223> OTHER INFORMATION: human PKR(2-174)

<400> SEQUENCE: 36

Met Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Ala Gly Asp Leu Ser Ala
1 5 10 15

Gly Phe Phe Met Glu Glu Leu Asn Thr Tyr Arg Gln Lys Gln Gly Val
20 25 30

Val Leu Lys Tyr Gln Glu Leu Pro Asn Ser Gly Pro Pro His Asp Arg
35 40 45

Arg Phe Thr Phe Gln Val Ile Ile Asp Gly Arg Glu Phe Pro Glu Gly
50 55 60

Glu Gly Arg Ser Lys Lys Glu Ala Lys Asn Ala Ala Ala Lys Leu Ala
65 70 75 80

Val Glu Ile Leu Asn Lys Glu Lys Lys Ala Val Ser Pro Leu Leu Leu
85 90 95

Thr Thr Thr Asn Ser Ser Glu Gly Leu Ser Met Gly Asn Tyr Ile Gly
100 105 110

Leu Ile Asn Arg Ile Ala Gln Lys Lys Arg Leu Thr Val Asn Tyr Glu
115 120 125

Gln Cys Ala Ser Gly Val His Gly Pro Glu Gly Phe His Tyr Lys Cys
130 135 140

Lys Met Gly Gln Lys Glu Tyr Ser Ile Gly Thr Gly Ser Thr Lys Gln
145 150 155 160

Phe Met Glu Glu Leu Asn Thr Tyr Arg Gln Lys Gln Gly Val Val Leu
1 5 10 15

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Lys	Tyr	Gln	Glu	Leu	Pro	Asn	Ser	Gly	Pro	Pro	His	Asp	Arg	Arg	Phe
		20						25					30		
Thr	Phe	Gln	Val	Ile	Ile	Asp	Gly	Arg	Glu	Phe	Pro	Glu	Gly	Glu	Gly
		35					40					45			
Arg	Ser	Lys	Lys	Glu	Ala	Lys	Asn	Ala	Ala	Ala	Lys	Leu	Ala	Val	Glu
		50				55					60				
Ile	Leu	Asn	Lys	Glu	Lys	Lys	Ala	Val	Ser	Pro	Leu	Leu	Leu	Thr	Thr
65					70					75					80
Thr	Asn	Ser	Ser	Glu	Gly	Leu	Ser	Met	Gly	Asn	Tyr	Ile	Gly	Leu	Ile
			85						90					95	
Asn	Arg	Ile	Ala	Gln	Lys	Lys	Arg	Leu	Thr	Val	Asn	Tyr	Glu	Gln	Cys
			100					105					110		
Ala	Ser	Gly	Val	His	Gly	Pro	Glu	Gly	Phe	His	Tyr	Lys	Cys	Lys	Met
		115					120					125			
Gly	Gln	Lys	Glu	Tyr	Ser	Ile	Gly	Thr	Gly	Ser	Thr	Lys	Gln	Glu	Ala
	130					135					140				
Lys	Gln	Leu	Ala	Ala	Lys	Leu	Ala	Tyr	Leu	Gln	Ile	Leu	Ser	Glu	
145					150					155					

<210> SEQ ID NO 38
 <211> LENGTH: 173
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: (full) dsRNA-binding domain of PKR - M1

<400> SEQUENCE: 38

Ala	Gly	Asp	Leu	Ser	Ala	Gly	Phe	Phe	Met	Glu	Glu	Leu	Asn	Thr	Tyr
1				5					10					15	
Arg	Gln	Lys	Gln	Gly	Val	Val	Leu	Lys	Tyr	Gln	Glu	Leu	Pro	Asn	Ser
		20						25					30		
Gly	Pro	Pro	His	Asp	Arg	Arg	Phe	Thr	Phe	Gln	Val	Ile	Ile	Asp	Gly
		35					40					45			
Arg	Glu	Phe	Pro	Glu	Gly	Glu	Gly	Arg	Ser	Lys	Lys	Glu	Ala	Lys	Asn
		50				55					60				
Ala	Ala	Ala	Lys	Leu	Ala	Val	Glu	Ile	Leu	Asn	Lys	Glu	Lys	Lys	Ala
65					70					75					80
Val	Ser	Pro	Leu	Leu	Leu	Thr	Thr	Thr	Asn	Ser	Ser	Glu	Gly	Leu	Ser
			85						90					95	
Met	Gly	Asn	Tyr	Ile	Gly	Leu	Ile	Asn	Arg	Ile	Ala	Gln	Lys	Lys	Arg
		100						105					110		
Leu	Thr	Val	Asn	Tyr	Glu	Gln	Cys	Ala	Ser	Gly	Val	His	Gly	Pro	Glu
		115					120					125			
Gly	Phe	His	Tyr	Lys	Cys	Lys	Met	Gly	Gln	Lys	Glu	Tyr	Ser	Ile	Gly
	130					135					140				
Thr	Gly	Ser	Thr	Lys	Gln	Glu	Ala	Lys	Gln	Leu	Ala	Ala	Lys	Leu	Ala
145					150					155					160
Tyr	Leu	Gln	Ile	Leu	Ser	Glu	Glu	Thr	Ser	Val	Lys	Ser			
			165					170							

<210> SEQ ID NO 39
 <211> LENGTH: 2769
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:

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<223> OTHER INFORMATION: expression cassette for reporter gene
      NLS-iRFP-Flag-GFPY39->TAG-6His
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1)..(852)
<223> OTHER INFORMATION: promoter
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (949)..(984)
<223> OTHER INFORMATION: NLS
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (988)..(1929)
<223> OTHER INFORMATION: iRFP
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1930)..(2004)
<223> OTHER INFORMATION: 25 Helix
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (2011)..(2034)
<223> OTHER INFORMATION: Flag
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (2035)..(2748)
<223> OTHER INFORMATION: GFPY39 CDS
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (2149)..(2151)
<223> OTHER INFORMATION: Y39-encoding triplet
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (2749)..(2766)
<223> OTHER INFORMATION: 6His

<400> SEQUENCE: 39

ggcattgatt attgactagt tattaatagt aatcaattac ggggtcatta gttcatagcc      60
catatatgga gttccgcgtt acataactta cggtaaattgg cccgcctggc tgaccgccca      120
acgacccccg cccattgacg tcaataatga cgtatgttcc catagtaacg ccaataggga      180
ctttccattg acgtcaatgg gtggagtatt tacggtaaac tgcccacttg gcagtacatc      240
aagtgtatca tatgccaaagt ccgcccccta ttgacgtcaa tgacggtaaa tggcccgctt      300
ggcattatgc ccagtacatg accttacggg actttcctac ttggcagtac atctacgtat      360
tagtcatcgc tattaccatg gtgatgcggt tttggcagta caccaatggg cgtggatagc      420
ggtttgactc acggggattt ccaagtctcc accccattga cgtcaatggg agtttgtttt      480
ggcaccaaaa tcaacgggac ttccaaaat gtcgtaataa ccccgccccg ttgacgcaaa      540
tgggcggtag gcgtgtacgg tgggaggtct atataagcag agctcgttta gtgaaccgtc      600
agatcactag aagctttatt gcggtagttt atcacagtta aattgctaac gcagtcagtg      660
cttctgacac aacagtctcg aacttaagct gcagaagttg gtcgtgaggc actgggcagg      720
taagtatcaa gggtacaaga cagggttaag gagaccaata gaaactgggc ttgtcgagac      780
agagaagact cttgcgtttc tgataggcac ctattggtct tactgacatc cactttgcct      840
ttctctccac aggtgtccac tcccagttca attacagctc ttaaggctag agtacttaat      900
acgactcact ataggctagc caccatgggc cgcctggaaa gcaccccgcc gaaaaaaaaa      960
cgcaaagtgg aagatagcgc gagcgcggaa ggatccgctg ccaggcagcc tgacctcttg      1020
acctgcgacg atgagccgat ccatatcccc ggtgccatcc aaccgcatgg actgctgctc      1080
gcccctcgcc cgcacatgac gatcgttgcc ggcagcgaca accttcccga actcaccgga      1140
ctggcgatcg gcgcctgat cgcccgctct gcggccgatg tcttcgactc ggagacgcac      1200

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aaccgtctga cgatcgccctt ggccgagccc ggggcggccg tcggagcacc gatcactgtc 1260
ggcttcacga tgcgaaagga cgcaggcttc atcggtcctt ggcatcgcca tgatcagctc 1320
atcttctctg agctcgagcc tccccagcgg gacgtcgccg agccgcaggc gttcttcctg 1380
cgcaccaaca ggcgcctcgg ccgcctgcag gccgcgaaa ccttgaaaag cgctcgcc 1440
gccgcggcgc aagaggtgcg gaagattacc ggcttcgatc ggggtgatgat ctatcgcttc 1500
gcctccgact tcagcggcga agtgatcgca gaggatcggg gcgcgaggt cgagtcaaaa 1560
ctaggcctgc actatcctgc ctcaaccgtg ccggcgcagg cccgtcgggt ctataccatc 1620
aacccggtac ggatcattcc cgatatcaat tatcggccgg tgccggtcac cccagacctc 1680
aatccggtca ccgggcggcc gattgatctt agcttcgcca tcctgcgcag cgtctcgccc 1740
gtccatctgg aattcatgcg caacataggc atgcacggca cgatgtcgat ctcgattttg 1800
cgcggcgagc gactgtgggg attgatcgtt tgccatcacc gaacgcgta ctacgtcgat 1860
ctcgatggcc gccaacctg cgagctagtc gccagggtc tggcctggca gatcggcgtg 1920
atggaagagc tggcggaagc ggcggccaaa gaagcggcgg ccaaagaagc ggcggccaaa 1980
gaagcggcgg ccaaagcggc gcccggtacc gattacaaag atgatgatga taaagtgagc 2040
aagggcgagg agctgttcac cgggggtggt cccatcctgg tcgagctgga cggcgacgta 2100
aacggccaca agttcagcgt gtcggcgag gccgaggcgg atgccaccta gggcaagctg 2160
accctgaagt tcattctgac caccggcaag ctgcccgctc cctggccac cctcgtgacc 2220
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gacggcaact acaagaccgg cgcggagggt aagttcgagg gcgacacct ggtgaaccgc 2400
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tacaactaca acagccacaa cgtctatata atggccgaca agcagaagaa cggcatcaag 2520
gccaaettca agatccgcca caacatcgag gacggcagcg tgcagctcgc cgaccactac 2580
cagcagaaca ccccatcgg cgacggcccc gtgctgctgc ccgacaacca ctacctgagc 2640
acccagtccg ccctgagcaa agaccccaac gagaagcggc atcacatggt cctgctggag 2700
ttcgtgaccg ccgcgggat cactctcggc atggacgagc tgtacaagca tcaccatcac 2760
catcactga 2769

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<210> SEQ ID NO 40
<211> LENGTH: 3341
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: T7 RNA polymerase expression cassette for
        constitutive expression via CMV promoter
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1)..(588)
<223> OTHER INFORMATION: CMV promoter
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (690)..(3341)
<223> OTHER INFORMATION: CDS

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<400> SEQUENCE: 40

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gttgacattg attattgact agttattaat agtaatcaat tacgggggtca ttagttcata 60

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gcccataat	ggagttccgc	gttacataac	ttacggtaaa	tggcccgcct	ggtgaccgc	120
ccaacgaccc	cgcgccattg	acgtcaataa	tgacgtatgt	tcccatagta	acgccaatag	180
ggactttcca	ttgacgtcaa	tgggtggact	atttacggta	aactgcccac	ttggcagtac	240
atcaagtgtg	tcatatgcc	agtacgcccc	ctattgacgt	caatgacggg	aaatggccc	300
cctggcatta	tgcccagtac	atgaccttat	gggactttcc	tacttggcag	tacatctacg	360
tattagtcat	cgctattacc	atgggtgatg	ggttttggca	gtacatcaat	gggcgtggat	420
agcggtttga	ctcacgggga	tttccaagtc	tccaccccat	tgacgtcaat	gggagtttgt	480
tttggcacca	aaatcaacgg	gactttccaa	aatgtcgtaa	caactccgcc	ccattgacgc	540
aaatgggccc	taggcgtgta	cgggtggagg	tctatataag	cagagctctc	tggttaacta	600
gagaaccac	tgcttactgg	cttatcgaaa	ttaatacgac	tactatagg	gagaccaag	660
ctggctagcg	tttaacttta	agcttcacca	tgaacacgat	taacatcgct	aagaacgact	720
tctctgacat	cgaactggct	gctatcccg	tcaacactct	ggctgaccat	tacggtgagc	780
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tccgcaagat	gtttgagcgt	caacttaaa	ctgggtgagg	tgcggataac	gctgccgcca	900
agcctctcat	cactacccta	ctccctaaga	tgattgcacg	catcaacgac	tggtttgagg	960
aagtgaagc	taagcgcggc	aagcggccga	cagccttcca	gttcctgcaa	gaaatcaagc	1020
cggagccgt	agcgtacatc	accattaaga	ccactctggc	ttgcctaacc	agtgtgaca	1080
atacaaccgt	tcaggctgta	gcaagcgcaa	tgggtcgggc	cattgaggac	gaggctcgct	1140
tcggtcgtat	ccgtgacctt	gaagctaagc	acttcaagaa	aaacgttgag	gaacaactca	1200
acaagcgcgt	agggcacgct	tacaagaaag	catttatgca	agttgtcgag	gctgacatgc	1260
tctctaaggg	tctactcggt	ggcgaggcgt	ggtcttcgtg	gcataaggaa	gactctattc	1320
atgtaggagt	acgctgcac	gagatgctca	ttgagtcaac	cggaatgggt	agcttacacc	1380
gccccaatgc	tggcgtagta	ggccaagact	ctgagactat	cgaactcgca	cctgaatacg	1440
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gcgtagtcc	tcctaagccg	tggactggca	ttactgggtg	tggctattgg	gctaacggtc	1560
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acaagaaagt	cctagcgggc	gccaacgtaa	tcaccaagtg	gaagcattgt	ccggctcgagg	1740
acatccctgc	gattgagcgt	gaagaactcc	cgatgaaacc	ggaagacatc	gacatgaatc	1800
ctgaggctct	caccgcgtgg	aaacgtgctg	ccgctgctgt	gtaccgcaag	gacaaggctc	1860
gcaagtctcg	ccgtatcagc	cttgagttca	tgcttgagca	agccaataag	tttgctaacc	1920
ataaggccat	ctggttcctt	tacaacatgg	actggcgccg	tcgtgtttac	gctgtgtcaa	1980
tgttcaaccc	gcaaggtaac	gatatgacca	aaggactgct	tacgtggcgc	aaaggtaaac	2040
caatcggtaa	ggaaggttac	tactggctga	aaatccacgg	tgcaaacgtg	gcgggtgtcg	2100
ataaggttcc	gttccttgag	cgcacaaagt	tcattgagga	aaaccacgag	aacatcatgg	2160
cttgcgctaa	gtctccactg	gagaacactt	ggtgggctga	gcaagattct	ccgttctgct	2220
tccttgctgt	ctgctttgag	tacgtggggg	tacagcacca	cggcctgagc	tataactgct	2280
cccttcgct	ggcgtttgac	gggtcttgct	ctggcatcca	gcacttctcc	gcgatgctcc	2340

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acgggattgt tgctaagaaa gtcaacgaga ttctacaagc agacgcaatc aatgggaccg 2460
ataacgaagt agttaccgtg accgatgaga acactgggtg aatctctgag aaagtcaagc 2520
tgggcactaa ggcactggct ggtcaatggc tggtctacgg tgttactcgc agtgtgacta 2580
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tggaagatac cattcagcca gctattgatt cgggcaaggg tctgatgttc actcagccga 2700
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ctgcggttga agcaatgaac tggcttaagt ctgctgctaa gctgctggct gctgagggtca 2820
aagataagaa gactggagag attcttcgca agcgttgcgc tgtgcattgg gtaactctg 2880
atggtttccc tgtgtggcag gaatacaaga agcctattca gacgcgttg aacctgatgt 2940
tcctcgggtca gttccgctta cagcctacca ttaacaccaa caaagatagc gagattgatg 3000
cacacaaaca ggagtctggt atcgtctcta actttgtaca cagccaagac ggtagccacc 3060
ttcgtaaagc tgtagtgtgg gcacacgaga agtacggaat cgaatctttt gcaactgattc 3120
acgactcctt cggtaaccatt ccggctgacg ctgcgaacct gttcaaagca gtgcgcgaaa 3180
ctatgggtga cacatatgag tcttgtgatg tactgggtga tttctacgac cagttcgtg 3240
accagttgca cgagtctcaa ttggacaaaa tgccagcact tccggctaaa ggtaacttga 3300
acctccgtga catcttagag tcggacttcg cgttcgcgta a 3341

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<210> SEQ ID NO 41
<211> LENGTH: 3494
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: NES-T7 RNA polymerase expression cassette
        containing "T-REx" promoter comprising two tetO sequences
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1)..(584)
<223> OTHER INFORMATION: promoter
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (586)..(604)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (607)..(625)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (798)..(845)
<223> OTHER INFORMATION: NES
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (846)..(3494)
<223> OTHER INFORMATION: CDS

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<400> SEQUENCE: 41

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gacattgatt attgactagt tattaatagt aatcaattac ggggtcatta gttcatagcc 60
catatatgga gttccgcgtt acataactta cggtaaatgg cccgcctggc tgaccgccca 120
acgacccccg cccattgacg tcaataatga cgtatgttcc catagtaacg ccaataggga 180
ctttccattg acgtcaatgg gtggagtatt tacggtaaac tgcccacttg gcagtacatc 240
aagtgtatca tatgccaagt acgcccccta ttgacgtcaa tgacggtaaa tggcccgct 300
ggcattatgc ccagtacatg accttatggg actttcctac ttggcagtac atctacgtat 360

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tagtcatcgc	tattaccatg	gtgatgcggt	tttggcagta	catcaatggg	cgtggatagc	420
ggtttgactc	acggggattt	ccaagtctcc	acccattga	cgtcaatggg	agtttgttt	480
ggcaccaaaa	tcaacgggac	tttccaaaat	gtcgtaacaa	ctccgcccc	ttgacgcaaa	540
tgggcggtag	gcgtgtacgg	tgggaggtct	atataagcag	agctctccct	atcagtgata	600
gagatctccc	tatcagtgat	agagatcgtc	gacgagctcg	tttagtgaac	cgtcagatcg	660
cctggagacg	ccatccacgc	tgttttgacc	tccatagaag	acaccgggac	cgatccagcc	720
tccggactct	agcgtttaaa	catcacaagt	ttgtacaaaa	aagcaggcgc	tagcgtttaa	780
acttaagctt	caccatgtgc	ccggtgccgc	tgcagctgcc	gccgctggaa	cgctgaccc	840
tggataaac	gattaacatc	gctaagaacg	acttctctga	catcgaactg	gctgctatcc	900
cgttcaacac	tctggctgac	cattacggtg	agcgtttagc	tcgcgaacag	ttggcccttg	960
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ttaacattgc	gcaaaacacc	gcatggaaaa	tcaacaagaa	agtcctagcg	gtcgccaacg	1860
taatcacc	gtggaagcat	tgtccggtcg	aggacatccc	tgcgattgag	cgtgaagaac	1920
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agaacactgg tgaatctct gagaaagtca agctgggcac taaggcactg gctggccaat 2700
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ggtccaaaga gttcgggttc cgtcaacaag tgctggaaga taccattcag ccagctattg 2820
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tcgcgttcgc gtaa 3494

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<210> SEQ ID NO 42
<211> LENGTH: 3169
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: NES-T7 RNA polymerase expression cassette
        containing "Tet-On" promoter comprising 8 tetO sequences
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1)..(458)
<223> OTHER INFORMATION: promoter
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (18)..(36)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (54)..(72)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (90)..(108)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (126)..(144)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (162)..(180)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (198)..(216)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (234)..(252)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (270)..(288)
<223> OTHER INFORMATION: tetO
<220> FEATURE:

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<221> NAME/KEY: misc_feature
 <222> LOCATION: (473)..(520)
 <223> OTHER INFORMATION: NES
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (521)..(3169)
 <223> OTHER INFORMATION: CDS

<400> SEQUENCE: 42

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cagtgataga gaacgtatga agagtttact ccctatcagt gatagagaac gtatgcagac	120
tttactccct atcagtgata gagaacgtat aaggagttta ctccctatca gtgatagaga	180
acgtatgacc agtttactcc ctatcagtga tagagaacgt atctacagtt tactccctat	240
cagtgataga gaacgtatat ccagtttact ccctatcagt gatagagaac gtatacgctt	300
taggcgtgta cgggtggcgc ctataaaagc agagctcgtt tagtgaaccg tcagatcgcc	360
tggagcaatt ccacaacact tttgtcttat accaacttcc cgtaccactt cctaccctcg	420
aaagtccata tggctagcct gcagggatcc agatatcaag cttgccacca tgtgcccggt	480
gcgctgcag ctgcgcgcgc tggaaacgcct gacctggat aacacgatta acatcgctaa	540
gaacgacttc tctgacatcg aactggctgc tatcccgctc aacactctgg ctgaccatta	600
cggtgagcgt ttagctcgcg aacagttggc ccttgagcat gagtcttacg agatgggtga	660
agcacgcttc cgcaagatgt ttgagcgtca acttaaagct ggtgaggttg cggataacgc	720
tgccgccaaag cctctcatca ctaccctact ccctaagatg attgcacgca tcaacgactg	780
gtttgaggaa gtgaaagcta agcgcggcaa gcgcccgaca gccttcaggt tcctgcaaga	840
aatcaagccg gaagccgtag cgtacatcac cattaagacc actctggctt gcctaaccag	900
tgctgacaat acaaccgttc aggtctgtac aagcgcaatc ggtcgggcca ttgaggacga	960
ggctcgcttc ggtcgtatcc gtgacctga agctaagcac ttcaagaaaa acgttgagga	1020
acaactcaac aagcgcgtag ggcacgtcta caagaaagca tttatgcaag ttgtcgaggc	1080
tgacatgctc tctaagggtc tactcgggtg cgaggcgtgg tcttcgtggc ataaggaaga	1140
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cgaagacgtt tacatgcctg aggtgtacaa agcgattaac attgcgcaaa acaccgcatg	1500
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tgctaaccat aagggcatct ggttccctta caacatggac tggcgcgggtc gtgtttacgc	1800
tgtgtcaatg ttcaaccgcg aaggtaacga tatgacccaa ggactgctta cgctggcgaa	1860
aggtaaacca atcggtaagg aaggttacta ctggctgaaa atccacgggtg caaactgtgc	1920
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cctgatgttc ctccgtcagt tccgcttaca gcctaccatt aacaccaaca aagatagcga	2820
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<210> SEQ ID NO 43

<211> LENGTH: 3389

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: NES-T7 RNA polymerase expression cassette for constitutive expression via CMV promoter

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (1)..(588)

<223> OTHER INFORMATION: CMV promoter

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (693)..(740)

<223> OTHER INFORMATION: NES

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (741)..(3389)

<223> OTHER INFORMATION: CDS

<400> SEQUENCE: 43

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ccaacgaccc ccgcccattg acgtcaataa tgacgtatgt tcccatagta acgccaatag	180
ggactttcca ttgacgtcaa tgggtgggact atttacggta aactgcccac ttggcagtac	240
atcaagtgtg tcatatgcca agtacgcccc ctattgacgt caatgacggg aaatggcccc	300
cctggcatta tgcccagtac atgaccttat gggactttcc tacttggcag tacatctacg	360
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agcggtttga ctcacgggga ttccaagtc tccaccccat tgacgtcaat gggagtttgt	480
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aaatgggcgg taggcgtgta cgggtgggagg tctatataag cagagctctc tggctaacta	600
gagaaccac tgcttactgg cttatcgaaa ttaatacgac tcaactatagg gagacccaag	660
ctggctagcg tttaaactta agcttcacca tgtgcccggg gccgctgcag ctgccgccgc	720
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aacagttggc ccttgagcat gagtcttacg agatgggtga agcacgcttc cgcaagatgt	900
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ttaccgtgac cgatgagaac actggtgaaa tctctgagaa agtcaagctg ggcactaagg	2580
cactggctgg tcaatggctg gcttacggtg ttactcgagc tgtgactaag cgttcagtca	2640
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tagtgtgggc acacgagaag tacggaatcg aatcttttgc actgattcac gactccttcg 3180
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tcttagagtc ggacttcgcg ttcgcgtaa 3389

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<210> SEQ ID NO 44
<211> LENGTH: 2054
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: PylRSF expression cassette for constitutive
expression via CMV promoter
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1)..(588)
<223> OTHER INFORMATION: CMV promoter
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (690)..(2054)
<223> OTHER INFORMATION: CDS
<400> SEQUENCE: 44

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ggactttcca ttgacgtcaa tgggtggact atttacggtg aactgcccac ttggcagtac 240
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cctggcatta tgcccagtac atgaccttat gggactttcc tacttggcag tacatctacg 360
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aaatggcgcg taggcgtgta cgggtgggag tctatataag cagagctctc tggctaacta 600
gagaaccac tgcttactgg cttatcgaaa ttaatacgac tcaatagtg gagaccacag 660
ctggctagcg tttaaactta agcttcacca tggataaaaa accgctgaat accctgatct 720
ctgctactgg tctgtggatg agtcgtaccg gaaccattca taaaatcaaa caccacgagg 780
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gcgctcttc tcgtacagca cgtgcactgc gtcaccacaa atatcgtaa acctgtaaac 900
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gaagcaaatt ctctccggcc attcctgttt ctaccaggga gtcggtttct gttccagcaa	1140
gtgtgagcac cagcattagc agtattagca ccggtgccac cgctagcgcc ctgggttaaag	1200
gcaataccaa tccgattaca agcatgtctg ccccggttca agcatcagct ccagcactga	1260
caaaatccca aaccgatcgt ctggagggtc tgctgaatcc gaaagacgaa atcagcctga	1320
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tgcaacaaat ctatgccgaa gaacgtgaga actatctggg gaaactggaa cgtgaaatca	1440
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gcattgactt caaaattgtg ggcgacagct gtatgggtgt tggcgacacc ctggatgtca	1860
tgcacggcga cctggaactg tctagtgccg ttgtggggcc aatcccgtg gatcgtgagt	1920
ggggatcga caaaccttg atcgggtcgg gttttggtct ggagcgtctg ctgaaagtaa	1980
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<210> SEQ ID NO 45
 <211> LENGTH: 2105
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: NES-PylRSF expression cassette for
 constitutive expression via CMV promoter
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (1)..(588)
 <223> OTHER INFORMATION: CMV promoter
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (696)..(743)
 <223> OTHER INFORMATION: NES
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (744)..(2105)
 <223> OTHER INFORMATION: CDS

<400> SEQUENCE: 45

gttgacattg attattgact agttattaat agtaatcaat tacgggggtca ttagttcata	60
gcccataatat ggagttccgc gttacataac ttacggtaaa tggcccgccct ggctgaccgc	120
ccaacgaccc ccgcccattg acgtcaataa tgacgtatgt tcccatagta acgccaatag	180
ggactttcca ttgacgtcaa tgggtgggact atttacggta aactgcccac ttggcagtac	240
atcaagtgtg tcatatgccca agtacgcccc ctattgacgt caatgacggg aaatggcccc	300
cctggcatta tgcccagtag atgaccttat gggactttcc tacttggcag tacatctacg	360
tattagtcat cgctattacc atgggtgatgc ggttttggca gtacatcaat gggcgtggat	420
agcgggtttga ctacggggga ttccaagtc tccaccccat tgacgtcaat gggagtttgt	480

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tttggcacca aaatcaacgg gactttccaa aatgtogtaa caactccgcc ccattgacgc 540
aaatgggcgg taggcgtgta cgggtgggagg tctatataag cagagctctc tggetaacta 600
gagaaccac tgcttactgg cttatcgaaa ttaatacgac tcaactatagg gagaccacaag 660
ctggctagcg tttaaactta agcttcacca tggcgtgccc ggtgccgctg cagctgccgc 720
cgctggaacg cctgaccctg gatgataaaa aaccgctgaa taccctgac tctgctactg 780
gtctgtggat gagtcgtacc ggaaccattc ataaatcaa acaccacgag gtttagccgtt 840
cgaaaatcta tattgagatg gcgtgtggcg atcatctggt tgtgaacaat agccgctctt 900
ctcgtacagc acgtgcactg cgtcaccaca aatatcgtaa aacctgtaa cgttgccgtg 960
tgtccgatga ggatctgaac aaattcctga caaaagccaa tgaggaccaa acaagcgtga 1020
aagtgaagt cgtagcgct cctaccgta ctaaaaaagc aatgccgaaa tccgttgctc 1080
gtgccctaa accactggaa aacctgaag cagcacaggc acagccgtct ggaagcaaat 1140
tctctccggc cattcctgtt tctaccagg agtccgttct tggtccagca agtgtgagca 1200
ccagcattag cagtattagc accggtgcc cgcgtagcgc cctgggttaa ggcaatacca 1260
atccgattac aagcatgtct gccccgggtc aagcatcagc tccagcactg acaaaatccc 1320
aaaccgatcg tctggagggt ctgctgaatc cgaaagacga aatcagcctg aattccggca 1380
aaccgtttcg tgaactggag agcgaactgc tgtcacgtcg taaaaagac ctgcaacaaa 1440
tctatgccga agaacgtgag aactatctgg ggaaactgga acgtgaaatc acccgctttt 1500
tcgtggatcg tggctttctg gagatcaaat ccccgattct gattcctctg gagtatatcg 1560
agcgtatggg catcgacaat gataccgaac tgagcaaaac aattttccgt gtggataaaa 1620
acttctgtct gcgccctatg ctagcaccaa atctggctaa ctatctgcgc aaactggacc 1680
gtgccctgcc tgatcctatc aaaatcttcg agatcgccc gtgttatcgt aaagagtcgc 1740
acggtaaaaga acatctggag gagtttacca tgctgaactt ttgccaaatg gggttcagggt 1800
gtactcgtga gaacctggaa agcatcatca ccgattttct gaaccacctg ggcattgact 1860
tcaaaattgt gggcgacagc tgtatggtgt ttggcgacac cctggatgtc atgcacggcg 1920
acctggaact gtctagtgc gttgtgggcc caatcccgct ggatcgtgag tggggtatcg 1980
acaaaccttg gatcggtgcg ggttttggtc tggagcgtct gctgaaagta aaacacgact 2040
tcaagaacat caaacgtgct gcacgttccg agtcctatta caatgggtatt tctactaacc 2100
tgtaa 2105

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<210> SEQ ID NO 46
<211> LENGTH: 2194
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: NES-PylRSF expression cassette containing
      "T-REx" promoter comprising two tetO sequences
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1)..(584)
<223> OTHER INFORMATION: promoter
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (586)..(604)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (607)..(625)

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<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (785)..(832)
<223> OTHER INFORMATION: NES
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (833)..(2194)
<223> OTHER INFORMATION: CDS

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<400> SEQUENCE: 46

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gacattgatt attgactagt tattaatagt aatcaattac ggggtcatta gttcatagcc      60
catatatgga gttccgcgtt acataactta cggtaaattgg cccgcctggc tgaccgccca      120
acgacccccg cccattgacg tcaataatga cgtatgttcc catagtaacg ccaataggga      180
ctttccattg acgtcaatgg gtggagtatt tacggtaaac tgcccacttg gcagtacatc      240
aagtgtatca tatgccaagt acgcccccta ttgacgtcaa tgacggtaaa tggccccgct      300
ggcattatgc ccagtacatg accttatggg actttectac ttggcagtac atctacgtat      360
tagtcatcgc tattaccatg gtgatgcggg ttggcagta catcaatggg cgtggatagc      420
ggtttgactc acggggattt ccaagtctcc accccattga cgtcaatggg agtttgtttt      480
ggcaccaaaa tcaacgggac ttccaaaat gtcgtaacaa ctccgcccca ttgacgcaaa      540
tgggcggtag gcgtgtacgg tgggaggtct atataagcag agctctccct atcagtgata      600
gagatctccc tatcagtgat agagatcgtc gacgagctcg tttagtgaac cgtcagatcg      660
cctggagacg ccataccacg tgttttgacc tccatagaag acaccgggac cgatccagcc      720
tccgactctc agcgtttaaa catcacaagt ttgtacaaaa aagcaggcgc tagacacccat      780
ggcgtgcccc gtgcgcgtgc agctgcgcgc gctggaacgc ctgaccctgg atgataaaaa      840
accgctgaat accctgatct ctgctactgg tctgtggatg agtcgtaccg gaaccattca      900
taaaatcaaa caccacgagg ttagccgttc gaaaatctat attgagatgg cgtgtggcga      960
tcatctggtt gtgaacaata gccgctcttc tcgtacagca cgtgcaactgc gtcaccacaa      1020
atatcgtaaa acctgtaaac gttgccgtgt gtccgatgag gatctgaaca aattcctgac      1080
aaaagccaat gaggacaaaa caagcgtgaa agtgaaagtc gttagcgctc ctaccgtac      1140
taaaaaagca atgccgaaat ccgttgctcg tgcccctaaa cactggaaa aactgaagc      1200
agcacaggca cagccgtctg gaagcaaat ctctccggcc attcctgttt ctaccagga      1260
gtccgtttct gttccagcaa gtgtgagcac cagcattagc agtattagca ccggtgccac      1320
cgctagcgcc ctggttaaa gcaataccaa tccgattaca agcatgtctg ccccggttca      1380
agcatcagct ccagcactga caaaatccca aaccgatcgt ctggaggttc tgetgaaatcc      1440
gaaagacgaa atcagcctga attccggcaa accgtttcgt gaactggaga gcgaactgct      1500
gtcacgtcgt aaaaaagacc tgcaacaaat ctatgccgaa gaacgtgaga actatctggg      1560
gaaactggaa cgtgaaatca cccgcttttt cgtggatcgt ggctttctgg agatcaaatc      1620
cccgattctg attcctctgg agtatatcga gcgtatgggc atcgacaatg ataccgaact      1680
gagcaaacaa attttccgtg tggataaaaa cttctgtctg cgccctatgc tagcaccaaa      1740
tctggetaac tatctgcgca aactggaccg tgccctgcct gatcctatca aaatcttcga      1800
gatcggcccc tgttatcgta aagagtccga cggtaaagaa catctggagg agtttaccat      1860
gctgaacttt tgccaaatgg gttcagggtg tactcgtgag aacctggaaa gcatcatcac      1920

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cgattttctg aaccacctgg gcattgactt caaaattgtg ggcgacagct gtatggtgtt	1980
tggcgacacc ctggatgtca tgcacggcga cctggaactg tctagtgcg ttgtggggcc	2040
aatcccgctg gatcgtgagt ggggtatcga caaaccttgg atcggtgcgg gttttggtct	2100
ggagcgtctg ctgaaagtaa aacacgactt caagaacatc aaacgtgctg cacgttccga	2160
gtcctattac aatggtatct ctactaacct gtaa	2194

<210> SEQ ID NO 47
 <211> LENGTH: 1883
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: NES-PylRSF expression cassette containing
 "tet-On" promoter comprising 8 tetO sequences
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (1)..(458)
 <223> OTHER INFORMATION: promoter
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (18)..(36)
 <223> OTHER INFORMATION: tetO
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (54)..(72)
 <223> OTHER INFORMATION: tetO
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (90)..(108)
 <223> OTHER INFORMATION: tetO
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (126)..(144)
 <223> OTHER INFORMATION: tetO
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (162)..(180)
 <223> OTHER INFORMATION: tetO
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (198)..(216)
 <223> OTHER INFORMATION: tetO
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (234)..(252)
 <223> OTHER INFORMATION: tetO
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (270)..(288)
 <223> OTHER INFORMATION: tetO
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (474)..(521)
 <223> OTHER INFORMATION: NES
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (522)..(1883)
 <223> OTHER INFORMATION: CDS

<400> SEQUENCE: 47

caattgctcg agtttactcc ctatcagtga tagagaacgt atgtcgactt tactccctat	60
cagtgataga gaacgtatga agagtttact ccctatcagt gatagagaac gtatgcagac	120
tttactccct atcagtgata gagaacgtat aaggagtta ctcctatca gtgatagaga	180
acgtatgacc agtttactcc ctatcagtga tagagaacgt atctacagtt tactccctat	240
cagtgataga gaacgtatat ccagtttact ccctatcagt gatagagaac gtatacgctt	300
taggcgtgta cggtgggcgc ctataaaagc agagctcggt tagtgaaccg tcagatcgcc	360

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tggagcaatt ccacaacact ttgtcttat accaactttc cgtaccactt cctaccctcg 420
aaagtccata tggctagcct gcagggatcc agatatcaag cttcaccatg gcgtgcccgg 480
tgccgctgca gctgccgccc ctggaacgcc tgaccctgga tgataaaaaa ccgctgaata 540
ccctgatctc tgctactggt ctgtggatga gtcgtaccgg aaccattcat aaaatcaaac 600
accacgaggt tagccgttcg aaaatctata ttgagatggc gtgtggcgat catctggttg 660
tgaacaatag ccgctcttct cgtacagcac gtgcactgcg tcaccacaaa tatcgtaaaa 720
cctgtaaacg ttgccgtgtg tccgatgagg atctgaacaa attcctgaca aaagccaatg 780
aggaccaaac aagcgtgaaa gtgaaagtcg ttagcgctcc taccgctact aaaaaagcaa 840
tgccgaaatc cggtgctcgt gccccataac cactggaaaa cactgaagca gcacaggcac 900
agccgtctgg aagcaaattc tctccggcca ttctgtttc taccaggag tccgtttctg 960
ttccagcaag tgtgagcacc agcattagca gtattagcac cggtgccacc gctagcgccc 1020
tggttaaagg caataccaat ccgattacaa gcatgtctgc cccggttcaa gcatcagctc 1080
cagcactgac aaaatcccaa accgatcgtc tggaggttct gctgaatccg aaagcggaaa 1140
tcagcctgaa ttccggcaaa ccgtttcgtg aactggagag cgaactgctg tcacgtcgta 1200
aaaaagacct gcaacaaatc tatgccgaag aacgtgagaa ctatctgggg aaactggaac 1260
gtgaaatcac ccgcttttct gtggatcgtg gctttctgga gatcaaatcc ccgattctga 1320
ttctctgga gtatatcgag cgtatgggca tcgacaatga taccgaactg agcaacaaaa 1380
ttttccgtgt ggataaaaac ttctgtctgc gccctatgct agcaccaaat ctggctaact 1440
atctgcgcaa actggaccgt gccctgctg atcctatcaa aatcttcgag atcgcccggt 1500
gttatcgtaa agagtccgac ggtaagaac atctggagga gtttaccatg ctgaactttt 1560
gcaaatggg ttcagggtgt actcgtgaga acctggaaag catcatcacc gattttctga 1620
accacctggg cattgacttc aaaattgtgg gcgacagctg tatgggtgtt ggcgacacc 1680
tggatgtcat gcacggcgac ctggaactgt ctagtgccgt tgtgggccc atcccgtgg 1740
atcgtgagtg gggatcgac aaacctgga tcggcgccgg ttttggctg gagcgtctgc 1800
tgaaagtaaa acacgacttc aagaacatca aacgtgctgc acgttccgag tcctattaca 1860
atggtatttc tactaacctg taa 1883

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<210> SEQ ID NO 48
<211> LENGTH: 226
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: trnAPyl expression cassette for expression via
T7 promoter
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1)..(17)
<223> OTHER INFORMATION: T7 promoter
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (18)..(86)
<223> OTHER INFORMATION: trnA(Pyl)
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (98)..(226)
<223> OTHER INFORMATION: T7term
<400> SEQUENCE: 48

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taatacgaact cactatagga aacctgatca tgtagatcga atggactcta aatccgttca	60
gccggggttag attcccgggg ttcccggttt tgccgcctg ctaacaaagc ccgaaaggaa	120
gctgagttgg ctgctgccac cgctgagcaa taactagcat aaccctctgg ggcctctaaa	180
cggtctctga ggggtttttt gctgaaagga ggaactatat ccgat	226

```

<210> SEQ ID NO 49
<211> LENGTH: 295
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: tRNAPyl-HDV ribozyme expression cassette for
expression via T7 promoter
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1)..(17)
<223> OTHER INFORMATION: T7 promoter
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (18)..(86)
<223> OTHER INFORMATION: tRNA(Pyl)
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (87)..(154)
<223> OTHER INFORMATION: HDV
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (167)..(295)
<223> OTHER INFORMATION: T7term

```

<400> SEQUENCE: 49

taatacgact	cactatagga	aacctgatca	tgtagatcga	atggactcta	aatccgttca	60
gccggggttag	attcccgggg	tttcggcta	gccatggtcc	cagcctcctc	gctggcggt	120
agtgggcaac	atgcttcggc	atggcgcaatg	ggactttaaa	ctcgagctgc	taacaaagcc	180
cgaagggaag	ctgagttggc	tgtgtgccacc	gctgagcaat	aactagcata	accctctggg	240
gcctctaaac	gggtcttgag	gggttttttg	ctgaaaggag	gaactataac	cggat	295

```
<210> SEQ ID NO 50
<211> LENGTH: 669
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: tRNAPyl expression cassette containing a
variant of the human U6 promoter modified to comprise 8 tetO
before and two tetO sequences after the U6 promoter
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (24)..(42)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (60)..(78)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (96)..(114)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (132)..(150)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (168)..(186)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
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<222> LOCATION: (204)..(222)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (240)..(285)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (276)..(294)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (337)..(537)
<223> OTHER INFORMATION: U6 promoter
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (540)..(558)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (568)..(586)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (589)..(657)
<223> OTHER INFORMATION: tRNA(Pyl)

<400> SEQUENCE: 50

gaattccaat tgctcgagtt tactccctat cagtgataga gaacgtatgt cgactttact      60
ccctatcagt gatagagaac gatatgaagag ttctactccct atcagtgata gagaacgtat      120
gcagacttta ctccctatca gtgatagaga acgtataagg agtttactcc ctatcagtga      180
tagagaacgt atgaccagtt tactccctat cagtgataga gaacgtatct acagtttact      240
ccctatcagt gatagagaac gatatatccag ttctactccct atcagtgata gagaacgtat      300
gtcgaggtag gcgtgtacgg tcaaggtcgg gcaggaagag ggcctatttc ccatgattcc      360
ttcatatttg catatacgat acaaggctgt tagagagata attagaatta atttgactgt      420
aaacacaaag atattagtac aaaatacgtg acgtagaaaag taataatttc ttgggtagtt      480
tgcagtttta aaattatggt ttaaaatgga ctatcatatg cttaccgtaa cttgaaagtt      540
ccctatcagt gatagagatt atatatctcc ctatcagtga tagagaccgg aaacctgac      600
atgtagatcg aatggactct aaatccgttc agccgggtta gattcccggg gtttccggtt      660
tttggatcc                                     669

<210> SEQ ID NO 51
<211> LENGTH: 178
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: tRNAPyl expression cassette containing a
variant of the human H1 promoter modified to comprise 2 tetO
sequences
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1)..(99)
<223> OTHER INFORMATION: H1 promoter
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (49)..(67)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (77)..(95)
<223> OTHER INFORMATION: tetO
<220> FEATURE:
<221> NAME/KEY: misc_feature

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<222> LOCATION: (100)..(168)
 <223> OTHER INFORMATION: tRNA(Pyl)

<400> SEQUENCE: 51

```
atatttgcac gtcgctatgt gttctgggaa atcaccataa acgtgaaatc cctatcagtg      60
atagagactt ataagttccc tatcagtgat agagatcccg gaaacctgat catgtagatc      120
gaatggactc taaatccggt cagccggggt agattcccg gggttccgaa gctttttt      178
```

<210> SEQ ID NO 52
 <211> LENGTH: 248
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: TetOn3G protein
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <223> OTHER INFORMATION: TetOn3G protein

<400> SEQUENCE: 52

```
Met Ser Arg Leu Asp Lys Ser Lys Val Ile Asn Ser Ala Leu Glu Leu
1          5          10          15
Leu Asn Gly Val Gly Ile Glu Gly Leu Thr Thr Arg Lys Leu Ala Gln
20          25          30
Lys Leu Gly Val Glu Gln Pro Thr Leu Tyr Trp His Val Lys Asn Lys
35          40          45
Arg Ala Leu Leu Asp Ala Leu Pro Ile Glu Met Leu Asp Arg His His
50          55          60
Thr His Ser Cys Pro Leu Glu Gly Glu Ser Trp Gln Asp Phe Leu Arg
65          70          75          80
Asn Asn Ala Lys Ser Tyr Arg Cys Ala Leu Leu Ser His Arg Asp Gly
85          90          95
Ala Lys Val His Leu Gly Thr Arg Pro Thr Glu Lys Gln Tyr Glu Thr
100         105         110
Leu Glu Asn Gln Leu Ala Phe Leu Cys Gln Gln Gly Phe Ser Leu Glu
115         120         125
Asn Ala Leu Tyr Ala Leu Ser Ala Val Gly His Phe Thr Leu Gly Cys
130         135         140
Val Leu Glu Glu Gln Glu His Gln Val Ala Lys Glu Glu Arg Glu Thr
145         150         155         160
Pro Thr Thr Asp Ser Met Pro Pro Leu Leu Lys Gln Ala Ile Glu Leu
165         170         175
Phe Asp Arg Gln Gly Ala Glu Pro Ala Phe Leu Phe Gly Leu Glu Leu
180         185         190
Ile Ile Cys Gly Leu Glu Lys Gln Leu Lys Cys Glu Ser Gly Gly Pro
195         200         205
Thr Asp Ala Leu Asp Asp Phe Asp Leu Asp Met Leu Pro Ala Asp Ala
210         215         220
Leu Asp Asp Phe Asp Leu Asp Met Leu Pro Ala Asp Ala Leu Asp Asp
225         230         235         240
Phe Asp Leu Asp Met Leu Pro Gly
245
```

<210> SEQ ID NO 53
 <211> LENGTH: 471
 <212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Methanosarcina mazei PylRS Y306A, Y384F mutant
(PylRSAF)_amino acid sequence

<400> SEQUENCE: 53

```

Met Ala Cys Pro Val Pro Leu Gln Leu Pro Pro Leu Glu Arg Leu Thr
 1           5           10           15
Leu Asp Asp Lys Lys Pro Leu Asn Thr Leu Ile Ser Ala Thr Gly Leu
 20           25           30
Trp Met Ser Arg Thr Gly Thr Ile His Lys Ile Lys His His Glu Val
 35           40           45
Ser Arg Ser Lys Ile Tyr Ile Glu Met Ala Cys Gly Asp His Leu Val
 50           55           60
Val Asn Asn Ser Arg Ser Ser Arg Thr Ala Arg Ala Leu Arg His His
 65           70           75           80
Lys Tyr Arg Lys Thr Cys Lys Arg Cys Arg Val Ser Asp Glu Asp Leu
 85           90           95
Asn Lys Phe Leu Thr Lys Ala Asn Glu Asp Gln Thr Ser Val Lys Val
100          105          110
Lys Val Val Ser Ala Pro Thr Arg Thr Lys Lys Ala Met Pro Lys Ser
115          120          125
Val Ala Arg Ala Pro Lys Pro Leu Glu Asn Thr Glu Ala Ala Gln Ala
130          135          140
Gln Pro Ser Gly Ser Lys Phe Ser Pro Ala Ile Pro Val Ser Thr Gln
145          150          155          160
Glu Ser Val Ser Val Pro Ala Ser Val Ser Thr Ser Ile Ser Ser Ile
165          170          175
Ser Thr Gly Ala Thr Ala Ser Ala Leu Val Lys Gly Asn Thr Asn Pro
180          185          190
Ile Thr Ser Met Ser Ala Pro Val Gln Ala Ser Ala Pro Ala Leu Thr
195          200          205
Lys Ser Gln Thr Asp Arg Leu Glu Val Leu Leu Asn Pro Lys Asp Glu
210          215          220
Ile Ser Leu Asn Ser Gly Lys Pro Phe Arg Glu Leu Glu Ser Glu Leu
225          230          235          240
Leu Ser Arg Arg Lys Lys Asp Leu Gln Gln Ile Tyr Ala Glu Glu Arg
245          250          255
Glu Asn Tyr Leu Gly Lys Leu Glu Arg Glu Ile Thr Arg Phe Phe Val
260          265          270
Asp Arg Gly Phe Leu Glu Ile Lys Ser Pro Ile Leu Ile Pro Leu Glu
275          280          285
Tyr Ile Glu Arg Met Gly Ile Asp Asn Asp Thr Glu Leu Ser Lys Gln
290          295          300
Ile Phe Arg Val Asp Lys Asn Phe Cys Leu Arg Pro Met Leu Ala Pro
305          310          315          320
Asn Leu Ala Asn Tyr Leu Arg Lys Leu Asp Arg Ala Leu Pro Asp Pro
325          330          335
Ile Lys Ile Phe Glu Ile Gly Pro Cys Tyr Arg Lys Glu Ser Asp Gly
340          345          350
Lys Glu His Leu Glu Glu Phe Thr Met Leu Asn Phe Cys Gln Met Gly
355          360          365

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Ser	Gly	Cys	Thr	Arg	Glu	Asn	Leu	Glu	Ser	Ile	Ile	Thr	Asp	Phe	Leu
370						375					380				
Asn	His	Leu	Gly	Ile	Asp	Phe	Lys	Ile	Val	Gly	Asp	Ser	Cys	Met	Val
385					390					395					400
Phe	Gly	Asp	Thr	Leu	Asp	Val	Met	His	Gly	Asp	Leu	Glu	Leu	Ser	Ser
			405						410					415	
Ala	Val	Val	Gly	Pro	Ile	Pro	Leu	Asp	Arg	Glu	Trp	Gly	Ile	Asp	Lys
			420					425					430		
Pro	Trp	Ile	Gly	Ala	Gly	Phe	Gly	Leu	Glu	Arg	Leu	Leu	Lys	Val	Lys
		435					440					445			
His	Asp	Phe	Lys	Asn	Ile	Lys	Arg	Ala	Ala	Arg	Ser	Glu	Ser	Tyr	Tyr
	450					455					460				
Asn	Gly	Ile	Ser	Thr	Asn	Leu									
465					470										

<210> SEQ ID NO 54

<211> LENGTH: 471

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Methanosarcina mazei PylRS N346A, C348A mutant (PylRSAA)_amino acid sequence

<400> SEQUENCE: 54

Met	Ala	Cys	Pro	Val	Pro	Leu	Gln	Leu	Pro	Pro	Leu	Glu	Arg	Leu	Thr
1				5					10					15	
Leu	Asp	Asp	Lys	Lys	Pro	Leu	Asn	Thr	Leu	Ile	Ser	Ala	Thr	Gly	Leu
	20						25					30			
Trp	Met	Ser	Arg	Thr	Gly	Thr	Ile	His	Lys	Ile	Lys	His	His	Glu	Val
	35					40						45			
Ser	Arg	Ser	Lys	Ile	Tyr	Ile	Glu	Met	Ala	Cys	Gly	Asp	His	Leu	Val
	50					55					60				
Val	Asn	Asn	Ser	Arg	Ser	Ser	Arg	Thr	Ala	Arg	Ala	Leu	Arg	His	His
65				70					75					80	
Lys	Tyr	Arg	Lys	Thr	Cys	Lys	Arg	Cys	Arg	Val	Ser	Asp	Glu	Asp	Leu
			85					90					95		
Asn	Lys	Phe	Leu	Thr	Lys	Ala	Asn	Glu	Asp	Gln	Thr	Ser	Val	Lys	Val
	100						105						110		
Lys	Val	Val	Ser	Ala	Pro	Thr	Arg	Thr	Lys	Lys	Ala	Met	Pro	Lys	Ser
	115						120					125			
Val	Ala	Arg	Ala	Pro	Lys	Pro	Leu	Glu	Asn	Thr	Glu	Ala	Ala	Gln	Ala
	130					135					140				
Gln	Pro	Ser	Gly	Ser	Lys	Phe	Ser	Pro	Ala	Ile	Pro	Val	Ser	Thr	Gln
145				150					155					160	
Glu	Ser	Val	Ser	Val	Pro	Ala	Ser	Val	Ser	Thr	Ser	Ile	Ser	Ser	Ile
			165					170					175		
Ser	Thr	Gly	Ala	Thr	Ala	Ser	Ala	Leu	Val	Lys	Gly	Asn	Thr	Asn	Pro
		180					185						190		
Ile	Thr	Ser	Met	Ser	Ala	Pro	Val	Gln	Ala	Ser	Ala	Pro	Ala	Leu	Thr
	195					200						205			
Lys	Ser	Gln	Thr	Asp	Arg	Leu	Glu	Val	Leu	Leu	Asn	Pro	Lys	Asp	Glu
	210					215					220				
Ile	Ser	Leu	Asn	Ser	Gly	Lys	Pro	Phe	Arg	Glu	Leu	Glu	Ser	Glu	Leu
225					230					235					240

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Leu Ser Arg Arg Lys Lys Asp Leu Gln Gln Ile Tyr Ala Glu Glu Arg
 245 250 255
 Glu Asn Tyr Leu Gly Lys Leu Glu Arg Glu Ile Thr Arg Phe Phe Val
 260 265 270
 Asp Arg Gly Phe Leu Glu Ile Lys Ser Pro Ile Leu Ile Pro Leu Glu
 275 280 285
 Tyr Ile Glu Arg Met Gly Ile Asp Asn Asp Thr Glu Leu Ser Lys Gln
 290 295 300
 Ile Phe Arg Val Asp Lys Asn Phe Cys Leu Arg Pro Met Leu Ala Pro
 305 310 315 320
 Asn Leu Tyr Asn Tyr Leu Arg Lys Leu Asp Arg Ala Leu Pro Asp Pro
 325 330 335
 Ile Lys Ile Phe Glu Ile Gly Pro Cys Tyr Arg Lys Glu Ser Asp Gly
 340 345 350
 Lys Glu His Leu Glu Glu Phe Thr Met Leu Ala Phe Ala Gln Met Gly
 355 360 365
 Ser Gly Cys Thr Arg Glu Asn Leu Glu Ser Ile Ile Thr Asp Phe Leu
 370 375 380
 Asn His Leu Gly Ile Asp Phe Lys Ile Val Gly Asp Ser Cys Met Val
 385 390 395 400
 Tyr Gly Asp Thr Leu Asp Val Met His Gly Asp Leu Glu Leu Ser Ser
 405 410 415
 Ala Val Val Gly Pro Ile Pro Leu Asp Arg Glu Trp Gly Ile Asp Lys
 420 425 430
 Pro Trp Ile Gly Ala Gly Phe Gly Leu Glu Arg Leu Leu Lys Val Lys
 435 440 445
 His Asp Phe Lys Asn Ile Lys Arg Ala Ala Arg Ser Glu Ser Tyr Tyr
 450 455 460
 Asn Gly Ile Ser Thr Asn Leu
 465 470

<210> SEQ ID NO 55

<211> LENGTH: 470

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Methanosarcina mazei PylRS Y306A, N346A, C348A,
 Y384Fmutant (PylRSAAAF)_amino acid sequence

<400> SEQUENCE: 55

Ala Cys Pro Val Pro Leu Gln Leu Pro Pro Leu Glu Arg Leu Thr Leu
 1 5 10 15
 Asp Asp Lys Lys Pro Leu Asn Thr Leu Ile Ser Ala Thr Gly Leu Trp
 20 25 30
 Met Ser Arg Thr Gly Thr Ile His Lys Ile Lys His His Glu Val Ser
 35 40 45
 Arg Ser Lys Ile Tyr Ile Glu Met Ala Cys Gly Asp His Leu Val Val
 50 55 60
 Asn Asn Ser Arg Ser Ser Arg Thr Ala Arg Ala Leu Arg His His Lys
 65 70 75 80
 Tyr Arg Lys Thr Cys Lys Arg Cys Arg Val Ser Asp Glu Asp Leu Asn
 85 90 95
 Lys Phe Leu Thr Lys Ala Asn Glu Asp Gln Thr Ser Val Lys Val Lys

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100					105					110					
Val	Val	Ser	Ala	Pro	Thr	Arg	Thr	Lys	Lys	Ala	Met	Pro	Lys	Ser	Val
	115						120					125			
Ala	Arg	Ala	Pro	Lys	Pro	Leu	Glu	Asn	Thr	Glu	Ala	Ala	Gln	Ala	Gln
	130					135					140				
Pro	Ser	Gly	Ser	Lys	Phe	Ser	Pro	Ala	Ile	Pro	Val	Ser	Thr	Gln	Glu
145					150					155					160
Ser	Val	Ser	Val	Pro	Ala	Ser	Val	Ser	Thr	Ser	Ile	Ser	Ser	Ile	Ser
				165					170					175	
Thr	Gly	Ala	Thr	Ala	Ser	Ala	Leu	Val	Lys	Gly	Asn	Thr	Asn	Pro	Ile
			180						185					190	
Thr	Ser	Met	Ser	Ala	Pro	Val	Gln	Ala	Ser	Ala	Pro	Ala	Leu	Thr	Lys
		195					200					205			
Ser	Gln	Thr	Asp	Arg	Leu	Glu	Val	Leu	Leu	Asn	Pro	Lys	Asp	Glu	Ile
	210					215					220				
Ser	Leu	Asn	Ser	Gly	Lys	Pro	Phe	Arg	Glu	Leu	Glu	Ser	Glu	Leu	Leu
225						230					235				240
Ser	Arg	Arg	Lys	Lys	Asp	Leu	Gln	Gln	Ile	Tyr	Ala	Glu	Glu	Arg	Glu
				245					250					255	
Asn	Tyr	Leu	Gly	Lys	Leu	Glu	Arg	Glu	Ile	Thr	Arg	Phe	Phe	Val	Asp
		260						265						270	
Arg	Gly	Phe	Leu	Glu	Ile	Lys	Ser	Pro	Ile	Leu	Ile	Pro	Leu	Glu	Tyr
		275					280					285			
Ile	Glu	Arg	Met	Gly	Ile	Asp	Asn	Asp	Thr	Glu	Leu	Ser	Lys	Gln	Ile
	290					295					300				
Phe	Arg	Val	Asp	Lys	Asn	Phe	Cys	Leu	Arg	Pro	Met	Leu	Ala	Pro	Asn
305						310					315				320
Leu	Ala	Asn	Tyr	Leu	Arg	Lys	Leu	Asp	Arg	Ala	Leu	Pro	Asp	Pro	Ile
				325					330					335	
Lys	Ile	Phe	Glu	Ile	Gly	Pro	Cys	Tyr	Arg	Lys	Glu	Ser	Asp	Gly	Lys
		340						345					350		
Glu	His	Leu	Glu	Glu	Phe	Thr	Met	Leu	Ala	Phe	Ala	Gln	Met	Gly	Ser
		355					360					365			
Gly	Cys	Thr	Arg	Glu	Asn	Leu	Glu	Ser	Ile	Ile	Thr	Asp	Phe	Leu	Asn
	370					375					380				
His	Leu	Gly	Ile	Asp	Phe	Lys	Ile	Val	Gly	Asp	Ser	Cys	Met	Val	Phe
385						390					395				400
Gly	Asp	Thr	Leu	Asp	Val	Met	His	Gly	Asp	Leu	Glu	Leu	Ser	Ser	Ala
				405					410					415	
Val	Val	Gly	Pro	Ile	Pro	Leu	Asp	Arg	Glu	Trp	Gly	Ile	Asp	Lys	Pro
				420				425						430	
Trp	Ile	Gly	Ala	Gly	Phe	Gly	Leu	Glu	Arg	Leu	Leu	Lys	Val	Lys	His
		435					440					445			
Asp	Phe	Lys	Asn	Ile	Lys	Arg	Ala	Ala	Arg	Ser	Glu	Ser	Tyr	Tyr	Asn
	450					455					460				
Gly	Ile	Ser	Thr	Asn	Leu										
465															470

<210> SEQ ID NO 56

<211> LENGTH: 4437

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: pCAGGS-T7-RNAP
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1)..(380)
<223> OTHER INFORMATION: CMV enhancer
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (382)..(659)
<223> OTHER INFORMATION: Chicken b-actin promoter
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (661)..(1678)
<223> OTHER INFORMATION: Chimeric intron chicken actin
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1789)..(4437)
<223> OTHER INFORMATION: T7-RNAP

<400> SEQUENCE: 56

gacattgatt attgactagt tattaatagt aatcaattac ggggtcatta gttcatagcc      60
catatatgga gttccgcgtt acataactta cggtaaatgg ccgcctggc tgaccgccca      120
acgacccccg cccattgacg tcaataatga cgtatgttcc catagtaacg ccaataggga      180
ctttccattg acgtcaatgg gtggagtatt tacggtaaac tgcccacttg gcagtacatc      240
aagtgtatca tatgccaagt acgcccccta ttgacgtcaa tgacggtaaa tggccccgct      300
ggcattatgc ccagtacatg accttatggg actttectac ttggcagtac atctacgtat      360
tagtcatcgc tattaccatg gtcgaggtga gcccacggtt ctgcttcaact ctccccatct      420
cccccccctc cccaccccga attttgtatt tatttatatt ttaattattt tgtgcagcga      480
tggtggcggg gggggggggg gggcgcgcg caggcggggc gggcgggggc gagggcgggg      540
gcggggcgag gcgggagagt gcggcggcag ccaatcagag cggcgcgctc cgaaagtttc      600
cttttatggc gaggcgcgcg cggcgcgcg cctataaaaa gcgaagcgcg cggcgggcgg      660
ggagtcgctg cgacgctgcc ttgcgccgtt gcccgcctcc gccgcgcctc cgcgcgcgcc      720
gcccgcgctc tgactgacgg cgttactccc acaggtgagc gggcgggacg gcccttctcc      780
tcgggctgtt aattagcgtt tgggtttaatg acggcttgtt tcttttctgt ggctgcgtga      840
aagccttgag gggctccggg agggcccttt gtgcgggggg agcggtcggg ggggtgcgtg      900
cgtgtgtgtg tgcgtgggga gcgcccgctg cggctccgcg ctgcccgcg gctgtgagcg      960
ctgcgggcgc ggcgcggggc tttgtgcgct ccgcagtggt cgcgagggga gcgcggccgg      1020
gggcgggtgcc ccgcggtgcg ggggggggctg cgaggggaac aaaggctgcg tgcgggggtg      1080
gtgcgtgggg ggggtgagcag ggggtgtggg cgcgtcggtc gggctgcaac cccccctgca      1140
ccccctccc cgagttgctg agcacggccc ggcttcgggt gcggggctcc gtacggggcg      1200
tggcgcgggg ctgcgcgtgc cgggcggggg gtggcggcag gtgggggtgc cgggcggggc      1260
ggggccgcct cgggcggggg agggctcggg ggaggggcgc ggcggccccc ggagcgccgg      1320
cggctgtcga ggcgcggcga gccgcagcca ttgcctttta tggtaatcgt gcgagagggc      1380
gcagggaactt cctttgtccc aaatctgtgc ggagccgaaa tctgggaggg gccgcgcac      1440
ccccctacg gggcgcgggg cgaagcggtg cggcgccggc aggaaggaaa tggcggggga      1500
gggccttcgt gcgtgcgcgc gccgcggtcc ccttctccct ctccagcctc ggggtgtgcc      1560
gcggggggac ggctgccttc ggggggggac gggcagggcg gggttcggct tctggcgtgt      1620
gaccggcggc tctagagcct ctgctaacca tgttcatgcc ttcttctttt tctacagct      1680

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cctgggcaac	gtgctggta	ttgtgctgtc	tcatcatttt	ggcaaagaat	tcgagctcat	1740
cggatccaga	tctgatccgg	atttactaac	tggaagaggc	actaaatgaa	cacgattaac	1800
atcgctaaga	acgacttctc	tgacatcgaa	ctggctgcta	tcccggtcaa	cactctggct	1860
gaccattacg	gtgagcggtt	agctcgcgaa	cagttggccc	ttgagcatga	gtcttacgag	1920
atgggtgaag	cacgcttccg	caagatgttt	gagcgtcaac	ttaaagctgg	tgagggtgcg	1980
gataacgctg	cgcgaagcc	tctcatcact	accctactcc	ctaagatgat	tgacgcgcatc	2040
aacgactggt	ttgaggaagt	gaaagctaag	cgcggcaagc	gcccgcacgc	cttcagttc	2100
ctgcaagaaa	tcaagccgga	agccgtagcg	tacatcacca	ttaagaccac	tctggcttgc	2160
ctaaccagtg	ctgacaatac	aacggttcag	gctgtagcaa	gcgcaatcgg	tcggggccatt	2220
gaggacgagg	ctcgtctcgg	tcgtatccgt	gaccttgaag	ctaagcactt	caagaaaaac	2280
gttgaggaaac	aactcaacaa	gcgcgtaggg	cacgtctaca	agaaagcatt	tatgcaagtt	2340
gtcgaggctg	acatgctctc	taagggtcta	ctcggtggcg	aggcgtggtc	ttcgtggcat	2400
aaggaagact	ctattcatgt	aggagtacgc	tgcatcgaga	tgctcattga	gtcaaccgga	2460
atggttagct	tacaccgcca	aaatgctggc	gtagtaggtc	aagactctga	gactatcgaa	2520
ctcgacctg	aatacgctga	ggctatcgca	accctgacg	gtgcgctggc	tgcatctct	2580
ccgatgttcc	aaccttgctg	agttctctct	aagccgtgga	ctggcattac	tggtggtggc	2640
tattgggcta	acggtcgctg	tcctctggcg	ctggtgcgta	ctcacagtaa	gaaagcactg	2700
atgcgctacg	aagacgttta	catgcctgag	gtgtacaaag	cgattaacat	tcgcgaaaac	2760
accgcatgga	aaatcaacaa	gaaagtctta	gcggctcgca	acgtaatcac	caagtggaa	2820
cattgtccgg	tcgaggacat	ccttcgcat	gagcgtgaag	aactcccgat	gaaaccggaa	2880
gacatcgaca	tgaatcctga	ggctctcacc	gcgtggaaac	gtgctgcgcg	tgctgtgtac	2940
cgaaggaca	aggctcgcaa	gtctcgccgt	atcagccttg	agttcatgct	tgagcaagcc	3000
aataagtttg	ctaaccataa	ggccatctgg	ttcccttaca	acatggactg	gcgcggctcg	3060
gtttacgctg	tgtcaatggt	caaccgcgaa	ggtaacgata	tgaccaaagg	actgcttacg	3120
ctggcgaaa	gtaaaccaat	cggtaaggaa	ggttactact	ggctgaaaa	ccacggtgca	3180
aactgtgcgg	gtgtcgataa	ggttcggtc	cctgagcgca	tcaagttcat	tgaggaaaac	3240
cacgagaaca	tcatggcttg	cgtaagtct	ccactggaga	acacttgggtg	ggctgagcaa	3300
gattctccgt	tctgttctct	tgctgtctgc	tttgagtacg	ctgggtgaca	gcaccacggc	3360
ctgagctata	actgtccct	tccgctggcg	tttgacgggt	cttgcctctg	catccagcac	3420
ttctccgca	tgctccgaga	tgaggtaggt	ggtcgcgcgg	ttaacttgct	tcctagttaa	3480
accgttcagg	acatctacgg	gattgttgct	aagaaagtca	acgagattct	acaagcagac	3540
gcaatcaatg	ggaccgataa	cgaagtagtt	accgtgaccg	atgagaacac	tggtgaaatc	3600
tctgagaaa	tcaagctggg	cactaaggca	ctggctggtc	aatggctggc	ttacggtgtt	3660
actcgacgtg	tgactaagcg	ttcagtcagt	acgctggctt	acgggtccaa	agagttcggc	3720
ttccgtcaac	aagtgtgga	agataccatt	cagccagcta	ttgattccgg	caagggtctg	3780
atgttcaact	agccgaatca	ggctgctgga	tacatggcta	agctgatttg	ggaatctgtg	3840
agcgtgacgg	tggtagctgc	ggttgaagca	atgaactggc	ttaagtctgc	tgctaagctg	3900
ctggctgctg	aggtaaaaga	taagaagact	ggagagattc	ttcgcaagcg	ttgcgctgtg	3960

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cattgggtaa ctctgatgg ttccctgtg tggcaggaat acaagaagcc tattcagacg 4020
cgcttgaacc tgatgttctt cggtcagttc cgettacagc ctaccattaa caccaacaaa 4080
gatagcgaga ttgatgcaca caaacaggag tctggtatcg ctctaactt tgtacacagc 4140
caagacggta gccaccttcg taagactgta gtgtgggcac acgagaagta cggaatcgaa 4200
tcttttgcac tgattcacga ctccctcggg accattccgg ctgacgctgc gaacctgttc 4260
aaagcagtg cgcgaaactat ggttgacaca tatgagtctt gtgatgtact ggctgatttc 4320
tacgaccagt tcgctgacca gttgcacgag tctcaattgg acaaaatgcc agcacttccg 4380
gctaaaggta acttgaacct cgtgacatc ttagagtcgg acttcgcgtt cgcgtaa 4437

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<210> SEQ ID NO 57
<211> LENGTH: 515
<212> TYPE: PRT
<213> ORGANISM: Mus musculus
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1) .. (515)
<223> OTHER INFORMATION: dsRNA-binding polypeptide (uniprot: Q03963)

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<400> SEQUENCE: 57

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Met Ala Ser Asp Thr Pro Gly Phe Tyr Met Asp Lys Leu Asn Lys Tyr
1      5      10      15
Arg Gln Met His Gly Val Ala Ile Thr Tyr Lys Glu Leu Ser Thr Ser
20     25     30
Gly Pro Pro His Asp Arg Arg Phe Thr Phe Gln Val Leu Ile Asp Glu
35     40     45
Lys Glu Phe Pro Glu Ala Lys Gly Arg Ser Lys Gln Glu Ala Arg Asn
50     55     60
Ala Ala Ala Lys Leu Ala Val Asp Ile Leu Asp Asn Glu Asn Lys Val
65     70     75     80
Asp Cys His Thr Ser Ala Ser Glu Gln Gly Leu Phe Val Gly Asn Tyr
85     90     95
Ile Gly Leu Val Asn Ser Phe Ala Gln Lys Lys Lys Leu Ser Val Asn
100    105    110
Tyr Glu Gln Cys Glu Pro Asn Ser Glu Leu Pro Gln Arg Phe Ile Cys
115    120    125
Lys Cys Lys Ile Gly Gln Thr Met Tyr Gly Thr Gly Ser Gly Val Thr
130    135    140
Lys Gln Glu Ala Lys Gln Leu Ala Ala Lys Glu Ala Tyr Gln Lys Leu
145    150    155    160
Leu Lys Ser Pro Pro Lys Thr Ala Gly Thr Ser Ser Ser Val Val Thr
165    170    175
Ser Thr Phe Ser Gly Phe Ser Ser Ser Ser Met Thr Ser Asn Gly
180    185    190
Val Ser Gln Ser Ala Pro Gly Ser Phe Ser Ser Glu Asn Val Phe Thr
195    200    205
Asn Gly Leu Gly Glu Asn Lys Arg Lys Ser Gly Val Lys Val Ser Pro
210    215    220
Asp Asp Val Gln Arg Asn Lys Tyr Thr Leu Asp Ala Arg Phe Asn Ser
225    230    235    240
Asp Phe Glu Asp Ile Glu Glu Ile Gly Leu Gly Gly Phe Gly Gln Val
245    250    255

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Phe Lys Ala Lys His Arg Ile Asp Gly Lys Arg Tyr Ala Ile Lys Arg
 260 265 270
 Val Lys Tyr Asn Thr Glu Lys Ala Glu His Glu Val Gln Ala Leu Ala
 275 280 285
 Glu Leu Asn His Val Asn Ile Val Gln Tyr His Ser Cys Trp Glu Gly
 290 295 300
 Val Asp Tyr Asp Pro Glu His Ser Met Ser Asp Thr Ser Arg Tyr Lys
 305 310 315 320
 Thr Arg Cys Leu Phe Ile Gln Met Glu Phe Cys Asp Lys Gly Thr Leu
 325 330 335
 Glu Gln Trp Met Arg Asn Arg Asn Gln Ser Lys Val Asp Lys Ala Leu
 340 345 350
 Ile Leu Asp Leu Tyr Glu Gln Ile Val Thr Gly Val Glu Tyr Ile His
 355 360 365
 Ser Lys Gly Leu Ile His Arg Asp Leu Lys Pro Gly Asn Ile Phe Leu
 370 375 380
 Val Asp Glu Arg His Ile Lys Ile Gly Asp Phe Gly Leu Ala Thr Ala
 385 390 395 400
 Leu Glu Asn Asp Gly Lys Ser Arg Thr Arg Arg Thr Gly Thr Leu Gln
 405 410 415
 Tyr Met Ser Pro Glu Gln Leu Phe Leu Lys His Tyr Gly Lys Glu Val
 420 425 430
 Asp Ile Phe Ala Leu Gly Leu Ile Leu Ala Glu Leu Leu His Thr Cys
 435 440 445
 Phe Thr Glu Ser Glu Lys Ile Lys Phe Phe Glu Ser Leu Arg Lys Gly
 450 455 460
 Asp Phe Ser Asn Asp Ile Phe Asp Asn Lys Glu Lys Ser Leu Leu Lys
 465 470 475 480
 Lys Leu Leu Ser Glu Lys Pro Lys Asp Arg Pro Glu Thr Ser Glu Ile
 485 490 495
 Leu Lys Thr Leu Ala Glu Trp Arg Asn Ile Ser Glu Lys Lys Lys Arg
 500 505 510
 Asn Thr Cys
 515

<210> SEQ ID NO 58
 <211> LENGTH: 552
 <212> TYPE: PRT
 <213> ORGANISM: Pan troglodytes
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1)..(552)
 <223> OTHER INFORMATION: dsRNA-binding polypeptide (uniprot: B9UKH7)

<400> SEQUENCE: 58

Met Ala Gly Asp Leu Ser Ala Gly Phe Phe Met Glu Glu Leu Asn Thr
 1 5 10 15
 Tyr Arg Gln Lys Gln Gly Val Val Leu Lys Tyr Gln Glu Leu Pro Asn
 20 25 30
 Ser Gly Pro Pro His Asp Arg Arg Phe Thr Phe Arg Val Ile Ile Asp
 35 40 45
 Gly Arg Glu Phe Pro Glu Gly Glu Gly Arg Ser Lys Lys Glu Ala Lys
 50 55 60

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Asn	Ala	Ala	Ala	Lys	Leu	Ala	Val	Glu	Ile	Leu	Asn	Lys	Glu	Lys	Lys
65					70					75					80
Ala	Val	Ser	Pro	Leu	Val	Leu	Thr	Thr	Thr	Asn	Ser	Ser	Glu	Gly	Leu
			85						90					95	
Ser	Met	Gly	Asn	Tyr	Ile	Gly	Leu	Ile	Asn	Arg	Ile	Ala	Gln	Lys	Lys
			100					105					110		
Arg	Leu	Thr	Val	Asn	Tyr	Glu	Gln	Cys	Ala	Ser	Gly	Val	His	Gly	Pro
		115					120					125			
Glu	Gly	Phe	His	Tyr	Lys	Cys	Lys	Met	Gly	Gln	Lys	Glu	Tyr	Ser	Ile
	130					135					140				
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Val	His	Tyr	Asn	Gly	Cys	Trp	Asp	Gly	Phe	Asp	Tyr	Asp	Pro	Glu	Thr
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Lys	Gly	Thr	Leu	Arg	Tyr	Met	Ser	Pro	Glu	Gln	Ile	Ser	Ser	Gln	Asp
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1. A eukaryotic cell comprising:
 - (a) a polynucleotide encoding a prokaryotic aminoacyl tRNA synthetase (pRS),
 - (b) a polynucleotide encoding a prokaryotic tRNA (ptRNA) and one or more than one ribozyme, wherein the pRS is capable of acylating the ptRNA, and wherein the sequences encoding the ptRNA and the ribozyme(s) are linked such that transcription produces an RNA molecule comprising both the ptRNA and the ribozyme(s) (ptRNA-ribozyme).
2. The eukaryotic cell of claim 1, wherein the ribozyme is selected from a hammerhead ribozyme, a hairpin ribozyme, a Varkud satellite ribozyme, a glucosamine-6-phosphate synthetase riboswitch and a hepatitis delta virus (HDV) ribozyme.
3. The eukaryotic cell of claim 1, wherein the ribozyme is a HDV ribozyme which, in the ptRNA-ribozyme, is directly linked to the 3' end of the ptRNA.
4. The eukaryotic cell of claim 1, wherein:
 - (i) transcriptional synthesis of the ptRNA-ribozyme is controlled by a tetracycline-responsive promoter element (TRE), or
 - (ii) the ptRNA-ribozyme is transcriptionally synthesized by an RNA polymerase, wherein expression of said RNA polymerase is controlled by a TRE.
5. The eukaryotic cell of claim 1, wherein the transcription of the pRS-encoding sequence is controlled by a tetracycline-responsive promoter element (TRE).
6. The eukaryotic cell of claim 4, further comprising a polynucleotide encoding a protein which binds to a tetracycline operator (tetO) sequence (tetO-binding protein), wherein the tetO-binding protein is selected from the group consisting of:
 - (i) a tetracycline repressor (tetR),
 - (ii) a tetracycline-controlled transcriptional activator (tTA), and
 - (iii) a reverse tTA (rtTA).
7. The eukaryotic cell of claim 1, wherein:
 - (i) transcriptional synthesis of the ptRNA-ribozyme is controlled by a T7 promoter, and/or
 - (ii) transcription of the pRS-encoding sequence is controlled by a T7 promoter, and wherein the eukaryotic cell further comprises
 - (c) a polynucleotide encoding a T7 RNA polymerase (T7RNAP).
8. The eukaryotic cell of claim 7, wherein expression of the T7RNAP is controlled by a TRE.
9. The eukaryotic cell of claim 7, wherein the T7RNAP carries
 - (i) a nuclear localization signal (NLS), or
 - (ii) a nuclear export signal (NES), or
 - (iii) none of NES and NLS.
10. The eukaryotic cell of claim 1, further comprising a polynucleotide encoding a dsRNA-binding polypeptide.
11. A polynucleotide encoding an RNA molecule comprising a ptRNA and one or more than one ribozyme (ptRNA-ribozyme) as defined in claim 1, wherein:
 - a) the ribozyme is a HDV ribozyme which, in the ptRNA-ribozyme, is directly linked to the 3' end of the ptRNA; or
 - b) the ptRNA is derived from a *Methanosarcina* species, in particular a pyrrolysyl tRNA of *M. mazei*; or
 - c) wherein a ptRNA molecule is encoded missing the 3'-terminal aminoacyl acceptor stem motif CCA; or
 - d) the ptRNA is encoded by the following nucleotide sequence (SEQ ID NO:60) ggaaacctgatcatgtagatcgaatggactnnnaatccgttcagccgggttagattcccggggtttccg wherein nnn encodes an anticodon.
12. A polynucleotide, or combination of two or more polynucleotides, comprising a nucleotide sequence encoding an RNA molecule comprising a ptRNA and one or more than one ribozyme (ptRNA-ribozyme) as defined in claim 1, and:
 - (i) a nucleotide sequence encoding a tetO-binding protein selected from the group consisting of: (a) a tetracycline repressor (tetR), (b) a tetracycline-controlled transcriptional activator (tTA), and (c), a reverse tTA (rtTA), or
 - (ii) a nucleotide sequence encoding a pRS as defined in claim 1, or
 - (ii) both of (i) and (ii).
13. A method for preparing a polypeptide of interest (POI) having one or more than one non-canonical amino acid (ncAA) residue in its amino acid sequence, wherein the method comprises:
 - (a) expressing, in a eukaryotic cell:
 - a prokaryotic aminoacyl tRNA synthetase (pRS) and an RNA molecule comprising a prokaryotic tRNA (ptRNA) and one or more than one ribozyme (ptRNA-ribozyme); and concomitantly or sequentially
 - (b) expressing the POI in the eukaryotic cell in the presence of one or more than one ncAA or salt thereof corresponding to the one or more than one ncAA residue of the POI; and
 - (c) optionally recovering the expressed POI; wherein the pRS is capable of acylating the ptRNA with the ncAA or the salt thereof, wherein the POI is encoded by a nucleotide sequence that comprises one or more than one selector codon encoding the one or more than one ncAA residue, and wherein the selector codon is the reverse complement of the anticodon of the ptRNA.
14. The method of claim 13, wherein the transcriptional synthesis of the ptRNA-ribozyme is controlled by a T7 promoter, and wherein the method further comprises expressing a T7 RNA polymerase (T7RNAP) in the eukaryotic cell prior to or concomitantly with the thereby controlled expression of the ptRNA-ribozyme (i) or the pRS (ii) or both (iii), respectively.
15. The method of claim 13, wherein:
 - (i) transcriptional synthesis of the ptRNA-ribozyme is controlled by a tetracycline-responsive promoter element (TRE), or

(ii) the ptRNA-ribozyme is transcriptionally synthesized by an RNA polymerase, wherein the expression of said RNA polymerase is controlled by a TRE;
wherein the method further comprises expressing a tetracycline repressor (tetR) in the eukaryotic cell at least prior to the expression of the ptRNA-ribozyme, and wherein expression of the ptRNA-ribozyme involves contacting the eukaryotic cell with tetracycline, doxycycline, or a functional analogue thereof, thereby inducing transcriptional synthesis of the ptRNA-ribozyme.

16. The method of claim **13**, wherein:

- (i) transcription of the pRS-encoding sequence is controlled by a tetracycline-responsive promoter element (TRE), or
- (ii) transcription of the pRS-encoding sequence is catalyzed by an RNA polymerase, wherein the expression of said RNA polymerase is controlled by a TRE;
wherein the method further comprises expressing a tetracycline repressor (tetR) in the eukaryotic cell at least prior to the expression of the pRS; and
wherein expression of the pRS involves contacting the eukaryotic cell with tetracycline, doxycycline, or a functional analogue thereof, thereby inducing transcription of the pRS-encoding sequence.

17. The method of claim **15**, wherein:

- (i) transcriptional synthesis of the ptRNA-ribozyme is controlled by a T7 promoter, or
- (ii) transcription of the pRS-encoding sequence is controlled by a T7 promoter, or
- (iii) both of (i) and (ii);

and wherein the method further comprises:

expressing a T7 RNA polymerase (T7RNAP) in the eukaryotic cell prior to or concomitantly with the thereby controlled expression of the ptRNA-ribozyme (i) or the pRS (ii) or both (iii), respectively, wherein expression of the T7RNAP is controlled by a TRE, and

expressing a tetR in the eukaryotic cell at least concomitantly with the expression of the T7RNAP;

wherein expression of the T7RNAP involves contacting the eukaryotic cell with tetracycline, doxycycline, or a functional analogue thereof, thereby inducing transcription of the T7RNAP-encoding sequence.

18. The method of claim **13**, wherein:

- (i) transcriptional synthesis of the ptRNA-ribozyme is controlled by a tetracycline-responsive promoter element (TRE), or
- (ii) the ptRNA-ribozyme is transcriptionally synthesized by an RNA polymerase, wherein expression of said RNA polymerase is controlled by a TRE;

wherein the method further comprises:

- (1) expressing a tetracycline-controlled transcriptional activator (tTA) in the eukaryotic cell prior to and concomitantly with the expression of the ptRNA-ribozyme, and
- (2) keeping the eukaryotic cell prior to the expression of the ptRNA-ribozyme in contact with tetracycline, doxycycline, or a functional analogue thereof, thereby suppressing the transcriptional synthesis of the ptRNA-ribozyme, and then reducing the concentration of, or preferably removing, said tetracycline, doxycycline, or functional analogue thereof so as to induce transcriptional synthesis of the ptRNA-ribozyme.

19. The method of claim **13**, wherein:

- (i) transcription of the pRS-encoding sequence is controlled by a TRE, or
- (ii) transcription of the pRS-encoding sequence is catalyzed by an RNA polymerase, wherein the expression of said RNA polymerase is controlled by a TRE;

wherein the method further comprises:

- (1) expressing a tetracycline-controlled transcriptional activator (tTA) in the eukaryotic cell prior to and concomitantly with the expression of the pRS, and
- (2) keeping the eukaryotic cell prior to the expression of the pRS in contact with tetracycline, doxycycline, or a functional analogue thereof, thereby suppressing transcription of the pRS-encoding sequence, and then reducing the concentration of, or preferably removing, said tetracycline, doxycycline, or functional analogue thereof so as to induce transcription of the pRS-encoding sequence.

20. The method of claim **18**, wherein:

- (i) transcriptional synthesis of the ptRNA-ribozyme is controlled by a T7 promoter, or
- (ii) transcription of the pRS-encoding sequence is controlled by a T7 promoter, or
- (iii) both of (i) and (ii);

wherein the method further comprises:

- (3) expressing a T7 RNA polymerase (T7RNAP) in the eukaryotic cell prior to or concomitantly with the thereby controlled expression of the ptRNA-ribozyme (i) or the pRS (ii) or both (iii), respectively, wherein expression of the T7RNAP is controlled by a TRE,
- (4) expressing a tTA in the eukaryotic cell prior to and concomitantly with the expression of the T7RNAP, and
- (5) keeping the eukaryotic cell prior to the expression of the T7RNAP in contact with tetracycline, doxycycline, or a functional analogue thereof, thereby suppressing transcription of the T7RNAP-encoding sequence, and then reducing the concentration of, or preferably removing, said tetracycline, doxycycline, or functional analogue thereof so as to induce transcription of the T7RNAP-encoding sequence.

21. The method of claim **13**, wherein:

- (i) transcriptional synthesis of the ptRNA-ribozyme is controlled by a tetracycline-responsive promoter element (TRE), or
- (ii) the ptRNA-ribozyme is transcriptionally synthesized by an RNA polymerase, wherein the expression of said RNA polymerase is controlled by a TRE;
wherein the method further comprises expressing a reverse tTA (rtTA) in the eukaryotic cell at least concomitantly with the expression of the ptRNA, and
wherein expression of the ptRNA-ribozyme involves contacting the eukaryotic cell with doxycycline, or a functional analogue thereof, thereby inducing transcriptional synthesis of the ptRNA-ribozyme.

22. The method of claim **13**, wherein:

- (i) transcription of the pRS-encoding sequence is controlled by a tetracycline-responsive promoter element (TRE), or
- (ii) transcription of the pRS-encoding sequence is catalyzed by an RNA polymerase, wherein the expression of said RNA polymerase is controlled by a TRE;
wherein the method further comprises expressing a reverse tTA (rtTA) in the eukaryotic cell at least concomitantly with the expression of the pRS; and

wherein expression of the pRS involves contacting the eukaryotic cell with doxycycline, or a functional analogue thereof, thereby inducing transcription of the pRS-encoding sequence.

23. A kit for preparing a polypeptide of interest (POI) having one or more than one non-canonical amino acid (ncAA) residue in its amino acid sequence, wherein the kit comprises one or more than one ncAA, or salt thereof, corresponding to the one or more than one ncAA residue of the POI, and:

(a) a polynucleotide of claim **11**.

24. A kit for preparing a polypeptide of interest (POI) having one or more than one non-canonical amino acid (ncAA) residue in its amino acid sequence, wherein the kit comprises one or more than one ncAA, or salt thereof, corresponding to the one or more than one ncAA residue of the POI, and:

(a) a polynucleotide, or combination of two or more polynucleotides, of claim **12**, each in combination with
(b) a polynucleotide encoding a dsRNA-binding polypeptide.

25. A kit for preparing a polypeptide of interest (POI) having one or more than one non-canonical amino acid (ncAA) residue in its amino acid sequence, wherein the kit comprises one or more than one ncAA, or salt thereof, corresponding to the one or more than one ncAA residue of the POI, and an eukaryotic cell of claim **1**.

* * * * *